



GE VERNOVA

ADMS

16.25.1

OMS Integration Programmer Guide

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1. Introduction

Important

It is recommended to use the Call Server and Crew Server HTTP interfaces instead of the legacy TCP/IP interfaces for communicating with external systems, such as External Interactive Voice Response (IVR), Automated Metering Infrastructure (AMI), Customer Information System (CIS), and Workforce Management System (WFM).

This section provides an overview of the ADMS Call server and Crew server interface design.

The programming paradigm for these interfaces is TCP or HTTP connections with XML messaging.

For purposes of this document and from the ADMS perspective, the third-party or integrated software component might be in the role of a client (i.e., an initiator of interactions) or a server (i.e., a recipient of interactions). Often an enterprise or integrator will develop a channel adapter component between systems. Therefore, in the examples in this document the third-party component is often referred to generically as a channel adapter, which might be developed by the customer, their integrator, or a software product company. In addition, some customer architectures will utilize an Enterprise Service Bus, so that component is included in some of the examples.

The ADMS product is made up of multiple components, including clients and servers. The interfaces relevant to this document are provided by the ADMS Call server and the ADMS Crew server. For the sake of simplicity, these are sometimes referred to as ADMS.

1.1. High Availability

1.1.1. ADMS Redundancy Management

Redundancy of ADMS and its processes is managed by two Habitat applications: CFGMAN and PROCMAN.

1.1.1.1. Configuration Management (CFGMAN)

The redundancy of ADMS is managed by the Habitat Configuration Management application (CFGMAN). CFGMAN monitors hardware and software failure, takes appropriate remedial actions, and provides overall system status to maintenance personnel.

CFGMAN assigns the overall role to the ADMS server. Three mutually exclusive roles can be assigned:

- **ENABLED:** Component is available and is actively performing. In a dual redundant server configuration, it is also called PRIMARY.

- **DISABLED:** Component is not performing any real-time function. The computer may be running any other kind of software but is not in state to become ENABLED should a failure of the present ENABLED computer occur. Sometimes the term OFFLINE is also used.
- **STANDBY:** Component is available and is ready to accept ENABLED role if required. In a dual redundant server configuration, the term BACKUP computer is also used.

1.1.1.2. Central Process Management (PROCMAN)

PROCMAN is the Habitat process management utility responsible for managing the entire collection of processes making up the ADMS solution. The term "management" in this context means deciding the order in which tasks are to be started, monitoring the status of these tasks, and deciding when they should be restarted, as well as sequencing the execution of internal functions within the tasks themselves.

PROCMAN can also stop or start a task depending on the overall role (enabled, disabled or standby) of the ADMS server.

1.1.2. High Availability of ADMS Interfaces Modules

Within ADMS, two applications manage the interfaces with WFM, CIS, IVR, and AMI:

- The Crew server (WFM interface)
- The Call server (CIS, IVR, and AMI interfaces)

Using both CFGMAN and PROCMAN processes, the high availability of the Crew server and Call server can be achieved as indicated below:

- When ADMS has been assigned the enabled (or Primary) role by CFGMAN, then both the Crew server and the Call server are started by PROCMAN.
- When ADMS has been assigned the standby/disabled role by CFGMAN, then both the Crew server and the Call server are stopped by PROCMAN.

The below image shows the high-availability design. On the client side, there can be a channel adapter, an enterprise service bus, or direct client code.

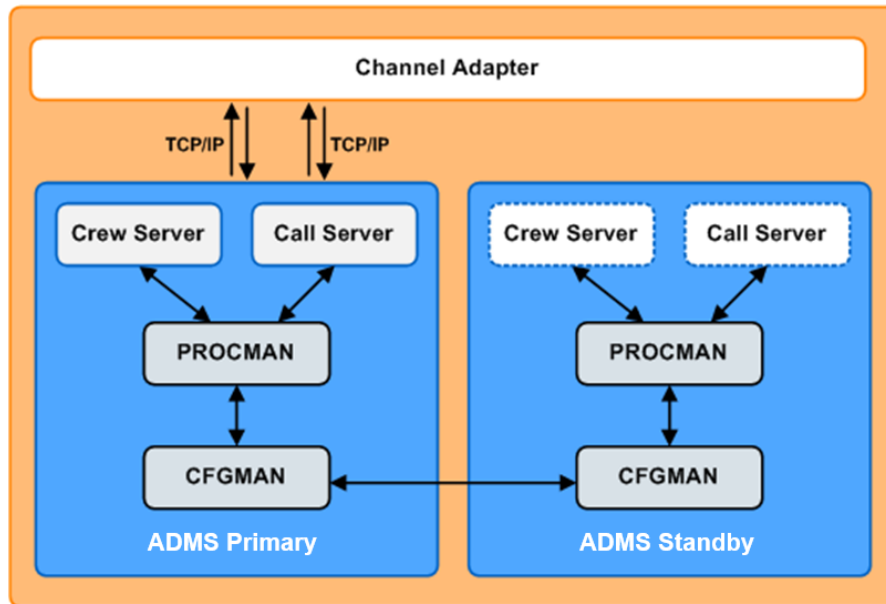


Figure 1. High Availability Design

1.1.3. Interface with the Channel Adapter or Other Client

Interface between the Channel Adapter (or other client) and the Crew/Call server modules is achieved through HTTP.

To interface with the ADMS interfaces modules in a high availability environment, the Channel Adapter must:

- Monitor the status of the ADMS interface modules (Crew server/Call server)
- Switch communication to another ADMS server after detecting that an ADMS interface module is down.

1.2. Security

All messages between an ADMS web API application and an external application (for example, WFM) can be exchanged using the HTTPS secure communications protocol (two way SSL certificates for authentication).

To provide additional security, you can configure WFM to require an authorization header for all requests from ADMS. The authorization header, combined with WFM web service configuration parameters, is used by WFM to verify that requests come from a trusted client (ADMS).

Requests from WFM to ADMS do not need to include an authorization header. WFM sends messages to ADMS only in response to message requests from ADMS. HTTPS ensures that the response from WFM (which contains requests for ADMS) is not tampered with during transmission.

ADMS web API application servers use TLS1.2.

1.2.1. TCP/IP

When an interface uses the TCP/IP protocol, security is achieved using firewall and adequate rules.

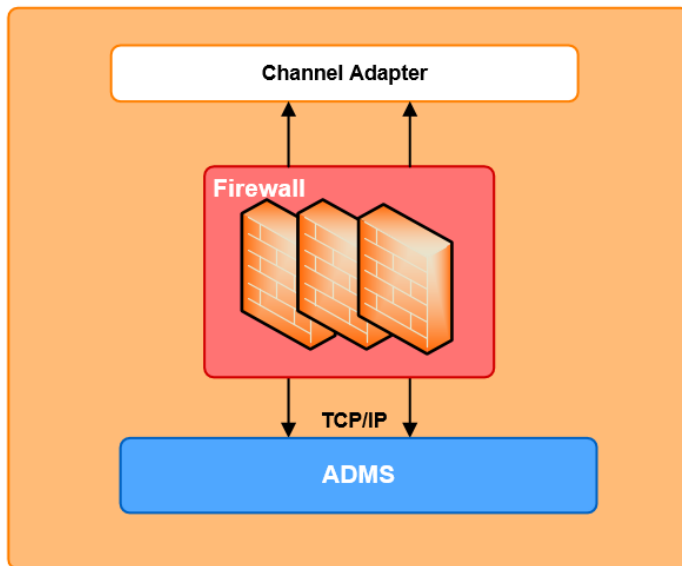


Figure 2. Firewall

There are numerous technologies and approaches possible within today's firewalls. Three common ones are:

- **IP Address Translation:** This is where the firewall hides the IP addresses of the machines that are inside the firewall from the outside. This can take the form of the firewall presenting only a single IP address to the outside firewall and mapping specific ports on the firewall to address / port pairs on the inside. It can also take the form of a simpler one-to-one mapping where node "X" has a different IP address on the outside of the firewall than it does on the inside (but its port numbers are unchanged).
- **Port Filtering:** Port filtering involves configuring the firewall so that it only allows connections and data across the firewall to specific well-known port numbers. All other attempts to send data or initiate connections to ports are rejected (actually, they are ignored or dropped, leaving it to the initiator to time out the requests).
- **Stateful Filtering:** This type of firewall generally allows the system inside the firewall free access to systems outside the firewall, but blocks any unsolicited connections from the outside to the system inside the firewall. Stateful firewalls generally allow port filtering such that specific ports may be opened to allow unsolicited access from outside systems.

1.2.2. Security Token

All messages in the ADMS interfaces are encoded as XML strings. The "message" refers to the payload that the application will use to pass the information. The message is combined with other information into a buffer that is sent or received between the two sides of the interface. The

"buffer" refers to the entire stream. All components of a buffer are strings using UTF-8 encoding.

The message structure includes five parts:

- A buffer header that describes the size and framing of the message.
- A security token based on the next three parts, plus a secret shared key.

The security token is created using the SHA256 hashing function, which generates a 256-bit hash value based on the XML message payload, user SID, current UTC time, and secret shared key.

- The XML message payload that the application uses to pass relevant information (described later in this document).
- A user SID.
- The time of the message, expressed in UTC format.

Note

Using the User Account and Authentication (UAA) feature is recommended instead of using a secret shared key. For more information about setting up UAA, refer to the *ADMS Software Installation and Maintenance Guide*.

1.2.2.1. Buffer Header

The buffer header describes the size and framing of the buffer. Version 1 of the buffer header is 40 bytes long, but future versions may be longer.

The values in the buffer header are sizes and are encoded based on their hex representation.

The eight-character representation of the XML message length allows for a message as large as 2 GB. However, both parties in the interface may agree upon a smaller number for a maximum message size. If this size is exceeded, then the link should be closed and reopened, to protect against a malicious sender that specifies a very large message size.

The buffer header is illustrated in the below image.

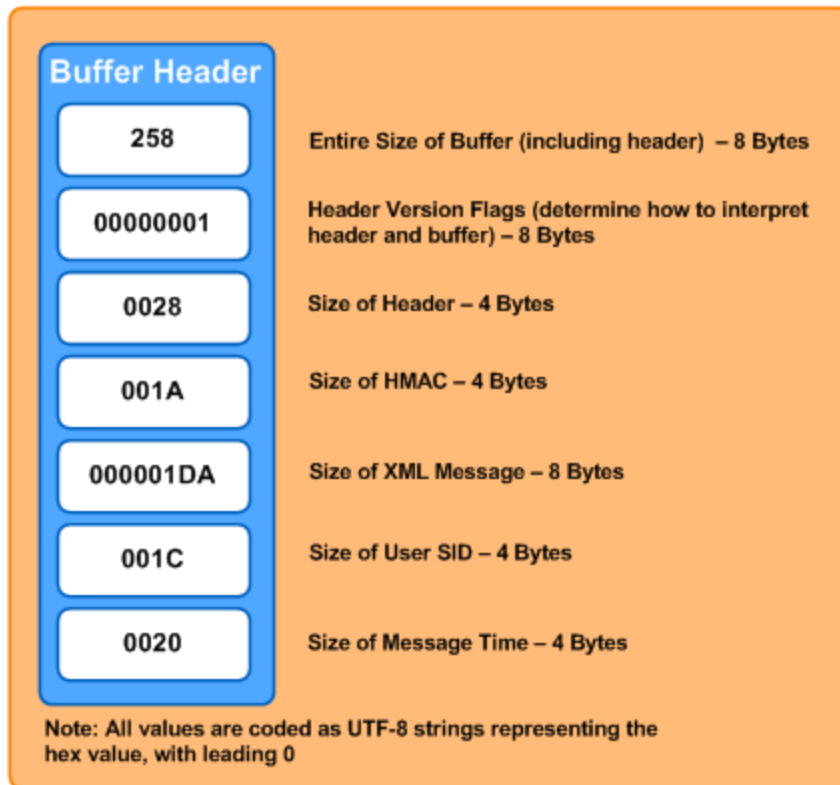


Figure 3. Buffer Header Format

1.2.2.2. Sending a Message

To send a message, the application will do the following:

1. Create the XML message.
2. Obtain the user SID (for example, "S-1-5-21-1085783642-1965037168-3936496997-86225").
3. Get the current time from the system, and convert to a string in the UTC format.
4. Calculate a security token using a SHA-256 algorithm, and convert this number to a string using its hex representation. The input to the algorithm is a single string obtained by concatenating:
 - The XML message.
 - The User SID.
 - The string representing the UTC current time.
 - A shared key. The shared key must be secure on both sides of the interface. This key is a string agreed upon by both sides of the interface. The key is encrypted and stored in the server configuration file. .NET ProtectedData (Triple-DES) is used to encrypt the key in the file. The same is done for the client.
5. Calculate a buffer header that specifies the size of the buffer and its parts. The header consists of a set of strings representing the size in bytes of the various parts.
6. Create a buffer with the header, the security token, the original XML message, the User SID, and

the message time.

7. Send the buffer.

This process is illustrated in the below image.

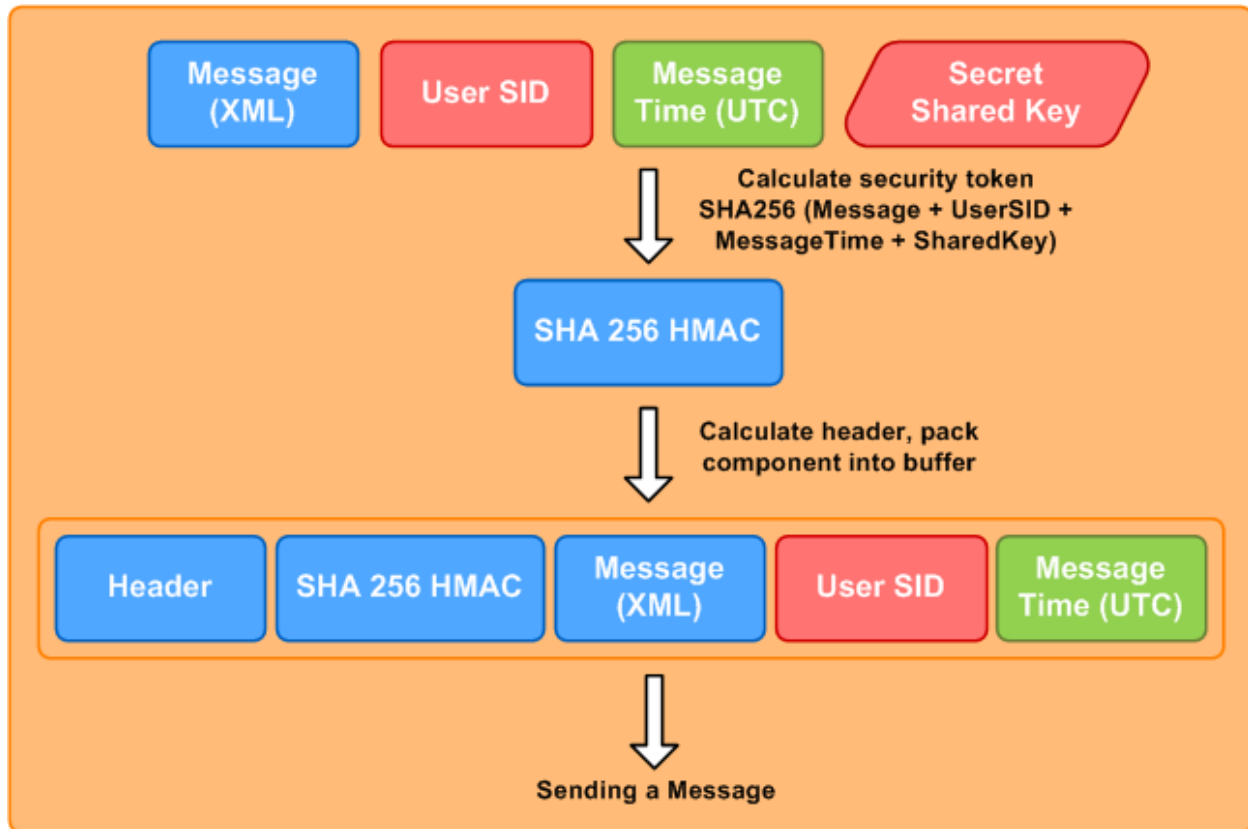


Figure 4. Sending a Message

1.2.2.3. Receiving a Message

When receiving a buffer, the receiver can read the message header, and then the rest of the buffer. The receiving application can check for a maximum message size before reading the entire buffer.

After reading the entire buffer, the receiving side must perform three checks:

- The security token passed in the buffer is compared with a security calculation made with the same arguments.
- The User SID is compared with a list of valid identifiers.
- The message time is compared with the current time to make sure that it is very recent (for example, within five minutes of the current time).

If all of the checks pass, the receiver can use the XML message.

This process is illustrated in the below image.

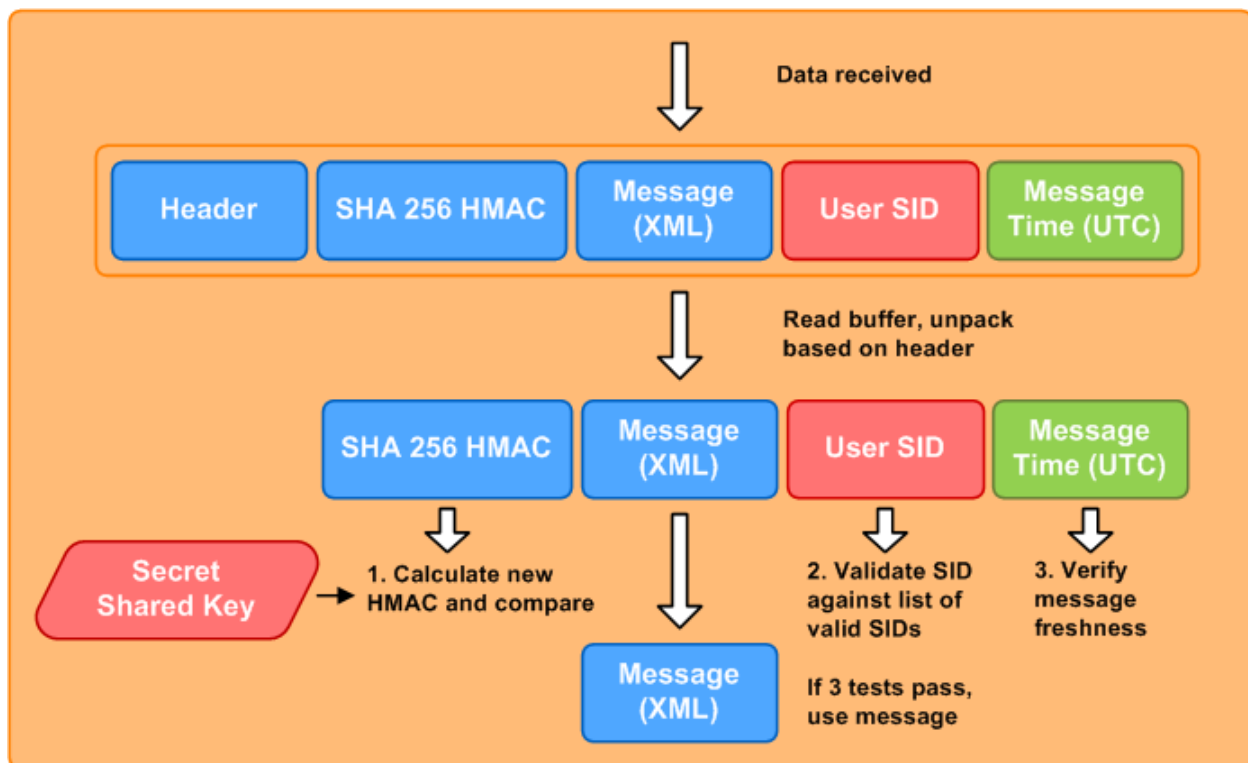


Figure 5. Receiving a Message

1.2.2.4. Valid SIDs and Shared Key Configuration

A user interface is provided in the Call server/Crew server/Relay server to configure the valid User SIDs from where the messages can be received by the respective applications. The same user interface is used to specify the shared secret key for message exchanges between respective applications. This file is named as ADMSTokenInfo.xml. The sample file is provided below:

```

<?xml version="1.0"?>
<TokenInfo xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns:xsd="http://www.w3.org/2001/XMLSchema">
  <ValidSids>
    <string>S-1-5-21-1085783642-1965037168-3936496997</string>
  </ValidSids>
  <CredentialInfo>AQAAANCMnd8BFdERjHoAwE/Cl+sBAAAA3eLLUZWyaEufNnf5JzTg6gQAAACAAAAAADZgAAQAAAAABAAADzMK
wTXJ/YJ5nfNoKp9ic2AAAAAASAAACgAAAAEAAAAAGmqIS9PotkLpxUzIY3+64k4AAAAATJauoUQucJHS5v1qOmVgf3UgnUZ+iVfK2IUk
ga7aj811khMrXB5W6LUxU9y511YWz9vNr9mXXiAUAAAAJAY+5HP3K8DtcZs2+JJbftXf6uk=</CredentialInfo>
  <SavedBy>S-1-5-21-1085783642-1965037168-3936496997</SavedBy>
  <SavedDate>2009-10-20T09:30:21.90625-07:00</SavedDate>
</TokenInfo>
  
```

The shared key is encrypted and stored in the file. The shared key is encrypted with a "machine" context. This means that the encrypted key can only be decrypted by a process running on the same node where the key was encrypted. Credential files are "tied" to the node where they are created.

It is possible to create a single file to use with all three applications (Call server, Crew server, and Relay server) if all three applications are hosted on the same node; credential files cannot be shared across nodes.

The IDS applications at the startup will load values of this file in the memory.

The sample user interface dialog to create this credential information file is provided in the below image.

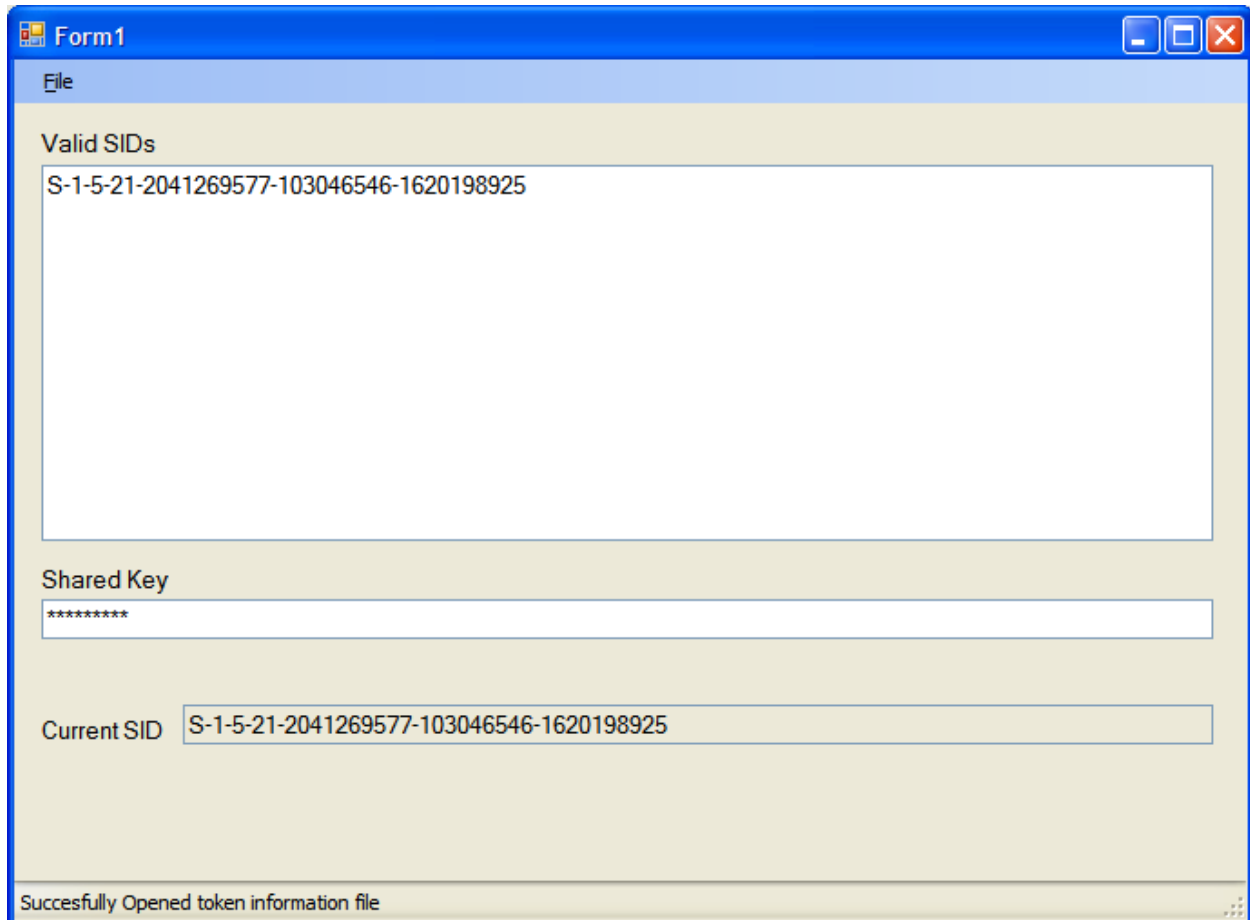


Figure 6. Creating a Credential Information File

1.2.2.5. Method Details: Sending a Message

1.2.2.5.1. Method GenerateCompositeMessage

User code will use this public method of the `IDMSTokenProvider`. `IDMSTokenProvider` class to create a composite message, i.e., the byte buffer sent to the remote host.

```
/// <summary>
/// Used for creating a byte array of the message, that can be sent over TCP/IP
/// to a client. If useSecurity flag is set, then it adds security sections to the
```

```

/// message which is described in Interface Design Document.
/// </summary>
/// <param name="messageBytes"></param>
/// Byte Array of the message that needs to be sent.
/// <param name="messageLength"></param>
/// Length of the original message in integer.
/// <param name="useSecurity"></param>
/// If this boolean security flag is set, then it add add security sections
/// to the message, as provided in the design document.
/// <returns></returns>
/// Byte array of the final message, that can be sent over TCP/IP link.

public byte[] GenerateCompositeMessage(byte[] messageBytes, int messageLength, bool useSecurity)

```

1.2.2.5.2. Checking for a Valid Composite Message Header

This method is used to determine if a valid composite message header can be created from an input byte array.

```

/// <summary>
/// This method will determine if a valid header can be formulated from the input byte array.
/// </summary>
/// <param name="rawBytes"></param>
/// The input byte array from which a valid header will be validated.
/// <returns></returns>
/// Returns 'true' if a valid header can be created, 'false' if a valid header can not be created.
public bool IsValidHeader(byte[] rawBytes)
{
    bool isValid = true;
    isValid = isValidHeader(rawBytes);
    return isValid;
}

```

1.2.2.6. Method Details: Receiving a Message

1.2.2.6.1. Method ValidateAndExtractReceivedMessageExt

```

/// <summary>
/// Extended version.
/// Validates and Extract a message received over TCP/IP. If the message is valid, then it returns
/// the raw message body of the actual message, which excludes the security related information and
/// the returned
/// message can be used by the receiving party for further processing.
/// </summary>
/// <param name="rawMessage"></param>
/// The incoming messages over TCP/IP
/// <param name="useSecurity"></param>
/// Whether to validate using security bytes or not

```

```

/// <param name="isValidPayload"></param>
/// Sets the flag to true, if it the correct number of payload bytes are present, as stipulated by the
header.
/// <param name="isValidHeader"></param>
/// Set to true if the header bytes are valid.
/// <param name="isValidSID"></param>
/// Set the flags to true if the message has vald SID.
/// <param name="isValidHash"></param>
/// Sets the flag to true if the message has valid hash.
/// <param name="isMessageAlive"></param>
/// Sets the integer value to 1 if the message is alive, sets it to -1, if in future and -2, if in
past.
/// <param name=" keyValidationException "></param>
/// If the private key could not be decrypted from the credential file an exception is passed to the
caller via this 'out' paramater, otherwise this parameter is null.
/// <returns></returns>
/// Returns the byte array of the actual message.
public byte[] ValidateAndExtractReceivedMessageExt(byte[] rawMessage,
bool useSecurity,
out bool isValidPayloadSiz,
out bool isValidHeader,
out bool isValidSID,
out bool isValidHash,
out int isMessageAlive,
out Exception keyValidationException)

```

1.2.2.7. Coding Examples

These examples are copied from the ADMS OMS communication connection class.

1.2.2.7.1. Writing a Composite Message Using Security Token

```

// Are we configured to use security tokens when communicating with this host?
bool useSecurity = _commParams[_commParamsIdx].UseSecurity;
// If using security check to see if the Hash provider encountered an exception while decrypting the
private key from the Credential file.
if (useSecurity)
{
    if (_writeHashProvider.EncryptionKeyException != null)
        throw new ConnectionException(this,
            "Exception while Decrypting Credential Key in: " + SecurityCredentialFile,
            _writeHashProvider.EncryptionKeyException);
}

// Create the Composite Message.
byte[] payloadByteArray = _writeHashProvider.GenerateCompositeMessage(msgByteArray,
msgByteArray.Length, useSecurity);
// Asynchronously send msg through the connection.
_writeAsyncClient.WriteAsync(_tcpClient, payloadByteArray, _writeAsyncClient);

```

1.2.2.7.2. Reading a Composite Message Using Security Token

```
// Loop until there aren't any more messages to deserialize.
while (true)
{
    // Are there at least enough bytes to form a 'IDMSMessageHeader' object to deserialize?
    // Each 'CompositeMessage' has a 'IDMSMessageHeader' at the begining.
    // '_receiveBuffer' is a FIFO buffer that is filled with bytes read from the TCP socket.
    if (_receiveBuffer.Peek() < _readHashProvider.CompositeMessageHeaderSize)
        break;
    // We should have a valid composite header at this point.
    if (!_readHashProvider.IsValidHeader(_receiveBuffer.Data))
    {
        throw new ...;
    }

    // Does the buffer contain the entire composite message??.
    int compositeMsgSiz = _readHashProvider.GetCompositeMessageSize(_receiveBuffer.Data);
    if (compositeMsgSiz < 1 || _receiveBuffer.Peek() < compositeMsgSiz)
        break;
    // Extract the CompositeMessage from the fifo buffer then validate it and extract // our payload.
    byte[] compositeMsg = new byte[compositeMsgSiz];
    _receiveBuffer.PopFront(ref compositeMsg, compositeMsg.Length);
    bool isValidSid, isValidHash, isValidPayloadSiz, isValidHeader, isValidMsg;
    int isMessageAlive;
    Exception keyValidationException;

    // Are we configured to use security tokens when communicating with this host?
    bool useSecurity = _commParams[_commParamsIdx].UseSecurity;
    byte[] myPayload;
    try
    {
        myPayload = _readHashProvider.ValidateAndExtractReceivedMessageExt(compositeMsg,
            useSecurity,
            out isValidPayloadSiz,
            out isValidHeader,
            out isValidSid,
            out isValidHash,
            out isMessageAlive,
            out keyValidationException);
    }
    catch (Exception e)
    {
        // Not good if we get an exception here, throw our own ConnectionException,
        // should result in connection closing.
        throw new ConnectionException(this, "Exception while extracting data received from remote connection
        using IDMSTokenProvider.ValidateAndExtractReceivedMessage()", e);
    }

    // Throw exceptions if data validations fail.
    if (!isValidHeader)
        throw new ConnectionException(this, "Invalid Composite Message HEADER received from remote
        connection!!");
}
```

```

if (!isValidPayloadSiz)
throw new ConnectionException(this, "Invalid Composite Message PAYLOAD SIZE received from remote
connection!!");
if (useSecurity)
{
if (keyValidationException != null)
throw new ConnectionException(this, "Exception while Decrypting Credential Key in: " +
SecurityCredentialFile, keyValidationException);
if (!isValidSid)
throw new ConnectionException(this, "Invalid Composite Message SID received from remote
connection!!");
if (!isValidHash)
throw new ConnectionException(this, "Invalid Composite Message HASH received from remote
connection!!");
if (isMessageAlive == IDMSHashProvider.MESSAGE_OLDER)
throw new ConnectionException(this, "Invalid Composite Message LIFE TIME (message is older than
allowed) for message received from remote connection!!");
}

// Convert the raw message to a NetMessageBase.
NetMessageBase netMsg = GetCorrectNetMessage(myPayload);
// Add the message to our NetMessageBase collection.
_receivedMsgs.Enqueue(netMsg);
}

```

1.2.2.8. Configuration

```

<CheckMessageSecurity>>false</CheckMessageSecurity>
<!--If set to true this parameter will enable checking of security token authenticity/expiration-->
<UseCredentialFile>>false</UseCredentialFile>
<!--If set to true this parameter will enable use of the Credential file for defining the private key
and user SIDs-->
<AllowUnregisteredSids>>false</AllowUnregisteredSids>
<!--If set to true this parameter will enable any user SID to be valid-->

```

The Credential File path configuration will remain the same.

There will be a configuration parameter named DisableSecurityToken, which allows the checking of security token to be disabled. Configuration variable CredentialInfoFile will specify the file name of the credential information generated. If this file is not available in the specified location or there is some problem in reading this file and DisableSecurityToken is set to false (meaning Security is enabled), then an error will be logged with the following message:

```
Security Token is enabled, but the Credential Info file is missing: Default secret key loaded.
```

In case the message is received with the wrong SID (security Identifier) or HMAC (hash-based message authentication code) and the credential file was loaded with default secret key, the following message will be logged in the application at the time of reception of message:

```
Message xxxx received from xxxx with unknown user SID. Probable Cause: Credential File Not found hence
```

using internal secret key.

1.3. Time Format

The Time stamps used through the messages of the different interfaces is expressed in UTC (Coordinated Universal Time).

1.4. Latitude/Longitude Format

The Latitude and Longitude used through the messages of the different interfaces is expressed in decimal degrees.

2. Interfaces - General Design

2.1. Introduction

The interfaces described in this chapter have been implemented on the ADMS side using .NET-based technology. The interfaces use XML messages for communication, which is the industry standard.

XML serialization and deserialization is done using .NET classes. One of the drawbacks of .NET-based deserialization is when an unknown enumeration is used in XML, it throws an exception. To avoid this scenario, the corresponding numeric value of the enumeration is preferred. Enumerations typically start with a numeric value of 0 and increment linearly (for example, 0, 1, 2, 3, and so on).

GE supplies the enumerations used for interfaces to external systems. When ADMS receives a message with an undefined enumeration numeric value, ADMS flags the message and sends a negative acknowledgement. The same behavior is expected from external systems.

Note

In messages, required fields (minOccurs="1" AND nillable != "true") must be provided with a valid value. Any non-required or nillable field (nillable="true") can be omitted or set to a blank value. In this case the system persists the existing value for these fields.

Each message type has an associated unique namespace. Each message has a version indicated at the end of the message type. For example, V1 refers to the message type version 1.

System capabilities of Call and Crew servers are a part of the configuration item. The supported system capabilities of the Call server are CallQuery, CallbackRequest, MeterEventQuery, MeterPingRequest, and CaseNoteQuery. The supported system capability of the Crew server is IsAvlCapable.

2.2. Hello Messages

Hello messages programmatically detect a stale link.

2.2.1. General Flow

The general flow of the Hello messages is the following:

- When the connection is made between ADMS and the Channel Adapter, both the Channel Adapter and ADMS send "Hello" messages.
- If the "Hello" messages have been correctly received and processed, the recipient sends a "Hello Response" message to the "Hello" message sender. The "Hello Response" message

includes information related to the identification of the expeditor of the message as well as some indications of its status. The system status can have three possible values (-1, 0, 1). For more information, see [System Status Indicator](#).

- After this first round of initial communication, both ADMS and the Channel Adapter send periodic "Hello" and "Hello Response" messages (just like a heartbeat), every 30 seconds (this duration is configurable). The recipient of the "Hello" message responds with a "Hello Response" message.
- The System sends the next "Hello" message only if it receives a "Hello Response" from the other system for its previous "Hello" message.
- If a recipient of a "Hello" message does not respond with a "Hello Response" within 90 seconds (configurable), the sender of the "Hello" message will consider the other system unavailable.
- A more detailed description of both "Hello" and "Hello Response" messages are provided in the sections below.

2.2.2. Hello Request

2.2.2.1. Characteristics

Message Type	XML
Frequency	30 seconds
Error Handling	Reflected in Hello Response Message
Required Information	• System identifier • System Status indicator • The system is capable of handling synchronization • Message base sequence
Optional Information	• Message base GUID

2.2.2.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/Hello" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/Hello" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="Hello" type="tns:HelloV1" />
<xs:complexType name="HelloV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<!--identifies system that sends hello message.-->
<xs:element minOccurs="0" maxOccurs="1" name="SystemId" type="xs:string" />
<!-- /// System Status will have right now three possible values (-1, 0, 1), which is described below
1: The connection to the forwarding entity is bad, so just keep sending Hello messages, till
you receive a value greater than -1. From ADMS prespective, this will be typically used in case
```

of Crew Interface Server (not connected to ADMS server), which doesn't have the buffering capability and so it doesn't expect any crew related messages from Channel Adapter. Once the connection status is good, then Crew Interface server will send the synchronization message to Channel Adapter.

0 : The connection status 0 means the connection is bad.

But, the messages are buffered, so the regular message communication is still allowed. From ADMS perspective, this is a situation, when Call Server is not connected to ADMS server, but since CallServer buffers the messages, these messages will eventually be forwarded to ADMS server. When the System Status value changes to 1, then resync messages should be sent.

1 : This is the ideal connection condition, when the data will be sent /received without any issues.-->

```
<xs:element minOccurs="1" maxOccurs="1" name="SystemStatus" type="xs:int" />
<!--True if system can send and receive synchronization messages.-->
<xs:element minOccurs="1" maxOccurs="1" name="IsSyncCapable" type="xs:boolean" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
</xs:schema>
```

2.2.2.3. Sample XML Message

```
<?xml version="1.0"?>
<Hello xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xmlns="http://www.esca.com/Hello">
<Sequence>58669</Sequence>
<Guid>d17f291a-0071-4e31-94d3-f8a110742e62</Guid>
<SystemId>ADMS Server</SystemId>
<SystemStatus>1</SystemStatus>
<IsSyncCapable>>false</IsSyncCapable>
</Hello>
```

2.2.3. Hello Response

2.2.3.1. Characteristics

Message Type	XML
Frequency	In response to a "Hello" message.
Error Handling	System Status Attribute will be 0 or -1.
Required Information	• MySystemId: System ID of the recipient. Identifies the system that sends hello response messages. • SystemIdEcho: System ID of the expeditor. Identifies the system that has sent hello messages. • SequenceEcho: Echo of sequence from the expeditor. Echo of Sequence is used to check for a stale link. For example, if ADMS does not receive the same echo in the Hello Response message for which it has sent a Hello message, it will disconnect and reconnect. • SystemStatus: Communication status. • Message base sequence.
Optional Information	• Message base Guid.

2.2.3.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/HelloResponse" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/HelloResponse" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="HelloResponse" type="tns:HelloResponseV1" />
<xs:complexType name="HelloResponseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<!--Identifies system that sends hello response message.-->
<xs:element minOccurs="0" maxOccurs="1" name="MySystemId" type="xs:string" />
<!--Identifies system that has sent hello message.-->
<xs:element minOccurs="0" maxOccurs="1" name="SystemIdEcho" type="xs:string" />
<!--Sequence of the hello message-->
<xs:element minOccurs="1" maxOccurs="1" name="SequenceEcho" type="xs:unsignedInt" />
<!-- /// System Status will have right now three possible values (-1, 0, 1), which is described below
1: The connection to the forwarding entity is bad, so just keep sending Hello messages, till
you receive a value greater than -1. From ADMS prespective, this will be typically used in case
of Crew Interface Server (not connected to ADMS server), which doesn't have the buffering
capability and so it doesn't expect any crew related messages from Channel Adapter.
Once the connection status is good, then Crew Interface server will send the synchronization
message to Channel Adapter.
0 : The connection status 0 means the connection is bad.
But, the messages are buffered, so the regular message communication is still allowed.
From ADMS perspective, this is a situation, when Call Server is not connected to ADMS
server, but since CallServer buffers the messages, these messages will eventually be
forwarded to ADMS server. When the System Status value changes to 1, then resync messages should be
```

```

sent.
1 : This is the ideal connection condition, when the data will be sent /received without any
issues.-->
<xs:element minOccurs="1" maxOccurs="1" name="SystemStatus" type="xs:int" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
</xs:schema>

```

2.2.3.3. Sample XML Message

```

<?xml version="1.0"?>
<HelloResponse xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/HelloResponse">
<Sequence>58662</Sequence>
<Guid>6ca4d330-e884-4ae5-8689-22cfe80601ef</Guid>
<MySystemId>Call Server</MySystemId>
<SystemIdEcho>Relay Server</SystemIdEcho>
<SequenceEcho>58662</SequenceEcho>
<SystemStatus>1</SystemStatus>
</HelloResponse>

```

2.3. Sequence Number

Each message will have a corresponding sequence number, which will increment linearly (for example, 1, 2, 3, and so on) and will be unique for the system sending it. This is now set as a data type unit and it should be reset back to 0 by the system, once it reaches the maximum value which is 4,294,967,295 for .NET. The sequence number will be used for information and for sending the negative acknowledgement message.

Sequence is reset to zero after reaching the maximum value or after restart of the application. The sequence number is specific to a link (one sequence number for the link with an AMI system and one sequence number for the link with a WFM system).

Detecting whether messages have been lost will be done using the echoing of the sequence number. This will be used for debugging purposes only.

2.4. System Status Indicator

System status indicates the status of the sender.

The system status can have three possible values (-1, 0, 1):

- **-1:** The connection to the forwarding entity is bad, so continue sending Hello messages until you receive a value greater than -1. From the ADMS perspective, this is used for the Crew Interface server (not connected to the ADMS server), which does not have the buffering capability and does not expect any crew related messages from the Channel Adapter. Once the connection status is good, the Crew Interface server will send the synchronization message to the Channel Adapter.
- **0:** The connection status is bad, but the messages are buffered, and the regular message communication is still allowed. From the ADMS perspective, this is a situation when the Call server is not connected to the ADMS server. However, since the Call server buffers the messages, these messages will eventually be forwarded to the ADMS server. When the System Status value changes to 1, the resync messages should be sent.
- **1:** This is the ideal connection condition, when the data will be sent /received without any issues.

2.5. Positive/Negative Acknowledgement Messages

These acknowledgment messages will be used for communication with the Channel Adapter only. The Positive Acknowledgement message will be sent on a demand basis, but the negative acknowledgement will be always sent when there is a problem with decoding the message content. For example, the CrewStatusWfmV1 expects the value of crew status to be one of the enumeration values mentioned in enum eWfmCrewStatus. However, if the Channel Adapter sends the value not specified in enum, a NACK message will be sent to the Channel Adapter. The message will also be logged on the ADMS side.

2.5.1. ACK/NACK Message Details

If an IncidentUpdate message gets acknowledged by the Channel Adapter, but there is an issue with WFM, the IncidentStatus message coming from WFM will have the Error Number and Error Code filled in the message. The message can have the ID of the Crew or Incident, so ADMS will be able to identify based on the sequence number a particular message with a problem. GE prefers to use CrewId for the CrewUpdate message and IncidentId for the IncidentUpdate message.

2.5.1.1. Characteristics

Message Type	XML
Frequency	In response to a message.

Error Handling	Nack message will be sent.
Required Information	• Is Ack/Nack • Sequence number of the received message (echo) • Error Number: 1. Unknown enum received. 2. Cannot forward the message. Resend when System Status is 1. • Message base sequence
Optional Information	• Error reason (to be logged). • ID of the crew/incident/vehicle associated with the message that is being acknowledged. • Additional error Information (to be logged or saved). • Message base Guid.

2.5.1.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/AckNack" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/AckNack" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="AckNack" type="tns:AckNackV1" />
<xs:complexType name="AckNackV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<!--True for Ack, false for Nack.-->
<xs:element minOccurs="1" maxOccurs="1" name="IsAck" type="xs:boolean" />
<!--Sequence number of the message that's being acknowledged.-->
<xs:element minOccurs="1" maxOccurs="1" name="MessageSequence" type="xs:unsignedInt" />
<!-- Error number, if any.-->
<xs:element minOccurs="1" maxOccurs="1" name="ErrNo" type="xs:int" />
<!--Human readable error reason, if negative. (To be logged.)-->
<xs:element minOccurs="0" maxOccurs="1" name="Reason" type="xs:string" />
<!--ID of the crew/incident/vehicle associated with message that's being acknowledged.-->
<xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
<!--Human readable name-->
<xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
<!--Additional error information. (To be logged or saved.)-->
<xs:element minOccurs="0" maxOccurs="1" name="AdditionalErrorInformation" type="xs:string" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
```

```
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
</xs:schema>
```

2.5.1.3. Sample XML Message

```
<?xml version="1.0"?>
<AckNack xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xmlns="http://www.esca.com/AckNack">
<Sequence>7710</Sequence>
<Guid>2b80acf8-ecc0-469b-bb1c-a6c3afbc7b40</Guid>
<IsAck>true</IsAck>
<MessageSequence>370255</MessageSequence>
<ErrNo>0</ErrNo>
<Reason />
<Id>5251</Id>
<Name />
<AdditionalErrorInformation />
</AckNack>
```

3. Workforce Management System Interfaces

! Important

It is recommended to use the Crew Server HTTP interface instead of the legacy TCP/IP interface for communicating with external systems.

3.1. Overview

ADMS interfaces with the Workforce Management System (WFM) through the Channel Adapter and the Enterprise Service Bus. Note that WFM middleware is not shown in the below image.

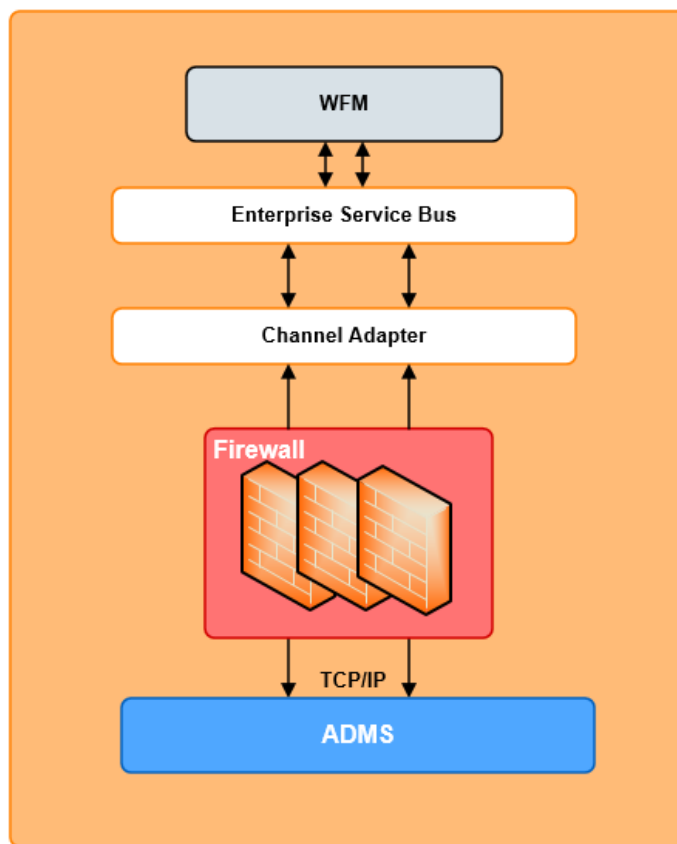


Figure 7. Workforce Management System Interface - Overview

Within ADMS, the Crew server is the component interfacing with the Channel Adapter as shown in the below image.

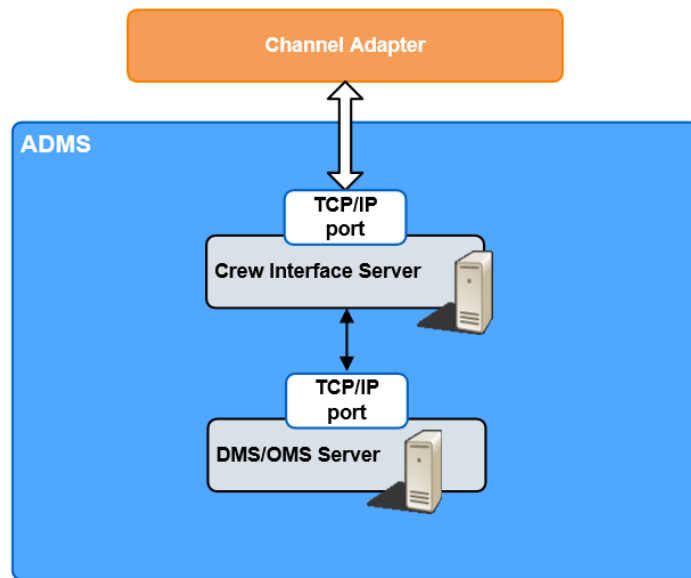


Figure 8. ADMS Components

The ADMS architecture provides the flexibility of running the Crew Interface server and the DMS/OMS server in two different physical servers. All the components of ADMS communicate with each other via TCP/IP ports. The installed configuration ensures that only one Crew Interface server is available to listen for connections and to transmit/receive messages at any given time. It is the responsibility of the Channel Adapter to re-connect to the redundant crew servers with a small delay of a few seconds between retries.

3.2. Interface Characteristics

The messages to the WFM interface will be both Query/Response type and publish/subscribe type.

The characteristics of the interface are the following:

Interface Type	XML Messaging
Protocol	HTTP

3.3. General Message Flow

3.3.1. General Message Convention

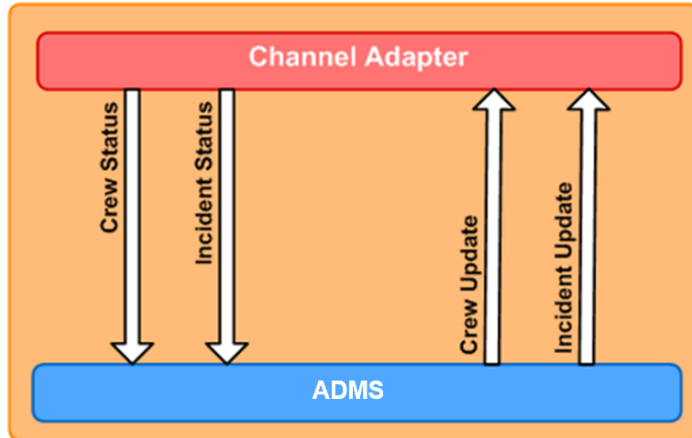


Figure 9. Message Convention

The message convention followed in this document is shown in the above image. An Incident Update and Crew Update message will be sent from ADMS to the Channel Adapter. Incident Status and Crew Status messages will be sent from the Channel Adapter to ADMS. There are separate messages for Incident Update and Crew Update, to allow sending only IncidentUpdate but no Crew Update. Channel Adapter holds the first IncidentUpdate message and merges the CrewUpdate message to send the merged message to WFM.

3.3.2. Use Cases

The following use cases are defined here to demonstrate the message flow:

- **Hello Messages:** Hello messages are used to programmatically detect a stale link.
- **Simple Assignment:** The operator assigns an incident to a crew and the crew completes the work.
- **No Trouble Found:** In this scenario, a crew is assigned to an incident, but the crew does not find any trouble after they get on-site. When the NoTroubleFound status is sent from the WFM the ADMS sets the incident to Pending Clear. The WFM can subsequently send an IncidentComplete status to clear the incident.
- **Canceling an Order:** In this scenario, the system assigns a crew to an incident, but the WFM operator cancels the order due to some unavoidable reason. The system unassigns the crew assigned by the operator and also includes crew comments, if they were sent in the interface status message.
- **Damage Assessment:** When a DamageAssessment status is sent from the WFM, the system unassigns the main crew from the incident.

- **Assignment to WFM:** The operator assigns an incident to the WFM (not to a specific crew). The WFM assigns a crew that finishes the work.
- **Device Re-Prediction:** The operator assigns a crew to an incident with a predicted distribution transformer out. While the crew is on the way to the predicted device location, more calls come in and the upstream fuse becomes predicted out. In this case, ADMS sends a DeviceReprediction update to WFM.

3.3.3. Hello Messages

The below image describes the normal flow of messages in the ideal environment, where ADMS connects to the Channel Adapter and messages are sent back and forth without any exception.

Please refer to [Hello Messages](#) for more details about Hello Messages.

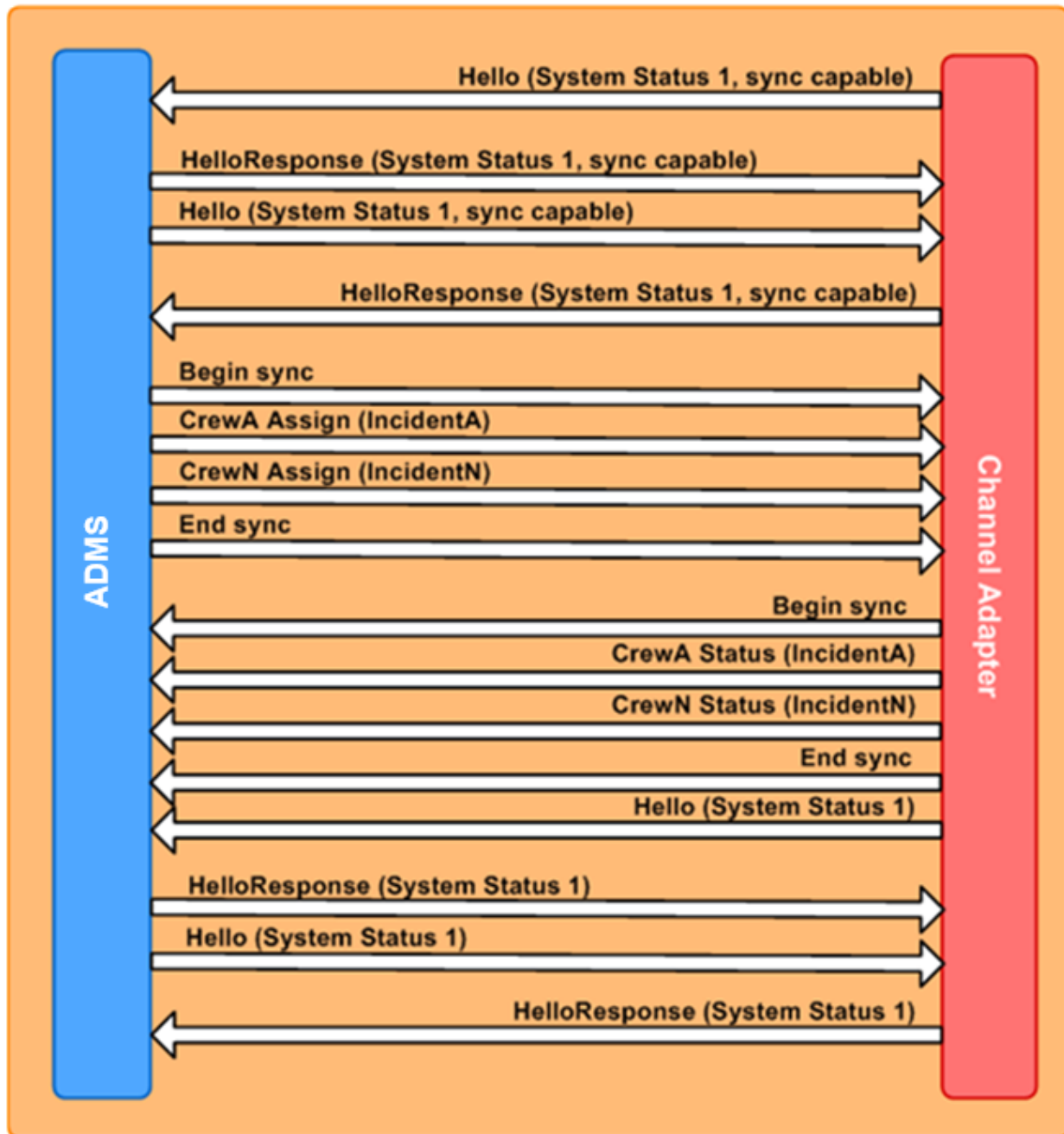


Figure 10. Hello Messages

3.3.4. Simple Assignment

In this scenario, the operator assigns an incident to a crew in ADMS. The crew travels to the site, clears the trouble, and energizes the customers.

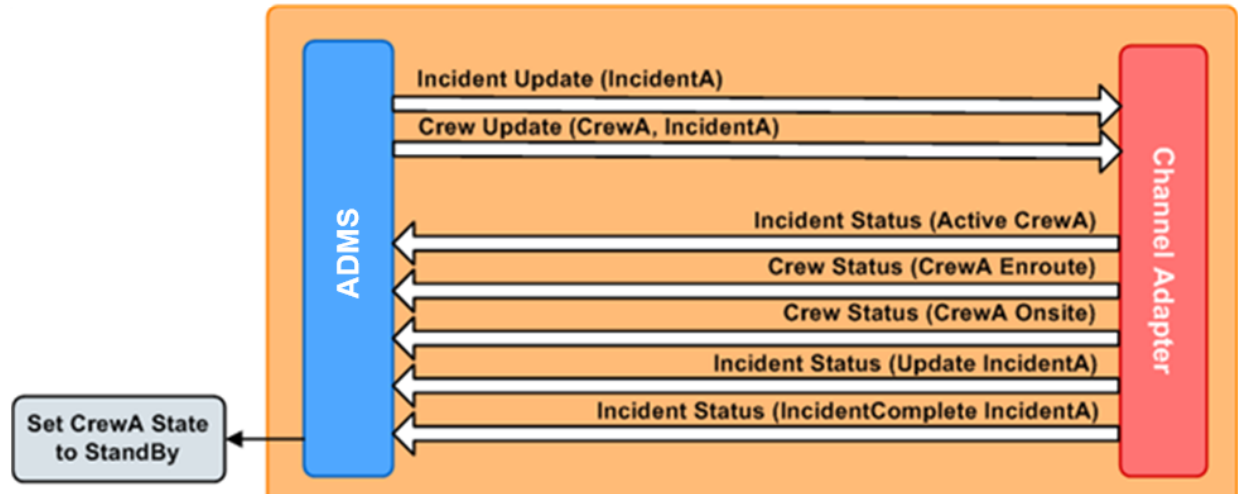


Figure 11. Simple Assignment

The ADMS system will first send the Incident Update message to inform the Channel Adapter about a new incident. The Crew Update message will follow containing the Crew Assignment information associated with the incident. After the crew is assigned successfully by WFM, an Incident Status message will be sent by the Channel Adapter to ADMS. The Channel Adapter keeps sending the Crew Statuses to ADMS.

When the order is complete, the Incident Complete message will be sent by WFM to ADMS. The incident will be flagged in ADMS to indicate that the incident is complete from a Work Management standpoint (but the incident must still be manually cleared).

3.3.5. No Trouble Found

In this scenario, a crew is assigned to an incident, but the crew does not find any trouble after they get on-site. When the NoTroubleFound status is sent from the WFM, the ADMS sets the incident to Pending Clear. The WFM can subsequently send an IncidentComplete status to clear the incident.

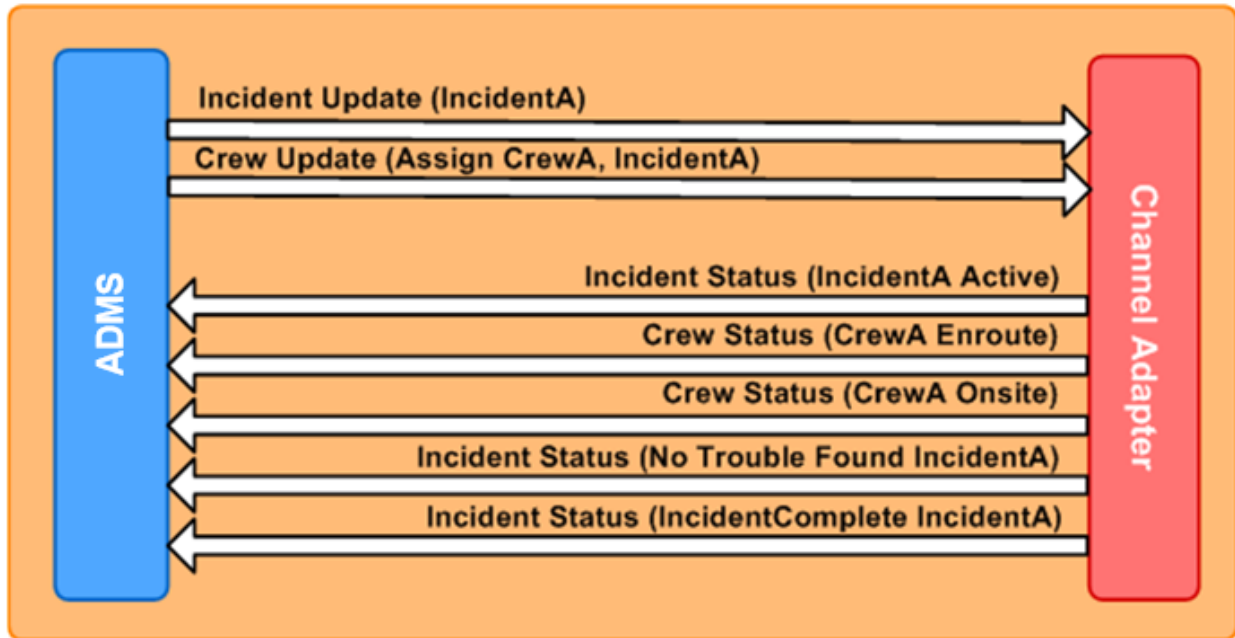


Figure 12. No Trouble Found

3.3.6. Order Re-Assignment by ADMS

In this scenario, a crew is assigned to an incident, but the crew determines they are not capable to fix the problem, because special equipment is needed. Another crew (that is capable of fixing the problem) is assigned. The first crew is un-assigned. Another crew, with special equipment, fixes the problem and reports to the operator. The operator marks the incident as solved, and un-assigns the crew.

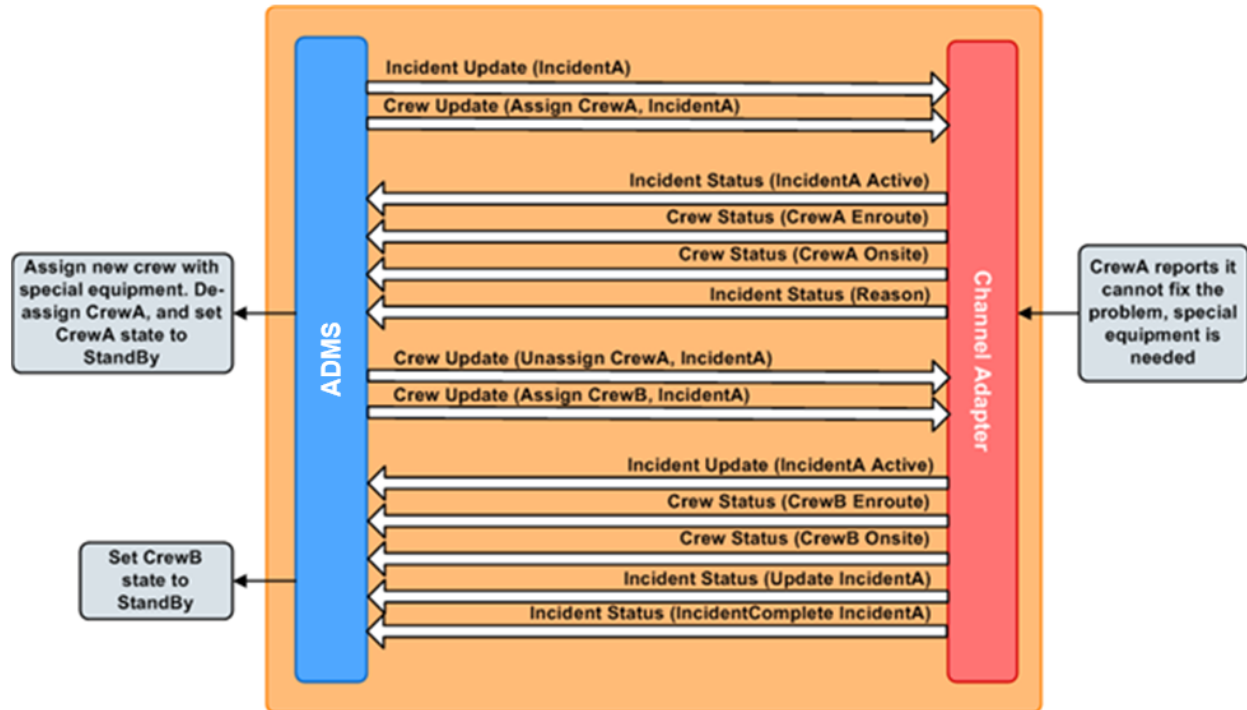


Figure 13. Order Re-Assignment by ADMS

3.3.7. Order Re-Assignment by Workforce Management

In this scenario, the operator assigns an incident to a crew. However, WFM assigns a different crew to do the work. The Channel Adapter sends a Crew Status message to ADMS to un-assign the crew. The Channel Adapter sends a Crew Status message to ADMS to assign a new crew. The ADMS system updates the crew assignment field and resets the TreeCrewNeeded or LineCrewNeeded flag, if it is already set.

If the assigned crew is not yet on site and the incident is cleared by the operator, the crew is de-assigned from the incident by WFM and gets the SignOn status.

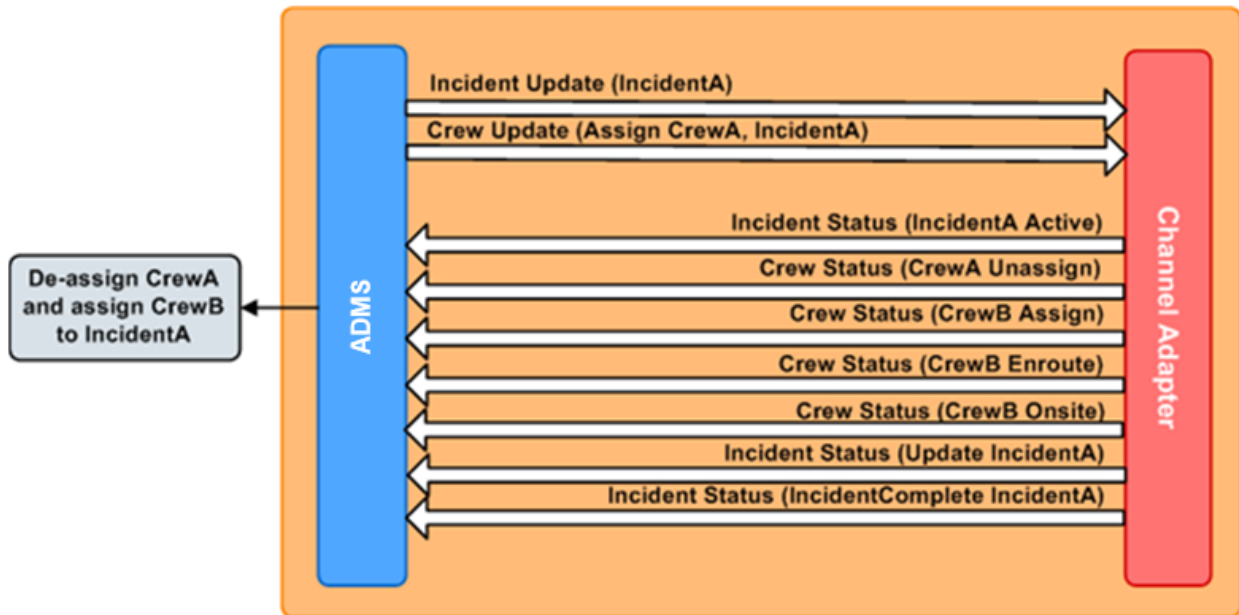


Figure 14. Order Re-Assignment by WFM

3.3.8. Damage Assessment/Patrolled

After a storm, a Damage Assessor/Evaluator could be assigned to an incident for assessing damage. The system allows for a temporary evaluator to be created on the fly from an existing crew. If an evaluator is assigned, the incident's working status changes to Evaluating. Once the Damage Assessment is received, the incident's working status changes to Evaluated and the evaluator is unassigned from the incident. The behavior of both Damage Assessment and Patrolled are the same; both are kept for backward compatibility.

3.3.9. Assignment with Assist

In this scenario, a crew is assigned to an incident, but the crew determines they need assistance when they find the problem. The operator assigns a second crew to help the first crew. The incident working status and crew working status are the same when the crew is in the Assigned, EnRoute, or OnSite state. It is possible to assign more than one crew to the same incident. The incident working status is set to the highest crew status of all the crews assigned. For example, if CrewA is assigned and they are EnRoute, and CrewB is assigned and they are OnSite, the incident working status is OnSite. All other crew statuses (for example, SignOn and SignOff) show the corresponding value in the system, but it has no effect on the incident's working status. The possible values for Crew Statuses and Incident Working Statuses are defined in the schema.

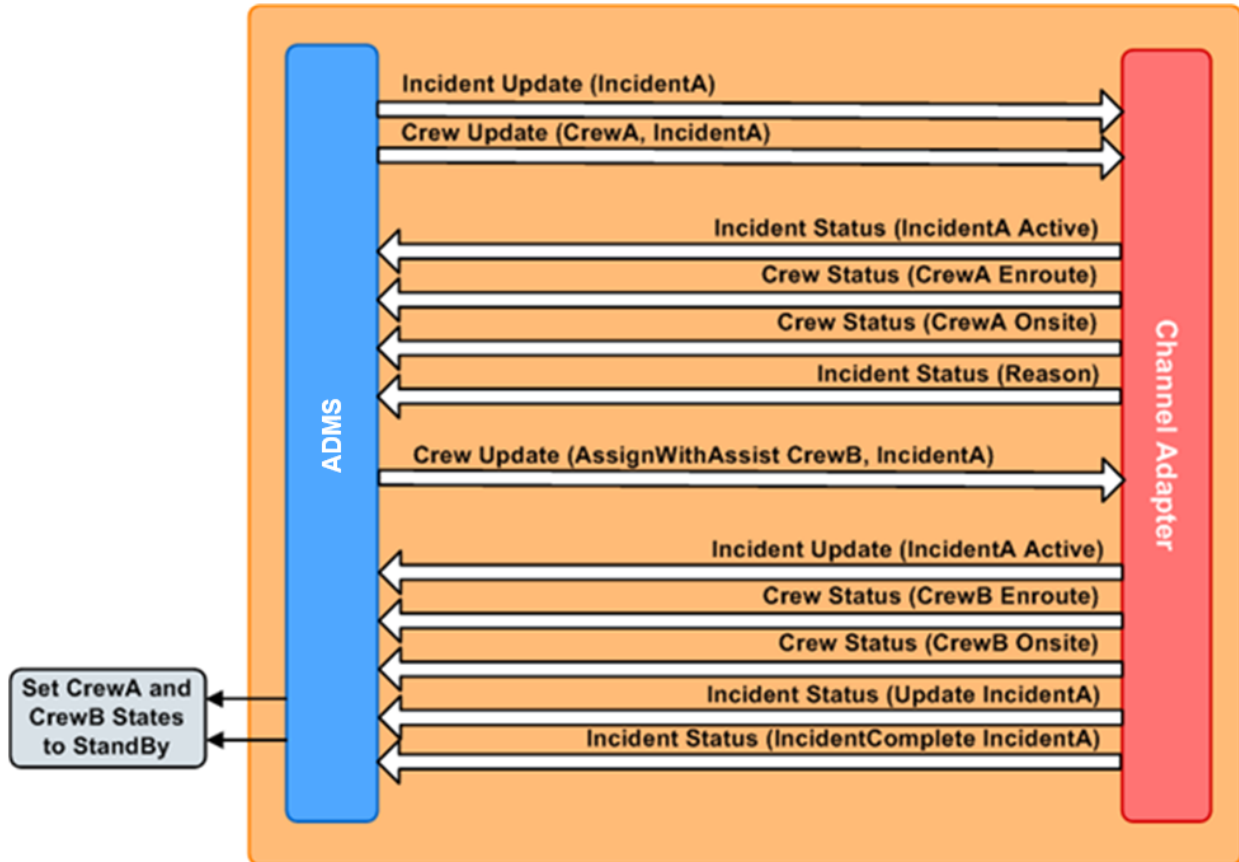


Figure 15. Crew Assignment with Assist

3.3.10. Assignments to Workforce Management

In this scenario, the incident is dispatched to the Workforce Management system, and the workforce management system assigns a crew to the incident.

The message for assigning to a non-specific crew has a field to indicate that there is no crew. That is, there is a positive indication that this should be dispatched by the Workforce Management system, rather than a reserved crew ID.

It is required that all crews are defined, even if WFM assigns them. In this case, IncidentUpdate has the IsNonSpecified flag set to true, and the following CrewUpdate message has an empty CrewId.

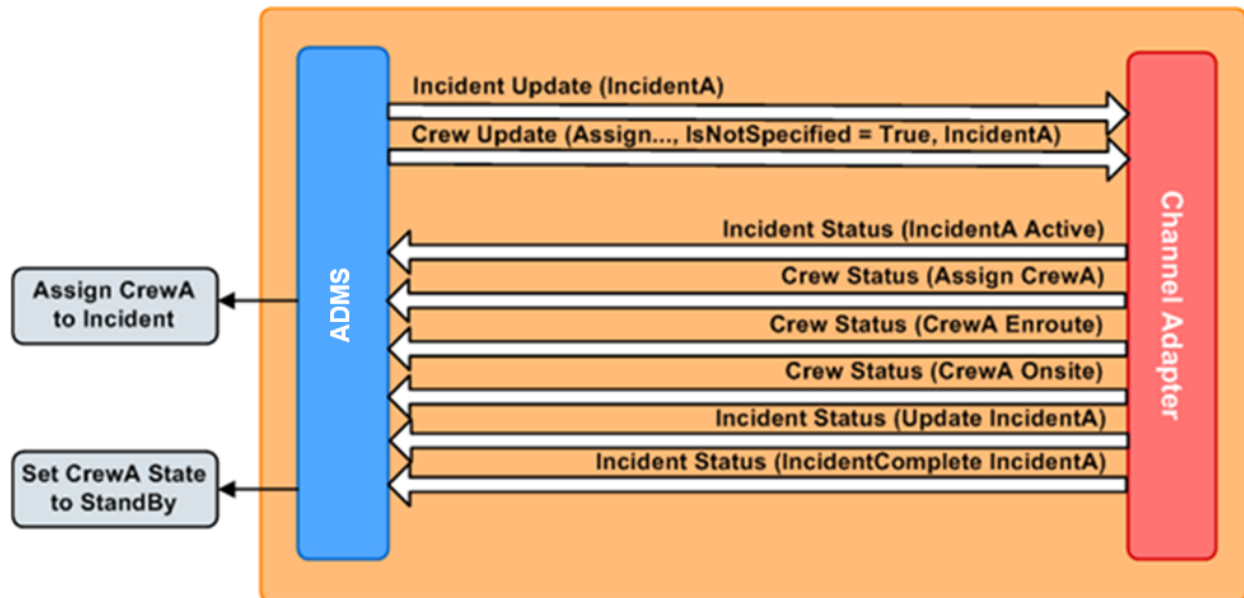


Figure 16. Assignment to WFM

3.3.11. Incident Roll-Up at ADMS with Crew Assigned

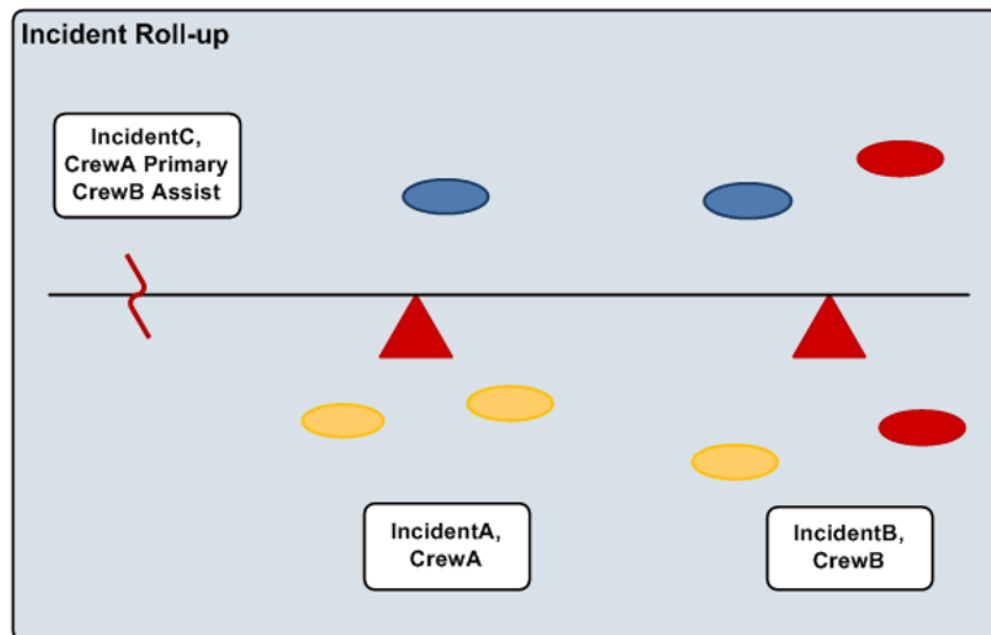


Figure 17. Incident Roll-Up Example

Chronology:

1. Yellow customers - three calls.
 - ADMS creates two Incidents: predicted and single customer. IncidentA is created before IncidentB.

- The operator assigns a WFM Crew to each Incident (CrewA, CrewB).
2. Blue customers call, Red customers call.
- ADMS predicts Fuse, rolls-up IncidentA and IncidentB to new IncidentC.
- WFM is now out of sync with ADMS.

To synchronize:

- ADMS keeps the order number of the earliest incident, which is IncidentA, and sends a DeviceRedirection UpdateType to CrewA.
- ADMS sends an Unassign order to CrewB and sends an Assign order for IncidentC.

i Note

When only single customer incidents are rolled up, an incident with a unique name is created. The crews are unassigned and a new assignment message is sent to the crews for the new incident number.

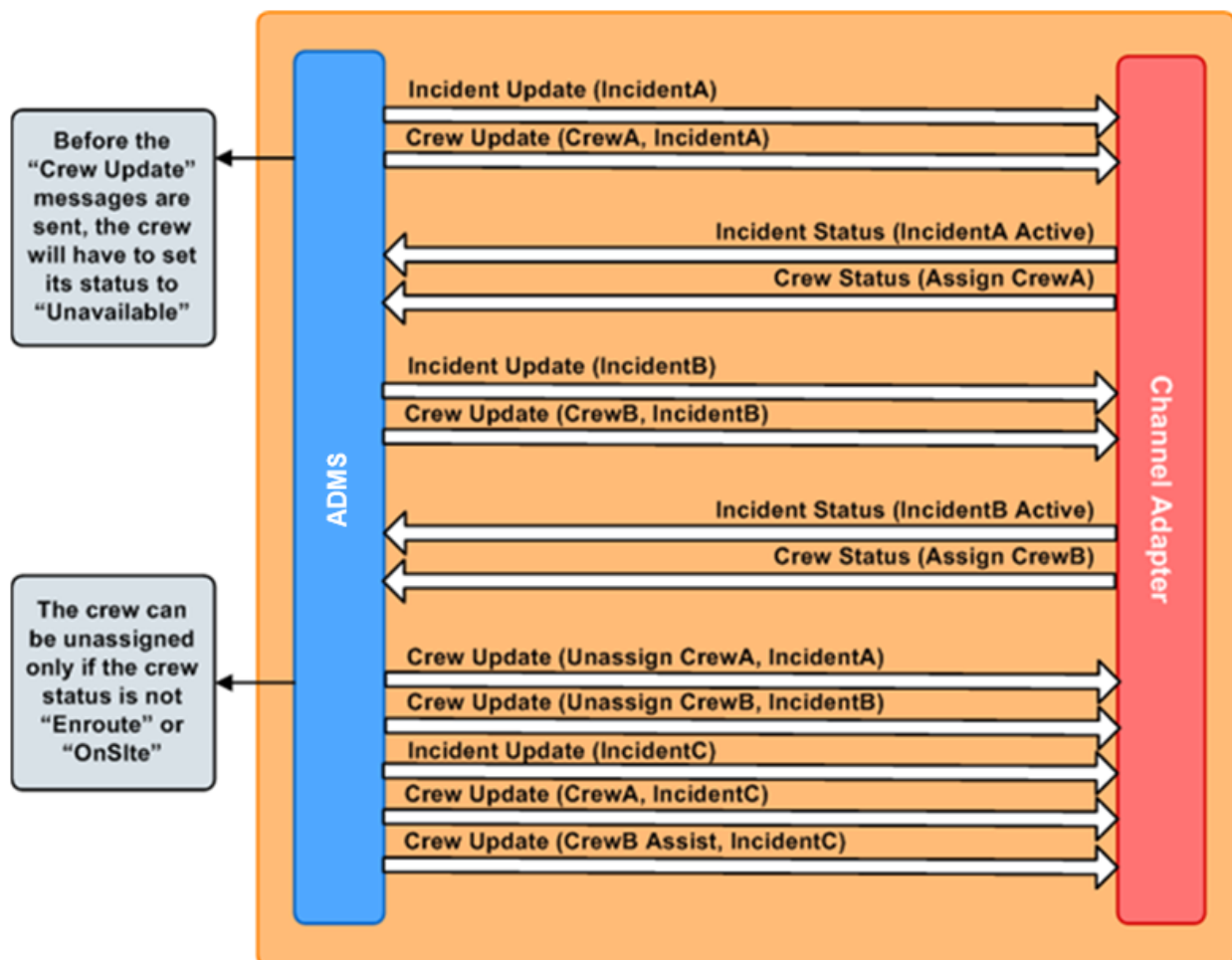


Figure 18. Incident Roll-Up Flowchart

3.4. Message Details

3.4.1. List of Messages

The list of supported messages is as follows:

- Hello
- Hello Response
- Begin Synchronization
- End Synchronization
- Crew Status
- Incident Status
- Incident Update
- Crew Update
- Ack/Nack
- Vehicle Location
- Vehicle Query
- Vehicle Subscribe
- Customer Message

Each message is defined by an XSD schema.

3.4.2. Hello

Characteristics, schema definition, and sample messages for the "Hello" message are provided in [Hello Messages](#).

3.4.3. Hello Response

Characteristics, schema definition and sample messages for the "Hello Response" message are provided in [Hello Response](#).

3.4.4. Begin Synchronization

This will be used only for communication with the Channel Adapter which will assess the messages to be sent after receiving the BeginSynchronization message from ADMS. After receiving this message, the Channel Adapter will send the current crew statuses of all the crews. The ADMS system will send incident updates of all the incidents. Synchronization messages will be triggered in

following three conditions:

- The first time connection is made after a start/restart of the system.
- The connection is dropped and reconnected.
- The system status indicator of the system goes from -1 and then comes back to 1.

3.4.4.1. Characteristics

Message Type	XML
Frequency	After restart and also after disconnection/connection phenomenon.
Error Handling	NACK message will be sent.
Required Information	• Echo of sequence from the hello response message that sent the system status value of 1 very first time after restart/reconnection. • Message base sequence.
Optional Information	• Message base Guid.

3.4.4.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/BeginSyncWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/BeginSyncWfm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="BeginSyncWfm" type="tns:BeginSyncWfmV1" />
<xs:complexType name="BeginSyncWfmV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<!-- Sequence echo from the hello message that requested the mismatch.-->
<xs:element minOccurs="1" maxOccurs="1" name="SequenceEcho" type="xs:long" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
</xs:schema>
```

3.4.4.3. Sample XML Message

```
<?xml version="1.0"?>
<BeginSyncWfm xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/BeginSyncWfm">
<Sequence>58</Sequence>
<Guid>344a0ba9-6412-4adb-8c7d-980f6a9238aa</Guid>
<SequenceEcho>0</SequenceEcho>
</BeginSyncWfm>
```

3.4.5. End Synchronization

3.4.5.1. Characteristics

Message Type	XML
Frequency	Immediately after all the synchronization messages have been sent.
Error Handling	NACK message will be sent.
Required Information	• Echo of sequence from the Begin Sync message that requested the mismatch. • Message base sequence.
Optional Information	• Message base GUID.

3.4.5.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/EndSyncWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/EndSyncWfm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="EndSyncWfm" type="tns:EndSyncWfmV1" />
<xs:complexType name="EndSyncWfmV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<!--Sequence echo from the Begin Sync message-->
<xs:element minOccurs="1" maxOccurs="1" name="SequenceEcho" type="xs:long" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
```

```

<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
</xs:schema>

```

3.4.5.3. Sample XML Message

```

<?xml version="1.0"?>
<EndSyncWfm xmlns:xsd="http://www.w3.org/2001/XMLSchema" xmlns:xsi="http://www.w3.org/2001/XMLSchema-
instance" xmlns="http://www.esca.com/EndSyncWfm">
<Sequence>6</Sequence>
<Guid>31f20423-68e4-4821-9ef8-7ac8521ab996</Guid>
<SequenceEcho>0</SequenceEcho>
</EndSyncWfm>

```

3.4.6. Crew Status

This is an inbound message for ADMS.

To trigger the Crew Server to send the IncidentUpdateWfmV1 message with the complete list of customer calls to the WFM, send a CrewStatusWfmV1 message with the "201" status. For example:

```

{
  "CrewId": "FI1940",
  "Status": 201,
  "IncidentId": "1870000121"
}

```

Note

In ADMS 3.13, WorkStatus, MobileCommunicationStatus, ShiftStatus, and AvailabilityStatus fields are added to provide detailed crew status information. The legacy Status field is still supported for backward compatibility. However, either the Status or WorkStatus value can be used. If both the Status and WorkStatus values are provided, the Status value is used and the WorkStatus value is ignored.

3.4.6.1. Characteristics

Message Type	XML
Frequency	After every status change of the crew from WFM.
Error Handling	NACK message will be sent.

Required Information	• Status of the crew: unavailable, sign off, sign on, on break, standby, dispatched, en route, onsite, waiting for equipment, released, unassigned, or available. • Crew job status • Crew communications status • Crew shift status • Crew availability status • WFM time stamp in UTC format. • Estimated time for arrival (ETA) time stamp • Message base sequence.
Optional Information	• Crew ID. • Assigned Incident ID (may be blank if not assigned). • Other crew information. • Comments from the crew. • Assigned Incident human readable name. • Updated vehicle information (list of vehicles associated with the crew). • Error number provided by the Channel Adapter. • Error code provided by the Channel Adapter. • Work order ID. • Crew area. • Additional data (custom fields) • Message base GUID.

3.4.6.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/CrewStatusWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/CrewStatusWfm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CrewStatusWfm" type="tns:CrewStatusWfmV1" />
  <xs:complexType name="CrewStatusWfmV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:MessageBaseV1">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
<!--Crew Status provided by Work Managment System.Follows enum eWfmCrewStatus, which is defined as
public enum eWfmCrewStatus
{
    NotSelected = -1,
    Unavailable = 0,
    SignOff = 1,
    SignOn = 2,
    OnBreak = 3,
    StandBy = 4,
    Dispatched = 5,
```

```

        EnRoute = 6,
        OnSite = 7,
        WaitingOnEquipment = 8,
        Released = 9,
        Unassigned = 10,
        Available = 11,
        OutOfCoverage = 12,
        Acknowledged = 13,
        Suspended = 14,
        PendingDispatch= 15
        Assigned = 16
    }

    -->
    <xs:element minOccurs="1" maxOccurs="1" name="WorkStatus" nillable="true" type="xs:int" />
<!--Crew job status provided by Work Managment System. Follows the CrewWorkStatus coded translation
item}-->
    <xs:element minOccurs="1" maxOccurs="1" name="MobileCommunicationStatus" nillable="true"
type="xs:int" />
<!--Crew job status provided by Work Managment System. Follows the MobileCommunicationStatus coded
translation item}-->
    <xs:element minOccurs="1" maxOccurs="1" name="ShiftStatus" nillable="true" type="xs:int" />
<!--Crew job status provided by Work Managment System. Follows the CrewShiftStatus coded translation
item}-->
    <xs:element minOccurs="1" maxOccurs="1" name="AvailabilityStatus" nillable="true"
type="xs:int" />
<!--Crew job status provided by Work Managment System. Follows the CrewAvailabilityStatus coded
translation item}-->
    <xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="OtherInformation" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Comments" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="IncidentName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Vehicle" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="ErrorNumber" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="ErrorCode" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="ARMSTimeStamp" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="WorkOrderId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="CrewArea" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="ETA" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData"
type="tns:ArrayOfPropertyUpdate" />
    </xs:sequence>
  </xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
  <xs:complexContent mixed="false">
    <xs:extension base="tns:NetMessageBase" />
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
  </xs:sequence>
</xs:complexType>

```

```

    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyUpdate">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyUpdate" nillable="true"
type="tns:PropertyUpdate" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyUpdate">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

3.4.6.2.1. Enum Schema Definition eWfmCrewStatus

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="eWfmCrewStatus" type="eWfmCrewStatus" />
  <xs:simpleType name="eWfmCrewStatus">
    <xs:restriction base="xs:string">
      <xs:enumeration value="NotSelected" />
      <xs:enumeration value="Unavailable" />
      <xs:enumeration value="SignOff" />
      <xs:enumeration value="SignOn" />
      <xs:enumeration value="OnBreak" />
      <xs:enumeration value="StandBy" />
      <xs:enumeration value="Dispatched" />
      <xs:enumeration value="EnRoute" />
      <xs:enumeration value="OnSite" />
      <xs:enumeration value="WaitingOnEquipment" />
      <xs:enumeration value="Released" />
      <xs:enumeration value="Unassigned" />
      <xs:enumeration value="Available" />
      <xs:enumeration value="OutOfCoverage" />
      <xs:enumeration value="Acknowledged" />
      <xs:enumeration value="Suspended" />
      <xs:enumeration value="PendingDispatch" />
      <xs:enumeration value="Assigned" />
    </xs:restriction>
  </xs:simpleType>
</xs:schema>

```

3.4.6.2.2. Enum Schema Definition eCrewWorkStatus

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="eCrewWorkStatus" type="eCrewWorkStatus" />

```

```

<xs:simpleType name="eCrewWorkStatus">
  <xs:restriction base="xs:string">
    <xs:enumeration value="NotSelected" />
    <xs:enumeration value="Unknown" />
    <xs:enumeration value="Idle" />
    <xs:enumeration value="Assigned" />
    <xs:enumeration value="Dispatched" />
    <xs:enumeration value="EnRoute" />
    <xs:enumeration value="OnSite" />
    <xs:enumeration value="NotSelected" />
    <xs:enumeration value="Unknown" />
    <xs:enumeration value="Idle" />
    <xs:enumeration value="Assigned" />
    <xs:enumeration value="Dispatched" />
    <xs:enumeration value="EnRoute" />
    <xs:enumeration value="OnSite" />
    <xs:enumeration value="Unassigned" />
    <xs:enumeration value="UnAssigning" />
    <xs:enumeration value="UnAssignQueued" />
    <xs:enumeration value="UnAssignSent" />
    <xs:enumeration value="UnAssignFailed" />
    <xs:enumeration value="Dispatching" />
    <xs:enumeration value="DispatchQueued" />
    <xs:enumeration value="Delivered" />
    <xs:enumeration value="DeliveryFailed" />
    <xs:enumeration value="TakingBack" />
    <xs:enumeration value="TakeBackSent" />
    <xs:enumeration value="TakeBackQueued" />
    <xs:enumeration value="TakenBack" />
    <xs:enumeration value="TakeBackFailed" />
    <xs:enumeration value="Accepted" />
    <xs:enumeration value="Rejected" />
    <xs:enumeration value="OnHold" />
    <xs:enumeration value="Started" />
    <xs:enumeration value="Suspended" />
    <xs:enumeration value="Finished" />
    <xs:enumeration value="Error" />
    <xs:enumeration value="AssignRequested" />
    <xs:enumeration value="UnassignRequested" />
    <xs:enumeration value="DispatchRequested" />
    <xs:enumeration value="TakebackRequested" />
  </xs:restriction>
</xs:simpleType>
</xs:schema>

```

3.4.6.3. Sample XML Message

```

<?xml version="1.0"?>
<CrewStatusWfm xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/CrewStatusWfm">
<Sequence>232</Sequence>

```

```
<Guid>eb519f60-48d2-4165-ad01-5b10886fab0d</Guid>
<CrewId>4454</CrewId>
<Status>2</Status>
<IncidentId />
<OtherInformation />
<Comments />
<IncidentName />
<Vehicle />
<ErrorNumber />
<ErrorCode />
<ARMSTimeStamp>0001-01-01T00:00:00</ARMSTimeStamp>
<WorkOrderId />
<CrewArea />
</CrewStatusWfm>
```

3.4.7. Incident Status

This is an inbound message for ADMS.

3.4.7.1. Characteristics

Message Type	XML
Frequency	After every order status change from WFM.
Error Handling	NACK message will be sent back.

Required Information

- Status
 - Actual restoration time of the incident, based on UTC format.
 - Real cause of the incident (This could be different from what the dispatcher has entered initially. In that case it will overwrite the Incident Cause.)
 - Sub-cause of the incident (This could be different from what the dispatcher has entered initially. In that case it will overwrite the Incident Sub-Cause.)
 - Outage type
 - Phase out
 - Failure Type (same as Affected Equipment in the Update Incident popup), defined as enum eFailureType.
 - Failed Equipment
 - Action taken
 - PowerQualityType
 - NonoutageType
 - Weather
 - FailureMode
 - Manufacturer
 - Animals
 - Plants
 - SystemAffected
 - Corrected customer count
 - ETR from WFM in UTC format. It should update the current ETR of the incident.
 - Number of poles down
 - Number of failed transformers
 - Number of failed conductors
 - Number of services down
 - Inaccessible flag
 - Follow Up Required flag
 - Line Crew Needed flag
 - Tree Crew Needed flag
 - First Responder Needed flag
 - Message base sequence
-

Optional Information

- Incident ID.
 - Associated currently assigned crew ID(s).
 - Error number provided by the Channel Adapter.
 - Error code provided by the Channel Adapter.
 - Comments about the order.
 - Point of Failure.
 - Crew ID.
 - Crew Type.
 - Number of assigned crews.
 - Work Order ID.
 - Message base GUID.
 - AdditionalData.
-

3.4.7.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/IncidentStatusWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/IncidentStatusWfm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="IncidentStatusWfm" type="tns:IncidentStatusWfmV1" />
  <xs:complexType name="IncidentStatusWfmV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:MessageBaseV1">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="ErrorNumber" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="ErrorCode" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Comment" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="ActualRestorationTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="CauseGroup" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="Cause" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="OutageType" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="PhasesOut" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="FailureType" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="FailedEquipment" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="ActionTaken" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="PowerQualityType" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="NonoutageType" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="Weather" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="FailureMode" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="Manufacturer" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="Animals" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="Plants" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="SystemAffected" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="CorrCustCount" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="WfmEtr" type="xs:dateTime" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>
```

```

        <xs:element minOccurs="1" maxOccurs="1" name="NoOfDevicesDown" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="NoOfPolesDown" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="NoOfTransformers" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="NoOfCrossArms" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="NoOfTreeTrimSpans" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="ConductorSpan" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="NoServicesDown" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="Inaccessible" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="FollowupRequired" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="LineCrewNeeded" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="TreeCrewNeeded" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="FirstResponderNeeded" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="EvaluatorNeeded" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="0" maxOccurs="1" name="PointOfFailure" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="WorkOrderId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData"
type="tns:ArrayOfPropertyValueV1" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
    <xs:complexContent mixed="false">
        <xs:extension base="tns:NetMessageBase" />
    </xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValueV1">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValueV1" nillable="true"
type="tns:PropertyValueV1" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValueV1">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
</xs:schema>

```

3.4.7.2.1. Enum Schema Definition eOrderStatus

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="eOrderStatus" type="eOrderStatus" />
  <xs:simpleType name="eOrderStatus">
    <xs:restriction base="xs:string">
      <xs:enumeration value="NotSelected" />
      <xs:enumeration value="NoTroubleFound" />
      <xs:enumeration value="DamageAssessment" />
      <xs:enumeration value="CancelOrder" />
      <xs:enumeration value="Complete" />
      <xs:enumeration value="Patrolled" />
      <xs:enumeration value="PendingAssigned" />
      <xs:enumeration value="BackToAssigned" />
      <xs:enumeration value="ConfirmedOpen" />
      <xs:enumeration value="Cleared" />
      <xs:enumeration value="Restored" />
      <xs:enumeration value="CancelIncident" />
    </xs:restriction>
  </xs:simpleType>
</xs:schema>
```

When this order status information is received by the OMS, the following actions are taken.

- **NotSelected:** This status is used to update incident properties without updating the incident status.
- **NoTroubleFound:** This kind of situation occurs when the incident is a single customer, isolated, or distribution TF predicted out. If the Incident Status message arrives with this type of OrderStatus and ActualRestoration time is filled in, the incident moves to the Complete working state.
- **DamageAssessment:** If more than one crew is assigned, unassign the primary crew and leave any remaining crew(s) assigned.
- **CancelOrder:** In this case, the OMS operator assigns a crew and WFM cancels the order for some valid reason. Ideally, there should be crew comments associated with canceling an order. ADMS adds a comment from the crew to the Crew Comments field of the incident and unassigns all the crews assigned to the incident.
- **Complete:** The incident's working status is set to "Complete". Based on the FIELD_CLEAR_DEVICE_LEVEL parameter, incidents associated with specified device types are moved to "Pending Clear".
- **Patrolled:** The incident's working status is set to "Patrolled".
- **PendingAssigned:** Changes the incident status to "Crew Dispatch is Still Pending".
- **BackToAssigned:** Changes the incident status to "Crew has been Dispatched to this Incident".
- **ConfirmedOpen:** Opens the device of record. This option is applicable only to predicted incidents on devices that do not have SCADA status measurements. It is available only if field clearing is enabled for this device of record type using the FIELD_CLEAR_DEVICE_LEVEL

parameter.

- **Cleared:** Automatically clears the pending clear incident. This option is applicable only to incidents on devices that do not have SCADA status measurements. It is available only if field clearing is enabled for this device of record type using the FIELD_CLEAR_DEVICE_LEVEL parameter.
- **Restored:** Closes the device of record and changes the incident status to Restored.
- **CancelIncident:** Cancels the incident in ADMS, unassigns crews, and sets the Cancel Restoration Step flag for the incident to exclude this outage from distribution statistics, such as CMI calculation.

3.4.7.2.2. Enum Schema Definition eIncidentWorkingStatus

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="eIncidentWorkingStatus" type="eIncidentWorkingStatus" />
<xs:simpleType name="eIncidentWorkingStatus">
<xs:restriction base="xs:string">
<xs:enumeration value="Unassigned" />
<xs:enumeration value="SentToWfm" />
<xs:enumeration value="PendingDispatched" />
<xs:enumeration value="Dispatched" />
<xs:enumeration value="EnRoute" />
<xs:enumeration value="OnSite" />
<xs:enumeration value="Cancel" />
<xs:enumeration value="Complete" />
<xs:enumeration value="PendingClear" />
<xs:enumeration value="Patrolled" />
<xs:enumeration value="Evaluating" />
<xs:enumeration value="Evaluated" />
<xs:enumeration value="Restored" />
<xs:enumeration value="Cleared" />
<xs:enumeration value="Assigned" />
<xs:enumeration value="MaxValue" />
</xs:restriction>
</xs:simpleType>
</xs:schema>
```

3.4.7.2.3. Enum Schema Definition ePhaseOut

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="ePhaseOut" type="ePhaseOut" />
<xs:simpleType name="ePhaseOut">
<xs:restriction base="xs:string">
<xs:enumeration value="ABC" />
<xs:enumeration value="AB" />
<xs:enumeration value="BC" />
</xs:restriction>
</xs:simpleType>
</xs:schema>
```

```

<xs:enumeration value="AC" />
<xs:enumeration value="A" />
<xs:enumeration value="B" />
<xs:enumeration value="C" />
<xs:enumeration value="None" />
</xs:restriction>
</xs:simpleType>
</xs:schema>

```

3.4.7.3. Sample XML Message

```

<?xml version="1.0"?>
<IncidentStatusWfm xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/IncidentStatusWfm">
<Sequence>1710</Sequence>
<Guid>81cba829-e479-460f-be78-c00d855ec696</Guid>
<IncidentId>3280000001</IncidentId>
<Status>100</Status>
<CrewId>FI1951</CrewId>
<ErrorNumber />
<ErrorCode />
<Comment />
<ActualRestorationTime>2017-11-24T08:00:00Z</ActualRestorationTime>
<Cause>0</Cause>
<OutageType>0</OutageType>
<PhasesOut>0</PhasesOut>
<FailureType>-1</FailureType>
<FailedEquipment>0</FailedEquipment>
<ActionTaken>2</ActionTaken>
<PowerQualityType>0</PowerQualityType>
<NonoutageType>0</NonoutageType>
<Weather>0</Weather>
<FailureMode>0</FailureMode>
<Manufacturer>0</Manufacturer>
<Animals>0</Animals>
<Plants>0</Plants>
<SystemAffected>0</SystemAffected>
<CorrCustCount>12</CorrCustCount>
<WfmEtr>2017-11-24T21:00:00Z</WfmEtr>
<NoOfPolesDown>1</NoOfPolesDown>
<NoOfTransformers>0</NoOfTransformers>
<NoOfCrossArms>0</NoOfCrossArms>
<NoOfTreeTrimSpans>0</NoOfTreeTrimSpans>
<ConductorSpan>0</ConductorSpan>
<NoServicesDown>0</NoServicesDown>
<Inaccessible>>false</Inaccessible>
<FollowupRequired>>false</FollowupRequired>
<LineCrewNeeded>>false</LineCrewNeeded>
<TreeCrewNeeded>>false</TreeCrewNeeded>
<FirstResponderNeeded>>false</FirstResponderNeeded>
<EvaluatorNeeded>>false</EvaluatorNeeded>

```

```
<PointOfFailure />
<WorkOrderId />
</IncidentStatusWfm>
```

3.4.8. Incident Update

This is an outbound message from ADMS. This message will indicate first transmittal of work from ADMS to WFM.

3.4.8.1. Characteristics

Message Type	XML
Frequency	• After order update of the incident from ADMS. • After incident roll down. • After trouble call conditions update. • After manually updating incident properties defined using the WFM_UPDATE_FIELDS parameter in NetServerOmsConfig.txt.
Error Handling	NACK message will be sent back.

Required Information

- Incident type
 - ETR
 - StartTime: StartTime of the incident in UTC format.
 - CritCust: Number of critical customers of type 1 affected.
 - CritCust2: Number of critical customers of type 2 affected.
 - CritCust3: Number of critical customers of type 3 affected.
 - CritCust4: Number of critical customers of type 4 affected.
 - CritCust5: Number of critical customers of type 5 affected.
 - PredCust: Number of active customers predicted out.
 - TotalPredictedCustomerCount: Number of active and inactive customers predicted out.
 - Device type: Switch (switch, fuse, breaker, or recloser etc.)
 - Device status: confirmed out, predicted out.
 - Priority: Priority of the incident.
 - IsNonSpecified: If true, then it means no crew has been specified for this incident. It is up to WFM to decide which crew needs to be assigned.
 - Case note number if any.
 - IsScada: If true, then it means that the incident is related to SCADA device outage.
 - IsUpdate
 - Update type
 - SystemAffected
 - NoOfPolesDown
 - NoOfTransformers
 - ConductorSpan
 - Material
 - NoServicesDown
 - FailedEquipment
 - Ranking
 - CrewAssignedCount
 - WeatherCode
 - Reason
 - ReasonGroup
 - RestorationTime
 - CallType: Shows the call type for non-outage incidents with a call type (set by the operator or extracted from a customer call). Shows "-1", for outage incidents and non-outage incidents without a call type. (A value of "-1" is used to signify a null or not set value; this value in messages is ignored.)
 - HazardousCustomer
-

Required Information (Cont...)

- HasAnUpstreamReclosingDisabledDevice
 - Message base sequence
 - X
 - Y
 - PhoneNumber
 - TroubleCode
 - CustomerList.Call.CallType: Shows the call type for each customer associated with an incident. This field appears only for outage incidents.
 - Priority
 - Critical
 - Critical2
 - Critical3
 - Critical4
 - Critical5
 - Condition 1
 - Condition 2
 - Description
 - Latitude
 - Longitude
 - PhoneCall
-

Optional Information:

- Incident ID. The default format is "dddnnnnnn" (Day of the Year + Sequence).
 - Name of the incident.
 - Device ID from GIS (or transformer ID for the single customer incident.). This is the device of record for the incident.
 - Device Name: Human readable name.
 - Device Size: The nominal rating of the device.
 - Substation Name: Substation that the device is nominally associated with.
 - Operated Substation Name: Substation that the device is dynamically associated with.
 - Feeder Name: Feeder name that the device is nominally associated with.
 - Operated Feeder Name: Feeder name that the device is dynamically associated with.
 - OpCenter: Service center the incident is associated with. This is determined based on the service center that the first customer belongs to. For deployments in which service centers do not form the lowest level of the organizational hierarchy, the appropriate hierarchy level can be specified using the WFM_ORG_LEVEL_FOR_OP_CENTER parameter in NetServerOmsConfig.txt. This configuration controls the org level that is to be communicated as the value for "OpCenter". An integer (n) from 1 to 9 specifies the ORGLEVELnNAME field to be passed as the value for OpCenter. If unspecified or invalid, the default is 1.
 - Driving directions
 - Sub Area (read from the first customer in the list – field GWA in the customer database).
 - Company ID for the first customer extracted from the customer file.
 - Lat/long of the device translated to Lat/Long in decimal degrees. All customer information has an associated lat/long. Channel Adapter can pick the first customer's Lat/Long to fill the information being sent to WFM.
 - Predicted phase where trouble is.
 - Additional operator comments.
 - Additional crew comments.
-

Optional Information: (Cont...)

- Customer list: Information about five customers associated with the incident. This is based on priority customers (see details in the schema).
 - Work order ID
 - Point of failure
 - Message base Guid
 - Name
 - Address
 - City
 - State
 - Zipcode
 - Remarks
 - MeterNumber
 - TfName
 - SwitchName
 - HazardousCondition
 - AccountNumber
 - Region information: region type, name, and ID
 - CustomerKey
 - CallReferenceId
 - PredictedFaultLocationX
 - PredictedFaultLocationY
 - AreaSettingCode (Integer): If the Incident is covered by an AreaSetting and the "Use Area Note" radio button on the AreaSetting Dialog is checked, this property will contain the value of the AreaSettingCode as defined in the Coded Translation. Otherwise it is null (not present in the Json).
 - AreaSettingText (String): If the Incident is covered by an AreaSetting, this property will contain either the text name of the AreaCode (as defined in the Coded Translation) if the "Use Area Note" radio button is checked, otherwise it will contain the "Free Form Remarks Text" if the "Use the Free Form Remarks" radio button is checked.
-

3.4.8.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/IncidentUpdateWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/IncidentUpdateWfm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="IncidentUpdateWfm" type="tns:IncidentUpdateWfmV1" />
  <xs:complexType name="IncidentUpdateWfmV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:MessageBaseV1">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentName" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="IncidentType" type="xs:int" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>
```

```

<xs:element minOccurs="1" maxOccurs="1" name="ETR" type="xs:dateTime" />
<xs:element minOccurs="0" maxOccurs="1" name="DevId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DevName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DevSize" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="SubstationName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="OperatedSubstationName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="FdrName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="OperatedFdrName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="OpCenter" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DrivingDirections" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="SubArea" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CompanyId" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="StartTime" type="xs:dateTime" />
<xs:element minOccurs="1" maxOccurs="1" name="CritCust" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="CritCust2" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="CritCust3" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="CritCust4" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="CritCust5" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="PredCust" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="TotalPredictedCustomerCount" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="DeviceType" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="DeviceStatus" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Priority" type="xs:int" />
<xs:element minOccurs="0" maxOccurs="1" name="LatLong" type="tns:PointD" />
<xs:element minOccurs="1" maxOccurs="1" name="IsNonSpecified" type="xs:boolean" />
<xs:element minOccurs="0" maxOccurs="1" name="TrblPredPhase" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="CaseNoteCode" type="xs:int" />
<xs:element minOccurs="0" maxOccurs="1" name="Comments" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CrewComments" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="IsScada" type="xs:boolean" />
<xs:element minOccurs="0" maxOccurs="1" name="CustomerList" type="tns:ArrayOfCustMsgV1" />
<xs:element minOccurs="1" maxOccurs="1" name="IsUpdate" type="xs:boolean" />
<xs:element minOccurs="1" maxOccurs="1" name="UpdateType" type="xs:int" />

```

<!--Numeric value of enum eUpdateType:

{None=0, DispatchCrew = 1, AssignWithAssist = 2, Unassign = 3, CancelIncident = 4, DeviceReprediction = 5, DamageAssessment = 6, ClearIncident = 7, AssignCrew = 8 } -->

```

<xs:element minOccurs="0" maxOccurs="1" name="WorkOrderId" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="SystemAffected" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="NoOfPolesDown" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="NoOfTransformers" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="NoOfCrossArms" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="NoOfTreeTrimSpans" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="ConductorSpan" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Material" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="NoServicesDown" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="FailedEquipment" type="xs:int" />
<xs:element minOccurs="0" maxOccurs="1" name="PointOfFailure" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CrewType" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="CrewsAssignedCount" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Ranking" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="ReasonGroup" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Reason" type="xs:int" />

```

```

        <xs:element minOccurs="1" maxOccurs="1" name="WeatherCode" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="RestorationTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="CallType" type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="Regions" type="tns:ArrayOfRegionLevelForV1" />
        <xs:element minOccurs="1" maxOccurs="1" name="HazardousCustomer" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="HasAnUpstreamReclosingDisabledDevice"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="PredictedFaultLocationX" nillable="true"
type="xs:double" />
        <xs:element minOccurs="1" maxOccurs="1" name="PredictedFaultLocationY" nillable="true"
type="xs:double" />
        <xs:element minOccurs="1" maxOccurs="1" name="AreaSettingCode" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="0" maxOccurs="1" name="AreaSettingText" type="xs:string" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
    <xs:complexContent mixed="false">
        <xs:extension base="tns:NetMessageBase" />
    </xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PointD">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="X" type="xs:double" />
        <xs:element minOccurs="1" maxOccurs="1" name="Y" type="xs:double" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfCustMsgV1">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="CustMsgV1" nillable="true"
type="tns:CustMsgV1" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="CustMsgV1">
    <xs:attribute name="Name" type="xs:string" />
    <xs:attribute name="PhoneNumber" type="xs:long" use="required" />
    <xs:attribute name="Address" type="xs:string" />
    <xs:attribute name="City" type="xs:string" />
    <xs:attribute name="State" type="xs:string" />
    <xs:attribute name="Zipcode" type="xs:string" />
    <xs:attribute name="TroubleCode" type="xs:string" />
    <xs:attribute name="Remarks" type="xs:string" />
    <xs:attribute name="MeterNumber" type="xs:string" />
    <xs:attribute name="TfName" type="xs:string" />
    <xs:attribute name="SwitchName" type="xs:string" />

```

```

<xs:attribute name="CallType" type="xs:int" use="required" />
<xs:attribute name="Priority" type="xs:int" use="required" />
<xs:attribute name="Critical" type="xs:int" use="required" />
<xs:attribute name="Critical2" type="xs:int" use="required" />
<xs:attribute name="Critical3" type="xs:int" use="required" />
<xs:attribute name="Critical4" type="xs:int" use="required" />
<xs:attribute name="Critical5" type="xs:int" use="required" />
<xs:attribute name="Condition_1" type="xs:int" use="required" />
<xs:attribute name="Condition_2" type="xs:int" use="required" />
<xs:attribute name="Description" type="xs:int" use="required" />
<xs:attribute name="HazardousCondition" type="xs:string" />
<xs:attribute name="Latitude" type="xs:double" use="required" />
<xs:attribute name="Longitude" type="xs:double" use="required" />
<xs:attribute name="AccountNumber" type="xs:string" />
<xs:attribute name="PhoneCall" type="xs:boolean" use="required" />
<xs:attribute name="CustomerKey" type="xs:string" />
<xs:attribute name="CallReferenceId" type="xs:string" />
</xs:complexType>
<xs:complexType name="ArrayOfRegionLevelForV1">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="RegionLevelForV1" nillable="true"
type="tns:RegionLevelForV1" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="RegionLevelForV1">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="RegionType" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="RegionName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="RegionID" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

3.4.8.3. Enum Schema Definition eUpdateType

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="eUpdateType" type="eUpdateType" />
<xs:simpleType name="eUpdateType">
<xs:restriction base="xs:string">
<xs:enumeration value="None" />
<xs:enumeration value="DispatchCrew" />
<xs:enumeration value="AssignWithAssist" />
<xs:enumeration value="Unassign" />
<xs:enumeration value="CancelIncident" />
<xs:enumeration value="DeviceReprediction" />
<xs:enumeration value="DamageAssessment" />
<xs:enumeration value="ClearIncident" />
<xs:enumeration value="AssignCrew" />
</xs:restriction>
</xs:simpleType>

```

```
</xs:schema>
```

3.4.8.4. Enum Schema Definition eIncidentType

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="eIncidentType" type="eIncidentType" />
  <xs:simpleType name="eIncidentType">
    <xs:restriction base="xs:string">
      <xs:enumeration value="SingleCustomer" />
      <xs:enumeration value="Momentary" />
      <xs:enumeration value="Predicted" />
      <xs:enumeration value="Confirmed" />
      <xs:enumeration value="Isolated" />
      <xs:enumeration value="ManuallyConfirmed" />
      <xs:enumeration value="Planned" />
      <xs:enumeration value="NonOutage" />
      <xs:enumeration value="IsolatedOutage" />
      <xs:enumeration value="AreaOutage" />
      <xs:enumeration value="Emergency" />
      <xs:enumeration value="PowerQuality" />
    </xs:restriction>
  </xs:simpleType>
</xs:schema>
```

3.4.8.5. Enum Schema Definition eDeviceStatus

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="eDeviceStatus" type="eDeviceStatus" />
  <xs:simpleType name="eDeviceStatus">
    <xs:restriction base="xs:string">
      <xs:enumeration value="ConfirmedOut" />
      <xs:enumeration value="PredictedOut" />
    </xs:restriction>
  </xs:simpleType>
</xs:schema>
```

3.4.8.6. Enum Schema Definition ePlannedOutageType

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="ePlannedOutageType" type="ePlannedOutageType" />
  <xs:simpleType name="ePlannedOutageType">
    <xs:restriction base="xs:string">
      <xs:enumeration value="Regular" />
      <xs:enumeration value="Emergent" />
    </xs:restriction>
  </xs:simpleType>
</xs:schema>
```

```

        <xs:enumeration value="LoadShed" />
    </xs:restriction>
</xs:simpleType>
</xs:schema>

```

3.4.8.7. Enum Schema Definition ePlannedOutageStatus

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="ePlannedOutageStatus" type="ePlannedOutageStatus" />
  <xs:simpleType name="ePlannedOutageStatus">
    <xs:restriction base="xs:string">
      <xs:enumeration value="Draft" />
      <xs:enumeration value="Cancelled" />
      <xs:enumeration value="Active" />
      <xs:enumeration value="ReadyToNotify" />
      <xs:enumeration value="Review" />
      <xs:enumeration value="Approved" />
      <xs:enumeration value="Completed" />
      <xs:enumeration value="PendingReschedule" />
      <xs:enumeration value="Rescheduled" />
    </xs:restriction>
  </xs:simpleType>
</xs:schema>

```

3.4.8.8. Enum Schema Definition eConfirmedIncidentSubtypes

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="eConfirmedIncidentSubtypes" type="eConfirmedIncidentSubtypes" />
  <xs:simpleType name="eConfirmedIncidentSubtypes">
    <xs:restriction base="xs:string">
      <xs:enumeration value="Confirmed" />
      <xs:enumeration value="FoundOpen" />
      <xs:enumeration value="Commanded" />
      <xs:enumeration value="LoadShed" />
    </xs:restriction>
  </xs:simpleType>
</xs:schema>

```

3.4.8.9. Sample XML Message

```

<?xml version="1.0"?>
<IncidentUpdateWfm xmlns:xsd="http://www.w3.org/2001/XMLSchema"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/IncidentUpdateWfm">
  <Sequence>204</Sequence>
  <Guid>50a5794b-4e97-4b4d-a102-79c904319afc</Guid>

```

```

<IncidentId>3280000001</IncidentId>
<IncidentName />
<IncidentType>3</IncidentType>
<ETR>2017-11-24T03:00:00.8171381-08:00</ETR>
<DevId>22234220</DevId>
<DevName>50878</DevName>
<SubstationName>PEBBLE BEACH</SubstationName>
<FdrName>F6533</FdrName>
<OpCenter>WST</OpCenter>
<DrivingDirections />
<SubArea />
<CompanyId>COMPANY</CompanyId>
<StartTime>2017-11-24T08:32:08.8171381Z</StartTime>
<CritCust>3</CritCust>
<CritCust2>0</CritCust2>
<CritCust3>0</CritCust3>
<CritCust4>0</CritCust4>
<CritCust5>0</CritCust5>
<PredCust>1214</PredCust>
<DeviceType>4</DeviceType>
<DeviceStatus>1</DeviceStatus>
<Priority>0</Priority>
<LatLong>
<X>943070.9375</X>
<Y>813821.4375</Y>
</LatLong>
<IsNonSpecified>true</IsNonSpecified>
<TrblPredPhase>0</TrblPredPhase>
<CaseNoteCode>0</CaseNoteCode>
<Comments />
<IsScada>false</IsScada>
<CustomerList>
<CustMsgV1 Name="Bedell Tim C" PhoneNumber="5555692527" Address="8097 STONE PL" City="" State=""
Zipcode="" TroubleCode="10000" Remarks="" MeterNumber="545613Q0" TfName="" SwitchName="" CallType="0"
Priority="0" Critical="0" Critical2="0" Critical3="0" Critical4="0" Critical5="0"
HazardousCondition="" Latitude="0" Longitude="0" />
</CustomerList>
<IsUpdate>false</IsUpdate>
<UpdateType>1</UpdateType>
<WorkOrderId />
<SystemAffected>1</SystemAffected>
<NoOfPolesDown>0</NoOfPolesDown>
<NoOfTransformers>0</NoOfTransformers>
<NoOfCrossArms>0</NoOfCrossArms>
<NoOfTreeTrimSpans>0</NoOfTreeTrimSpans>
<ConductorSpan>0</ConductorSpan>
<Material>0</Material>
<NoServicesDown>0</NoServicesDown>
<FailedEquipment>0</FailedEquipment>
<PointOfFailure />
</IncidentUpdateWfm>

```

3.4.9. Crew Update

This is an outbound message from ADMS. This message will be for the Crew Updates from ADMS to WFM.

3.4.9.1. Characteristics

Message Type	XML
Frequency	• After order update of the incident from ADMS. • After manually updating incident properties defined using the WFM_UPDATE_FIELDS parameter in NetServer0msConfig.txt if WFM_CREWUPDATE_FOR_INCIDENT_CHANGE = true.
Error Handling	NACK message will be sent back.
Required Information	• UpdateType: Numeric value of eUpdateType. • Crew Type: Numeric value of the CrewType coded translation item. • CrewTypeName: Text value of the CrewType coded translation Item. • IsAssisted: Set to true if assisting another crew already assigned. • Message base sequence.
Optional Information	• Incident ID: Incident GUID. • Incident name: Human readable ID in the format "dddnnnnnn") Day of the Year + Sequence. • CrewID: ID of the crew assigned. • Name: Crew name. • Vehicle: Vehicle information, if any. • Message base GUID.

3.4.9.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/CrewUpdateWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/CrewUpdateWfm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CrewUpdateWfm" type="tns:CrewUpdateWfmV1" />
  <xs:complexType name="CrewUpdateWfmV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:MessageBaseV1">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="UpdateType" type="xs:int" />
        </xs:sequence>
        <!--Numeric value of enum eUpdateType:
        {None = 0, DispatchCrew = 1, AssignWithAssist = 2, Unassign = 3, CancelIncident = 4,
        DeviceReprediction = 5, DamageAssessment = 6, ClearIncident = 7, AssignCrew = 8 } -->
        <xs:element minOccurs="1" maxOccurs="1" name="CrewType" type="xs:int" />
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>
```

```

        <xs:element minOccurs="1" maxOccurs="1" name="IsAssisted" type="xs:boolean" />
        <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Vehicle" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="UserId" type="xs:string" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
    <xs:complexContent mixed="false">
        <xs:extension base="tns:NetMessageBase" />
    </xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
    </xs:sequence>
</xs:complexType>
</xs:schema>

```

3.4.9.3. Sample XML Message

```

<?xml version="1.0"?>
<CrewUpdateWfm xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/CrewUpdateWfm">
<Sequence>277</Sequence>
<Guid>8774df11-d206-4da8-be19-78c8a6207862</Guid>
<IncidentId>3280000002</IncidentId>
<CrewId>4412</CrewId>
<UpdateType>1</UpdateType>
<IsAssisted>>false</IsAssisted>
<Name>Wooster [4412] - AUTO</Name>
<Vehicle>VH10002,VH10004</Vehicle>
</CrewUpdateWfm>

```

3.4.10. Ack/Nack

Characteristics, schema definitions, and sample messages for the "Ack/Nack" message are provided in [Positive/Negative Acknowledgement Messages](#).

3.4.11. Vehicle Location (Optional)

3.4.11.1. Characteristics

Message Type	XML
--------------	-----

Frequency	If WMS system is AVL (Automatic Vehicle Location) capable, then ADMS sends message for Vehicle location every 30 seconds (Configurable) and this message will be sent by WFM system as a response of vehicle query message.
Error Handling	NACK message will be sent back.
Required Information	• Message base sequence • Latitude • Longitude • X • Y
Optional Information	• System ID (for future use) • Vehicle locations • Message base GUID • VehicleId • CrewId

3.4.11.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/VehicleLocationWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/VehicleLocationWfm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="VehicleLocationWfm" type="tns:VehicleLocationWfmV1" />
<xs:complexType name="VehicleLocationWfmV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<!--Not in use at all-->
<xs:element minOccurs="0" maxOccurs="1" name="SystemId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="VehicleLocations" type="tns:ArrayOfVehicleLocationV1" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfVehicleLocationV1">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="VehicleLocationV1" nillable="true"
type="tns:VehicleLocationV1" />
</xs:sequence>
</xs:complexType>
```

```

</xs:sequence>
</xs:complexType>
<xs:complexType name="VehicleLocationV1">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="1" name="VehicleId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="Latitude" type="xs:double" />
<xs:element minOccurs="1" maxOccurs="1" name="Longitude" type="xs:double" />
<xs:element minOccurs="1" maxOccurs="1" name="X" type="xs:double" />
<xs:element minOccurs="1" maxOccurs="1" name="Y" type="xs:double" />
</xs:sequence>
</xs:complexType>
</xs:schema>

```

3.4.11.3. Sample XML Message

```

<?xml version="1.0"?>
<VehicleLocationWfm xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/VehicleLocationWfm">
<Sequence>3755</Sequence>
<Guid>2be046af-b91a-422a-baef-e2435729a0ea</Guid>
<SystemId />
<VehicleLocations>
<VehicleLocationV1>
<VehicleId />
<CrewId>5291</CrewId>
<Latitude>0</Latitude>
<Longitude>0</Longitude>
<X>944834.875</X>
<Y>813288</Y>
</VehicleLocationV1>
</VehicleLocations>
</VehicleLocationWfm>

```

3.4.12. Vehicle Query (Optional)

3.4.12.1. Characteristics

Message Type	XML
Frequency	Sent by ADMS every 30 seconds.
Error Handling	NACK message will be sent back.
Required Information	• Message base sequence
Optional Information	• System ID • Message base GUID

3.4.12.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/VehicleQueryWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/VehicleQueryWfm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="VehicleQueryWfm" type="tns:VehicleQueryWfmV1" />
<xs:complexType name="VehicleQueryWfmV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="1" name="WmsSystemId" type="xs:string" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
</xs:schema>
```

3.4.12.3. Sample XML Message

```
<?xml version="1.0"?>
<VehicleQueryWfm xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/VehicleQueryWfm">
<Sequence>266</Sequence>
<Guid>1bd4e7ab-d25c-4da4-84cd-032803e6b1b8</Guid>
<WmsSystemId>Crew Server</WmsSystemId>
</VehicleQueryWfm>
```

3.4.13. Vehicle Subscribe (Optional)

3.4.13.1. Characteristics

Message Type	XML
Frequency	Sent by ADMS after the hello message, if WFM is capable of sending vehicle locations.
Error Handling	NACK message will be sent back.

Required Information	• Frequency • Message base sequence
Optional Information	• System ID • Message base GUID

3.4.13.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/VehicleSubscribeWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/VehicleSubscribeWfm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="VehicleSubscribeWfm" type="tns:VehicleSubscribeWfmV1" />
<xs:complexType name="VehicleSubscribeWfmV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="1" name="WmsSystemId" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="Frequency" type="xs:int" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
</xs:schema>
```

3.4.13.3. Sample XML Message

```
<?xml version="1.0"?>
<VehicleSubscribeWfm xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/VehicleSubscribeWfm">
<Sequence>2</Sequence>
<Guid>1af2d00c-b300-47b7-9b62-d05fb685d9c0</Guid>
<WmsSystemId>Crew Server</WmsSystemId>
<Frequency>30</Frequency>
</VehicleSubscribeWfm>
```

3.4.14. Damage Assessment Status

3.4.14.1. Characteristics

Message Type	XML
Frequency	Sent by WFM
Required Information	NetMessageBase: • Message base sequence AssessmentLocation: • Latitude • Longitude • XCoordinate • YCoordinate • Status • Radius • RepairHours • IsTruckAccessible • TreeTrimmingRequired • NumOfPolesDown • NumOfTransformers • NumOfCrossArms • NumOfTreeTrimSpans • NumOfServicesDown • NumOfDevicesDown • ConductorSpan
Optional Information	MessageBase: • Id • OrderNumber • Name • AssessmentLocations NetMessageBase: • GUID AssessmentLocation: • Id • Comments • ConductorSpan • AdditionalData

3.4.14.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/AssessmentStatusWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/AssessmentStatusWfm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="AssessmentStatusWfm" type="tns:AssessmentStatusWfmV1" />
  <xs:complexType name="AssessmentStatusWfmV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:MessageBaseV1">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="OrderNumber" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="AssessmentLocations"
type="tns:ArrayOfAssessmentLocation" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:NetMessageBase" />
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="NetMessageBase">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfAssessmentLocation">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="AssessmentLocation" nillable="true"
type="tns:AssessmentLocation" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="AssessmentLocation">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="Latitude" type="xs:double" />
      <xs:element minOccurs="1" maxOccurs="1" name="Longitude" type="xs:double" />
      <xs:element minOccurs="1" maxOccurs="1" name="XCoordinate" type="xs:double" />
      <xs:element minOccurs="1" maxOccurs="1" name="YCoordinate" type="xs:double" />
      <xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="Radius" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="RepairHours" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="IsTruckAccessible" nillable="true"
type="xs:boolean" />
      <xs:element minOccurs="1" maxOccurs="1" name="TreeTrimmingRequired" type="xs:int" />
      <xs:element minOccurs="0" maxOccurs="1" name="Comments" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="NumOfPolesDown" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="NumOfTransformers" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="NumOfCrossArms" type="xs:int" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

```

        <xs:element minOccurs="1" maxOccurs="1" name="NumOfTreeTrimSpans" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="NumOfServicesDown" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="NumOfDevicesDown" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="ConductorSpan" type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="ServiceCenter" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="tns:ArrayOfPropertyValueV1"
/>
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValueV1">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValueV1" nillable="true"
type="tns:PropertyValueV1" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValueV1">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
</xs:schema>

```

3.4.14.3. Sample XML Message

```

<AssessmentStatusWfm xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/AssessmentStatusWfm">
<Sequence>465679</Sequence>
<Guid>8461010a-95d0-40eb-b743-236dadf7279a</Guid>
<Id>34cd91a1-33a1-4a2d-afe2-dc6d3ece8a7e</Id>
<OrderNumber>0590000001</OrderNumber>
<Name>4249</Name>
<AssessmentLocations>
<AssessmentLocation>
<Id>bbad0cc6-0196-4b44-bc85-4ac9349153a6</Id>
<Latitude>0</Latitude>
<Longitude>0</Longitude>
<XCoordinate>940464</XCoordinate>
<YCoordinate>775562.2</YCoordinate>
<Status>0</Status>
<Radius>12</Radius>
<RepairHours>2</RepairHours>
<IsTruckAccessible>false</IsTruckAccessible>
<TreeTrimmingRequired>0</TreeTrimmingRequired>
<Comments />
<NumOfPolesDown>1</NumOfPolesDown>
<NumOfTransformers>1</NumOfTransformers>
<NumOfServicesDown>1</NumOfServicesDown>
<NumOfDevicesDown>1</NumOfDevicesDown>
<ConductorSpan>1</ConductorSpan>
</AssessmentLocation>

```

```
</AssessmentLocations>
</AssessmentStatusWfm>
```

3.4.15. Damages

This message is sent by WFM to add, update, or delete damage data in ADMS.

3.4.15.1. Characteristics

Message Type	XML
Frequency	Sent by WFM
Required Information	NetMessageBase: - Message base sequence Damages: - DamageId - DamageAssessmentLocationId - AssetType - AssetSubType - Quantity
Optional Information	MessageBase: - Id - OrderNumber - Name - AssessmentLocations NetMessageBase: - GUID Damages: - Description

3.4.15.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="Damage" nillable="true" type="Damage" />
  <xs:complexType name="Damage">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="DamageId" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="AssetType" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="AssetSubType" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

```

    <xs:element minOccurs="1" maxOccurs="1" name="Quantity" type="xs:int" />
    <xs:element minOccurs="0" maxOccurs="1" name="Description" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

3.4.16. Order Status

This is an outbound message from ADMS. This message is for follow-up work order statuses from ADMS to WFM.

3.4.16.1. Characteristics

Message Type	XML
Frequency	After every order status change from WFM
Error Handling	NACK message is returned
Required Information	• Organization • OrderType • Priority • Status • DeleteOrder • Message base sequence
Optional Information	• Id • Name • Comment • DeviceName • DeviceId • DeviceType • DeviceFeederName • DevicePhases • Aor • DeviceStation • CreatedBy • ModifiedBy • IncidentId • IncidentName • Message base GUID

3.4.16.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/OrderStatusWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/OrderStatusWfm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="OrderStatusWfm" type="tns:OrderStatusWfmV1" />
<xs:complexType name="OrderStatusWfmV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="Comment" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceType" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceFeederName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DevicePhases" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="Aor" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="Organization" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="OrderType" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Priority" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceStation" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CreatedBy" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="ModifiedBy" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="IncidentName" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="DeleteOrder" type="xs:boolean" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
</xs:schema>
```

3.4.17. Order Update

This is an outbound message from ADMS. This message is for follow-up work order updates from ADMS to WFM.

3.4.17.1. Characteristics

Message Type	XML
Frequency	After every order update of the incident from ADMS
Error Handling	NACK message is returned
Required Information	• Organization • OrderType • Priority • Status • Created • Modified • Message base sequence
Optional Information	• ID • Name • Comment • Device Name • Device ID • Device Type • Device Feeder Name • Device Phases • AOR • Device Station • Created By • Modified By • Incident ID • Incident Name • Customer Address • Message base GUID

3.4.17.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/OrderUpdateWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/OrderUpdateWfm" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="OrderUpdateWfm" type="tns:OrderUpdateWfmV1" />
<xs:complexType name="OrderUpdateWfmV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="Comment" type="xs:string" />
```

```

<xs:element minOccurs="0" maxOccurs="1" name="DeviceName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceType" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceFeederName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DevicePhases" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="Aor" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="Organization" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="OrderType" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Priority" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceStation" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CreatedBy" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="ModifiedBy" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="IncidentName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CustomerAddress" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="Created" type="xs:dateTime" />
<xs:element minOccurs="1" maxOccurs="1" name="Modified" type="xs:dateTime" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
</xs:schema>

```

3.4.17.3. Sample XML Message

```

<?xml version="1.0"?>
<OrderUpdateWfm xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/OrderUpdateWfm">
<Sequence>369</Sequence>
<Guid>f9e4fa51-cea4-4d17-89aa-cee63525d6e8</Guid>
<Id>25A7B357-5067-44E9-8847-4FC1967AD443</Id>
<Name>3280000001-1</Name>
<Comment />
<DeviceName>50878</DeviceName>
<DeviceId>22234220</DeviceId>
<DeviceType />
<DeviceFeederName />
<DevicePhases />
<Aor>WST</Aor>

```

```

<Organization>3</Organization>
<OrderType>0</OrderType>
<Priority>9</Priority>
<Status>0</Status>
<DeviceStation>PEBBLE BEACH</DeviceStation>
<CreatedBy>DMSALL</CreatedBy>
<ModifiedBy />
<IncidentId>97166AA3-C117-4677-9465-CDFB95F995BD</IncidentId>
<IncidentName>3280000001</IncidentName>
<Created>2017-11-24T00:51:48-08:00</Created>
<Modified>0001-01-01T00:00:00</Modified>
</OrderUpdateWfm>

```

3.4.18. Planned Outage Update

This is an outbound message from ADMS. This message sends planned outage updates from ADMS to WFM.

3.4.18.1. Characteristics

Message Type	XML
Frequency	On demand.
Error Handling	NACK message is returned
Required Information	• Status • CreatedTime • PrimaryStartTime • PrimaryEndTime • AlternateStartTime • AlternateEndTime • UseAlternateDate • DevicePtX • DevicePtY • DeviceType • SwitchType • Modified • PlannedOutageType • Message base Sequence

Optional Information

- Id
 - Name
 - EventType
 - DeviceId
 - DeviceName
 - DeviceTypeString
 - CustomerAccountsString
 - CustomerAccountsNotifyString
 - AddedCustomersString
 - AddedCustomersNotifyString
 - RemovedCustomersString
 - RemovedCustomersNotifyString
 - Station
 - RepresentativeName
 - RepresentativeNumber
 - RepresentativeEmail
 - RepresentativeAddress
 - OutageMessage
 - CreatorComments
 - AOR
 - FeederName
 - WorkOrderId
 - Message base GUID
-

3.4.18.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/PlannedOutageUpdateWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/PlannedOutageUpdateWfm"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="PlannedOutageUpdateWfm" type="tns:PlannedOutageUpdateWfmV1" />
  <xs:complexType name="PlannedOutageUpdateWfmV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:MessageBaseV1">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="CreatedTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="PrimaryStartTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="PrimaryEndTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="AlternateStartTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="AlternateEndTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="UseAlternateDate" type="xs:boolean" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>
```

```

<xs:element minOccurs="0" maxOccurs="1" name="EventType" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceName" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="DevicePtX" type="xs:float" />
<xs:element minOccurs="1" maxOccurs="1" name="DevicePtY" type="xs:float" />
<xs:element minOccurs="1" maxOccurs="1" name="DeviceType" type="xs:int" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceTypeString" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="SwitchType" type="xs:int" />
<xs:element minOccurs="0" maxOccurs="1" name="CustomerAccountsString" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CustomerAccountsNotifyString" type="xs:string"
/>

<xs:element minOccurs="0" maxOccurs="1" name="AddedCustomersString" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="AddedCustomersNotifyString" type="xs:string"
/>

<xs:element minOccurs="0" maxOccurs="1" name="RemovedCustomersString" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="RemovedCustomersNotifyString" type="xs:string"
/>

<xs:element minOccurs="0" maxOccurs="1" name="Station" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="RepresentativeName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="RepresentativeNumber" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="RepresentativeEmail" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="RepresentativeAddress" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="OutageMessage" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CreatorComments" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="AOR" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="Modified" type="xs:dateTime" />
<xs:element minOccurs="0" maxOccurs="1" name="FeederName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="WorkOrderId" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="PlannedOutageType" type="xs:int" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
  <xs:complexContent mixed="false">
    <xs:extension base="tns:NetMessageBase" />
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

3.4.19. Associate Damage Assessment File

Describes a file attached to a damage assessment location.

3.4.19.1. Characteristics

Message Type	XML
Frequency	Sent by WFM
Error Handling	NACK message is returned
Required Information	• Message base sequence • AssessmentId • HasCoords
Optional Information	Attached Files: • FileReference • FileName • FileType • XCoordinate • YCoordinate Message Base: • GUID

3.4.19.2. Sample XML Message

```
<?xml version="1.0"?>
<AssociateAssessmentFileWfm xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://www.esca.com/AssociateAssessmentFileWfm">
<Sequence>1582</Sequence>
<Guid>efeb864a-3b97-485b-a9d2-e56ca7e4714d</Guid>
<AssessmentId>af4f19ff-e457-4334-9a67-3b8de3ae21b5</AssessmentId>
<File>
<FileReference>22</FileReference>
<FileName />
<FileType />
<XCoordinate>0</XCoordinate>
<YCoordinate>0</YCoordinate>
<HasCoords>false</HasCoords>
</File>
</AssociateAssessmentFileWfm>
```

3.4.20. Remove Attached Damage Assessment File

Removes attached file from a damage assessment location.

3.4.20.1. Characteristics

Message Type	XML
Frequency	Sent by WFM
Error Handling	NACK message is returned
Required Information	• Message base sequence • AssessmentId • FileReference
Optional Information	Message Base: • Guid

3.4.20.2. Sample XML Message

```
<?xml version="1.0"?>
<DeleteAssessmentFileWfm xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://www.esca.com/DeleteAssessmentFileWfm">
<Sequence>923</Sequence>
<Guid>3d4015e7-2f7d-4848-8b30-62c06fae4f0f</Guid>
<AssessmentId>dc92f5e5-fd95-406f-aa7b-590bd518e639</AssessmentId>
<FileReference>213</FileReference>
</DeleteAssessmentFileWfm>
```

3.4.21. Device State Update

This is an inbound message for ADMS.

When a DeviceStateUpdateWfmV1 message is received by NetServer, the requested state change is validated against the current device state. If the requested state matches the current state, the request is dropped. This also applies to phases. For example, if the current device state is "AB" open and the request state is "BC" ConfirmClose, the request is not valid because the "C" phase is already closed.

All requests are logged in NetServer at the INFO level. The "UserName" field in the request is only used for logging purposes; it is not used in the device status change command that is sent to the DMS server. The UserID value is used in the request to the DMS server. The WFM trusted user name and ID need to be defined in the NetServerMainConfig_localhost.txt and NetServerOmsConfig_localhost.txt configuration files using the TRUSTED_USER_ID_WFM and TRUSTED_USER_NAME_WFM parameters.

Note

When sending a ConfirmOpen command, all downstream predicted incidents for the device will be marked as predicted stable and will not be merged in the newly created incident.

3.4.21.1. Characteristics

Message Type	XML
Frequency	Sent by WFM
Error Handling	NACK message is returned
Required Information	• Device State (defined in eConfirmDeviceState) • Phases (defined in ePhaseOut)
Optional Information	• User Name • Device ID • Station • Feeder

3.4.21.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/DeviceStateUpdateWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/DeviceStateUpdateWfm"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="DeviceStateUpdateWfm" type="tns:DeviceStateUpdateWfmV1" />
  <xs:complexType name="DeviceStateUpdateWfmV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:MessageBaseV1">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="UserName" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="DeviceId" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Station" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Feeder" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="DeviceState" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="Phases" type="xs:int" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:NetMessageBase" />
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="NetMessageBase">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

3.4.21.3. Enum Schema Definition eConfirmDeviceState

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="eConfirmDeviceState" type="eConfirmDeviceState" />
<xs:simpleType name="eConfirmDeviceState">
<xs:restriction base="xs:string">
<xs:enumeration value="ConfirmOpen" />
<xs:enumeration value="ConfirmClosed" />
<xs:enumeration value="FoundOpen" />
</xs:restriction>
</xs:simpleType>
</xs:schema>
```

3.4.22. Can Clear Incident

This is an inbound message for ADMS. When a CanClearIncident message is received by NetServer, the incident clearance validation is run for the incident to check whether the incident can be cleared.

3.4.22.1. Characteristics

Message Type	XML
Frequency	Sent by WFM
Error Handling	NACK message is returned
Required Information	• Message base sequence
Optional Information	• Incident ID • Message base GUID

3.4.22.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/DeviceStateUpdateWfm" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/DeviceStateUpdateWfm"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="CanClearIncidentRequestWfm" type="tns:CanClearIncidentRequestWfmV1" />
<xs:complexType name="CanClearIncidentRequestWfmV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
```

```
<xs:complexType name="MessageBaseV1">
  <xs:complexContent mixed="false">
    <xs:extension base="tns:NetMessageBase" />
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>
```

4. CIS and External IVR Interfaces

! Important

It is recommended to use the Call Server HTTP interface instead of the legacy TCP/IP interface for communicating with external systems.

4.1. Overview

The ADMS system interfaces with CIS and the External IVR through the Channel Adapter and the Enterprise Service Bus.

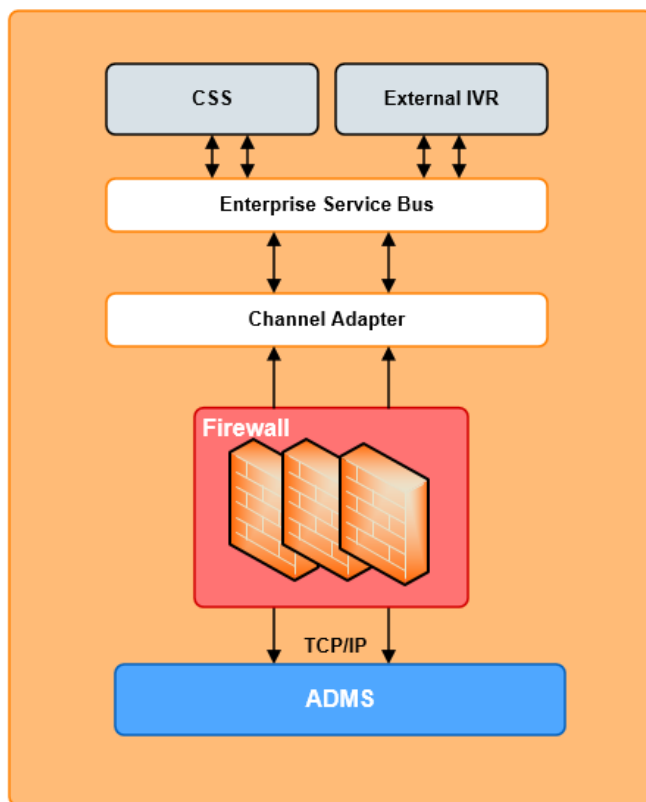


Figure 19. CIS and External IVR Interfaces - Overview

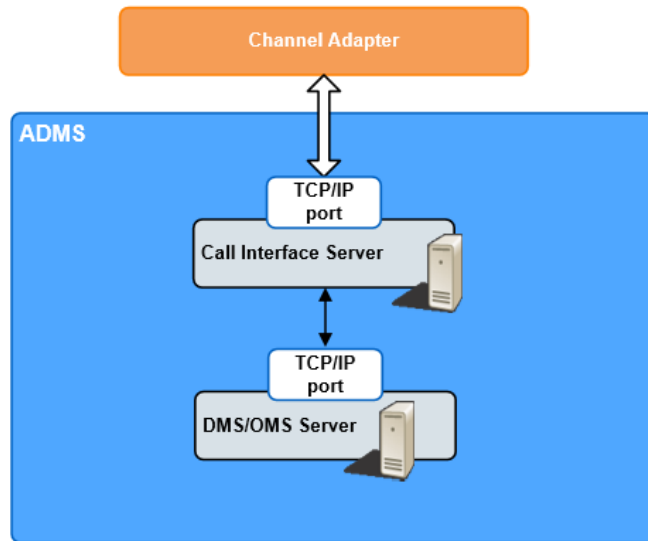


Figure 20. ADMS Components

The ADMS architecture provides the flexibility of running the Call server and the DMS/OMS server in two different physical servers as well. All the components of the ADMS communicate with each other on TCP/IP ports. The ADMS installation ensures that only one Crew Interface server is available to listen for connections and to transmit/receive messages at any given time. It is the responsibility of the Channel Adapter to re-connect to the redundant crew servers with a small delay of a few seconds between retries.

4.2. Interface Characteristics

The characteristics of the interface are the following:

Interface Type	XML Messaging
Protocol	TCP/IP

4.3. Message Flow

The interface to the Call server covers these functions:

- **Outage Calls (or "Calls"):** These are initiated by the power consumer to indicate that the power is out (or blinking, or dim lights, or there is an emergency). They are normally associated with a single customer. The message flow is from the CIS, or IVR to ADMS. Priority, unassociated calls are a special case of outage calls.
- **Call-backs:** These are initiated by ADMS and sent to the IVR when the customer wants a callback after a power restoration or ETR change.
- **Case Notes:** A case note is an indication of the trouble and an Estimated Time of Repair (ETR) that is maintained in ADMS. It is used to communicate outage related information to the

customer.

4.3.1. General Message Flow

The details of the Hello Messages are described in [Hello Messages](#).

The general message flow is shown in the below image.

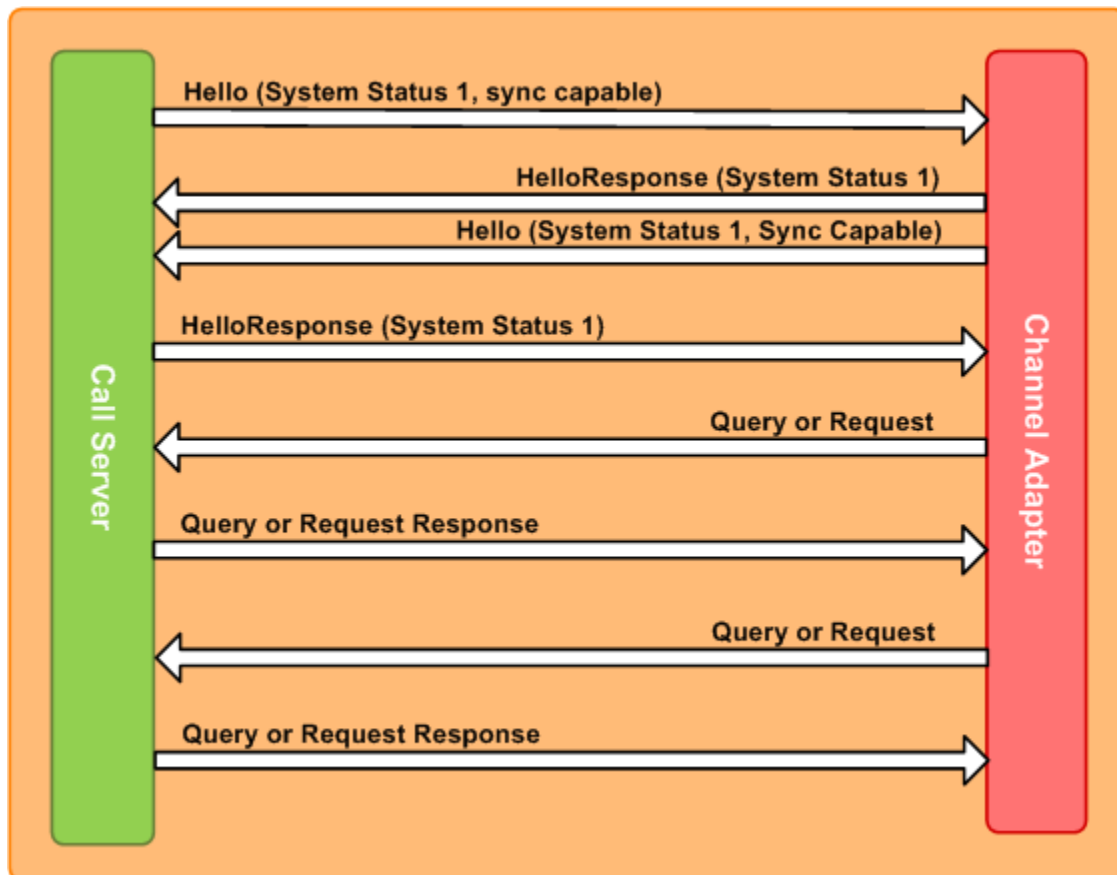


Figure 21. Call Server - General Message Flow

4.3.2. CIS Message Flow

In general, the CIS sends outage calls to the Call server and the adapter queries the Call server for case notes. The Call server queries the CIS for calls (nominally once every 10 seconds, which is configurable), and the adapter returns all trouble calls after a specified time in the query. If the connection is down for a period of time, the Call server asks for trouble calls since the last received call time.

The query for case notes can be for a single service center or an individual customer. Case note queries per customer are supported at the call rate (that is, for each customer call, it is possible for the CIS to query for a related case note).

The below image illustrates the general message flow for the CIS.

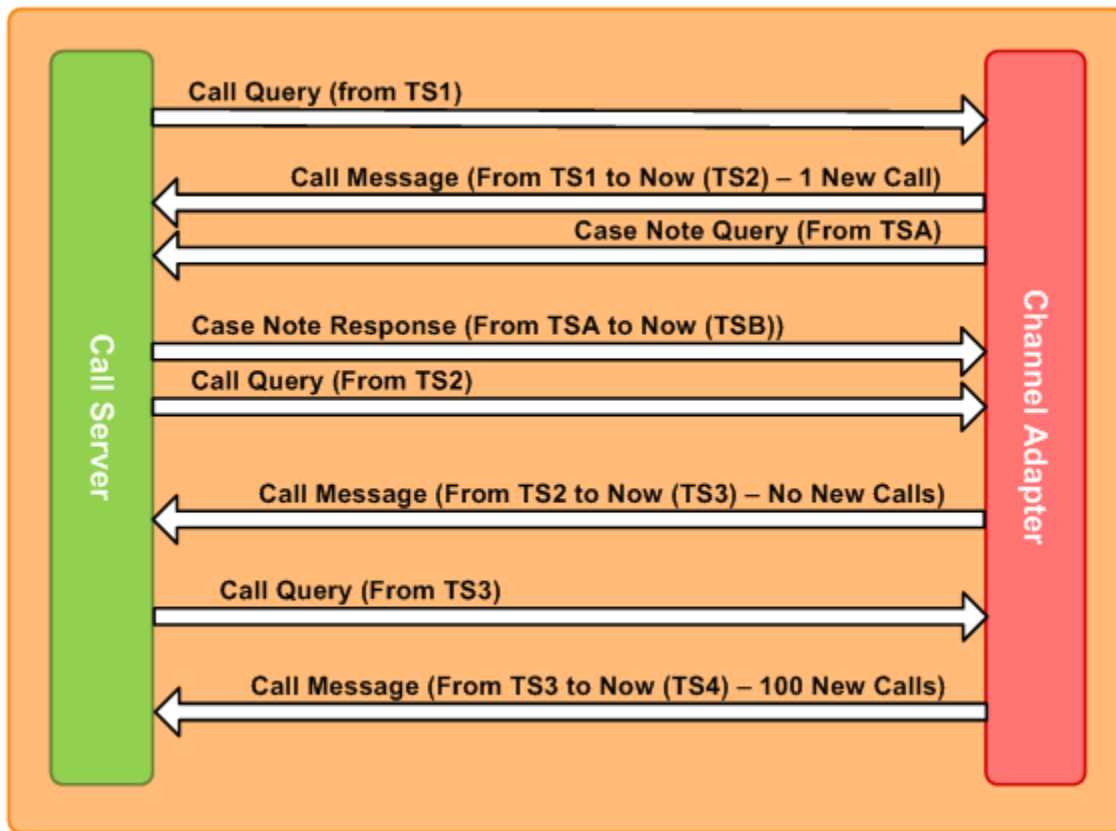


Figure 22. CIS Message Flow

4.3.3. IVR Message Flow

The IVR supports the same messages as the CIS and also supports callback messages.

The callback response message is just a positive acknowledgement that the message was received – it may be immediately returned from the adapter, or if the IVR supports it, this message may be returned later to indicate any problems.

i Note

The call back request message does not follow the general query and response paradigm. It is instead a push from ADMS system.

The message flow is illustrated in the below image.

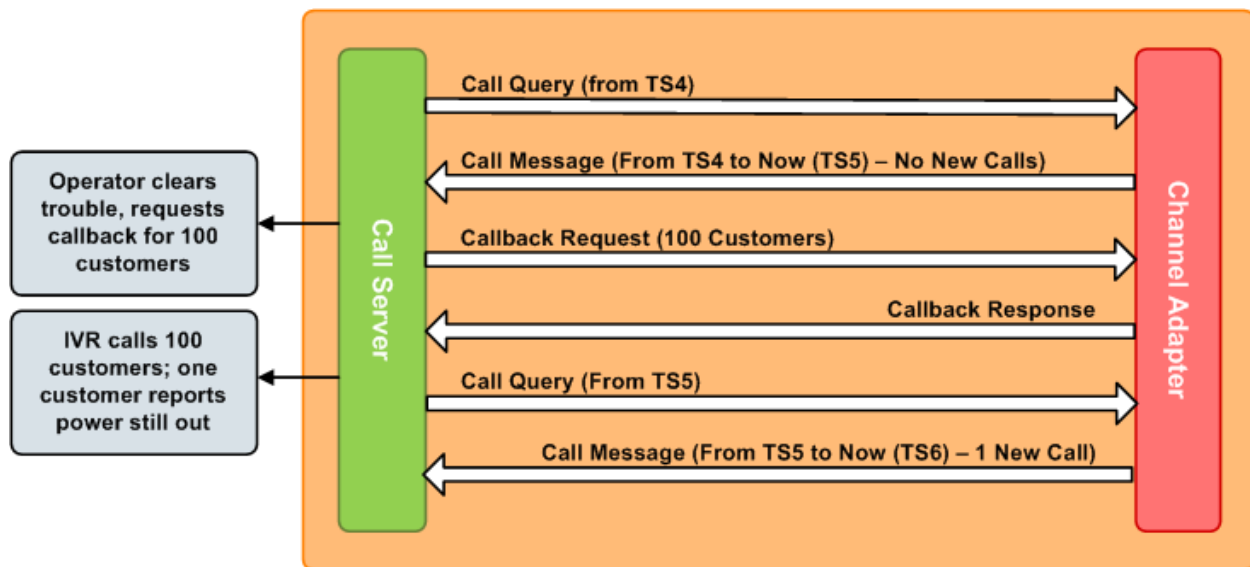


Figure 23. IVR Message Flow

4.4. Message Details

4.4.1. List of Messages

The list of supported messages is as follows:

- Hello
- Hello Response
- Ack/Nack
- Call query
- Call Message
- CallBack Request
- Case Note query
- Case Note Response
- Customer correction
- Customer correction acknowledgement

The following table describes the messages supported per each external system.

Table 1. CIS and IVR Supported Messages

Message	From CIS	From External IVR	To CIS	To External IVR
Hello	X	X	X	X

Message	From CIS	From External IVR	To CIS	To External IVR
Negative Acknowledgement	X	X	X	X
Call Query			X	X
Call Response	X	X		
Customer Correction			X	
Customer Correction Acknowledgement	X			
Callback Request				X
Case Note Query	X	X		
Case Note Response			X	X

4.4.2. Hello

Characteristics, schema definition, and sample messages for the "Hello" message are provided in [Hello Messages](#).

4.4.3. Hello Response

Characteristics, schema definition, and sample messages for the "Hello Response" message are provided in [Hello Response](#).

4.4.4. Call Query

This is a message that CallServer uses to get the latest calls from a CIS system. If there are no calls in the system, it starts the query with MinTime.

4.4.4.1. Characteristics

Message Type	XML
Frequency	30 seconds (configurable)
Error Handling	Timeout after a time.
Required Information	• Message base sequence • From Time Stamp: Request everything after (since) this time (echo of the timestamp last received from the call message).
Optional Information	• Message base GUID

4.4.4.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/CallQueryCiv" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/CallQueryCiv" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="CallQueryCiv" type="tns:CallQueryCivV1" />
<xs:complexType name="CallQueryCivV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="FromTimeStamp" type="xs:dateTime" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
</xs:schema>
```

4.4.4.3. Sample XML Message

```
<?xml version="1.0"?>
<CallQueryCiv xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/CallQueryCiv">
<Sequence>57644</Sequence>
<Guid>0032b4ac-2cab-4636-97bc-0da3623cb276</Guid>
<FromTimeStamp>0001-01-01T00:00:00</FromTimeStamp>
</CallQueryCiv>
```

4.4.5. Call Response

This is the schema in which CallServer expects a response back from the CIS system. Name, MeterNumber, DeviceId, FeederId, and Address fields are used only when AccountId is not modeled in the customer database used by the system.

4.4.5.1. Characteristics

Message Type	XML
--------------	-----

Frequency	As a response to CallQueryCivV1 message.
Error Handling	None
Required Information	• UpToTimeStamp: Requests for any changes before this time. • Message base sequence • Call Time: Time when the customer called. • Call Type^[1]: The type of call, which could be Lights Out, Area Out, etc. • Call duration • Condition_1¹: More information, such as Wires Down, Low Wires, etc. • Condition_2¹: More information, such as Pole Broken, Pole Down, etc. • Description¹: Description, such as Loud Noise, Saw Flash, etc. • Priority Code¹: None, Priority, Extreme. • Is Cancelled Call • Wants Call Back • Original Call Time • X: X-coordinate of the call. • Y: Y-coordinate of the call. • Latitude: Latitude of the call. • Longitude: Longitude of the call. • Is From Call Back • Trouble code • Is Unassociated • Query Time Stamp • Cancel Callback Request • Is Forwarded Call • Power Is On

Optional Information

- Message base GUID
 - Name: Name of the customer.
 - MeterNumber: Meter number of the customer.
 - Address: Address of the customer.
 - Reference ID: Reference ID of the customer.
 - Account ID: Account number of the customer.
 - Region_1: City of the call.
 - Region_2: Service center.
 - Alternate phone number
 - Comments
 - Feeder ID: Associated feeder name (currently not available, for future use).
 - Device ID: Associated device ID (currently not available, for future use).
 - Driving directions
 - Trouble code description
 - Callback Name
 - Alternate Name
 - Alternate Contact
 - User Name
 - Call Source
 - Call Source Type
 - Customer Key
 - Latest Callback Time
-

4.4.5.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/CallResponseCiv" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/CallResponseCiv" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CallResponseCiv" type="tns:CallResponseCivV1" />
  <xs:complexType name="CallResponseCivV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:MessageBaseV1">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="TroubleMessages"
type="tns:ArrayOfTroubleCallCivV1" />
          <xs:element minOccurs="1" maxOccurs="1" name="UpToTimeStamp" type="xs:dateTime" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:NetMessageBase" />
    </xs:complexContent>
  </xs:complexType>
```

```

</xs:complexType>
<xs:complexType name="NetMessageBase">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfTroubleCallCivV1">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="TroubleCallCivV1" nillable="true"
type="tns:TroubleCallCivV1" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="TroubleCallCivV1">
  <xs:attribute name="Name" type="xs:string" />
  <xs:attribute name="MeterNumber" type="xs:string" />
  <xs:attribute name="Address" type="xs:string" />
  <xs:attribute name="PhoneNumber" type="xs:string" />
  <xs:attribute name="ReferenceId" type="xs:string" />
  <xs:attribute name="AccountId" type="xs:string" />
  <xs:attribute name="CallTime" type="xs:dateTime" use="required" />
  <xs:attribute name="Region_1" type="xs:string" />
  <xs:attribute name="Region_2" type="xs:string" />
  <xs:attribute name="AltPhNum" type="xs:string" />
  <xs:attribute name="Comments" type="xs:string" />
  <xs:attribute name="CallType" type="xs:int" use="required" />
  <xs:attribute name="CallDuration" type="xs:int" use="required" />
  <xs:attribute name="Condition_1" type="xs:int" use="required" />
  <xs:attribute name="Condition_2" type="xs:int" use="required" />
  <xs:attribute name="Description" type="xs:int" use="required" />
  <xs:attribute name="PriorityCode" type="xs:int" use="required" />
  <xs:attribute name="FeederId" type="xs:string" />
  <xs:attribute name="DeviceId" type="xs:string" />
  <xs:attribute name="IsCancelledCall" type="xs:boolean" use="required" />
  <xs:attribute name="WantsCallback" type="xs:boolean" use="required" />
  <xs:attribute name="OriginalCallTime" type="xs:dateTime" use="required" />
  <xs:attribute name="X" type="xs:double" use="required" />
  <xs:attribute name="Y" type="xs:double" use="required" />
  <xs:attribute name="Latitude" type="xs:float" use="required" />
  <xs:attribute name="Longitude" type="xs:float" use="required" />
  <xs:attribute name="IsFromCallback" type="xs:boolean" use="required" />
  <xs:attribute name="DrivingDirections" type="xs:string" />
  <xs:attribute name="TroubleCode" type="xs:string" />
  <xs:attribute name="Unassociated" type="xs:boolean" use="required" />
  <xs:attribute name="QueryTimeStamp" type="xs:dateTime" use="required" />
  <xs:attribute name="CallbackName" type="xs:string" />
  <xs:attribute name="AlternateName" type="xs:string" />
  <xs:attribute name="AlternateContact" type="xs:string" />
  <xs:attribute name="CancelCallbackRequest" type="xs:boolean" use="required" />
  <xs:attribute name="UserName" type="xs:string" />
  <xs:attribute name="CallSource" type="xs:string" />
  <xs:attribute name="CallSourceType" type="xs:string" />
  <xs:attribute name="IsForwardedCall" type="xs:boolean" use="required" />

```

```

<xs:attribute name="BusinessName" type="xs:string" />
<xs:attribute name="CustomerKey" type="xs:string" />
<xs:attribute name="PowerIsOn" type="xs:boolean" use="required" />
<xs:attribute name="LatestCallbackTime" type="xs:string" />
</xs:complexType>
</xs:schema>

```

4.4.5.3. Sample XML Message

```

<?xml version="1.0"?>
<CallResponseCiv xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/CallResponseCiv">
<Sequence>58676</Sequence>
<Guid>ecc3ea81-3a4c-4a9a-80a3-e0a804c6b712</Guid>
<TroubleMessages />
<UpToTimeStamp>0001-01-01T00:00:00</UpToTimeStamp>
</CallResponseCiv>

```

4.4.6. Customer Correction Request

This message will be used to correct a wrongly coded customer. A delta file of customers will be updated in the ADMS system on a daily basis, which will include the information about these corrected customers. The format of the delta file will be the same as the main customer database file. In case of a situation where the customer database in the ADMS system is not updated and ADMS receives a customer call for a customer corrected in CIS, then it would still be handled with the old association.

4.4.6.1. Characteristics

Message Type	XML
Frequency	When the operator corrects the mismatch in ADMS.
Error Handling	None
Required Information	• Is Customer Relocated • Message base sequence

Optional Information

- Request ID
 - Premise Number
 - Customer Key
 - Account Number
 - Meter Number
 - Transformer Name
 - Transformer ID
 - Feeder Name
 - Company ID
 - Service Drop
 - Service Point ID
 - X Coordinate
 - Y Coordinate
 - Message base Guid
-

4.4.6.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/CustomerCorrectionCiv" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/CustomerCorrectionCiv"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="CustomerCorrectionCiv" type="tns:CustomerCorrectionCivV1" />
<xs:complexType name="CustomerCorrectionCivV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<!--Id of the request, which will be used to send the
acknowledgment back by the Channel adapter-->
<xs:element minOccurs="0" maxOccurs="1" name="RequestId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="PremiseNumber" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CustomerKey" type="xs:string" />
<!--//Account number of the customer-->
<xs:element minOccurs="0" maxOccurs="1" name="AccountNumber" type="xs:string" />
<!--Meter number of the customer-->
<xs:element minOccurs="0" maxOccurs="1" name="MeterNumber" type="xs:string" />
<!--Correct serving transformer-->
<xs:element minOccurs="0" maxOccurs="1" name="XfrmerName" type="xs:string" />
<!--Will be the PTAG of the serving transformer.-->
<xs:element minOccurs="0" maxOccurs="1" name="XfrmerId" type="xs:string" />
<!--Associated Feeder name-->
<xs:element minOccurs="0" maxOccurs="1" name="FdrName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CompanyId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="ServiceDrop" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="ServicePointID" type="xs:string" />
<!--X co-ordinate of the customer's location-->
<xs:element minOccurs="0" maxOccurs="1" name="XCoordinate" type="xs:string" />
<!--Y co-ordinate of the customer's location-->
```

```

<xs:element minOccurs="0" maxOccurs="1" name="YCoordinate" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="CustomerRelocated" type="xs:boolean" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
</xs:schema>

```

4.4.6.3. Sample XML Message

```

<?xml version="1.0"?>
<CustomerCorrectionCiv xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://www.esca.com/CustomerCorrectionCiv">
<Sequence>6782</Sequence>
<Guid>26b61bd5-8685-45c7-a370-ea07769ee17b</Guid>
<RequestId>bf068fad-e00a-41d3-be82-636e896c622b</RequestId>
<PremiseNumber>555129300555129300</PremiseNumber>
<CustomerKey />
<AccountNumber>555129300</AccountNumber>
<MeterNumber>3750520P</MeterNumber>
<XfrmerName>6-7512-9407-0-4</XfrmerName>
<XfrmerId>13298132</XfrmerId>
<FdrName>F8863</FdrName>
<CompanyId>6E</CompanyId>
<ServiceDrop />
<ServicePointID>%SERVICEPOINTID</ServicePointID>
<XCoordinate>937254.20</XCoordinate>
<YCoordinate>810344.80</YCoordinate>
<CustomerRelocated>false</CustomerRelocated>
</CustomerCorrectionCiv>

```

4.4.7. Callback Request Message

4.4.7.1. Characteristics

Message Type	XML

Frequency	Initiated when Dispatcher decides to send callbacks for the customers.
Error Handling	None
Required Information	• Sequence • Account ID of the customer • Is it a case note related call back? • ETR • Last call time of the customer • Trouble code
Optional Information	• GUID • Reference ID: To uniquely identify the callback request. • Account ID • CustomerKey • Call back phone number • Callback name • Service center code • Company code • Case note message

4.4.7.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/CallbackRequestCiv" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/CallbackRequestCiv" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="CallbackRequestCiv" type="tns:CallbackRequestCivV1" />
<xs:complexType name="CallbackRequestCivV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<!--To uniquely identify the callback request. (This is in addition to the sequence number.-->
<xs:element minOccurs="0" maxOccurs="1" name="CallbackRequestId" type="xs:string" />
<!--List of automated callback requests. (See additional information in the Supplemental Information
Section.)-->
<xs:element minOccurs="0" maxOccurs="1" name="CallbackRequests" type="tns:ArrayOfCallbackCivV1" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
```

```

<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfCallbackCivV1">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="CallbackCivV1" nillable="true"
type="tns:CallbackCivV1" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="CallbackCivV1">
<!--REFERENCE_NUM , Char 10-->
<xs:attribute name="ReferenceId" type="xs:string" />
<!--ACCT_ID, Char 10-->
<xs:attribute name="AccountId" type="xs:string" />
<xs:attribute name="CustomerKey" type="xs:string" />
<!--SVCNTR_CODE, CHAR 3-->
<xs:attribute name="OperatingCenterCode" type="xs:string" />
<!--COMPANY_CODE, CHAR 3-->
<xs:attribute name="CompanyCode" type="xs:string" />
<!--Ten digit phone number-->
<xs:attribute name="CallbackPhoneNumber" type="xs:string" />
<!--This will be true if the Call Back request is due to a
change in case note.-->
<xs:attribute name="CallbackName" type="xs:string" />
<!--Customer name to call-->
<xs:attribute name="IsCaseNoteChange" type="xs:boolean" use="required" />
<!--The Case Note Message.-->
<xs:attribute name="CaseNoteMessage" type="xs:string" />
<!--New estimated restoration time of the outage.-->
<xs:attribute name="ETR" type="xs:dateTime" use="required" />
<xs:attribute name="LastCallTime" type="xs:dateTime" use="required" />
<xs:attribute name="TroubleCode" type="xs:string" />
</xs:complexType>
</xs:schema>

```

4.4.7.3. Sample XML Message

```

<?xml version="1.0"?>
<CallbackRequestCiv xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/CallbackRequestCiv">
<Guid>1009a857-2e46-41af-924d-f4a033b8c2a2</Guid>
<Sequence>0</Sequence>
<CallbackRequestId>4b982c39-b3e1-4975-8406-65118859aeef</CallbackRequestId>
<CallbackRequests>
<CallbackCivV1 ReferenceId="" AccountId="555133731" OperatingCenterCode="" CompanyCode=""
CallbackPhoneNumber="5556345" IsCaseNoteChange="false" CaseNoteMessage="" ETR="0001-01-01T00:00:00"
LastCallTime="2012-03-07T15:10:21" />
</CallbackRequests>
</CallbackRequestCiv>

```

4.4.8. Case Note Query

4.4.8.1. Characteristics

Message Type	XML
Frequency	Initiated by channel adapter.
Error Handling	None
Required Information	• FromTimeStamp: Requests for any changes since this time (need to specify type). • QueryType: 1 = global (internal use) 2 = service center 3 = customer 4 = service point • Message base sequence
Optional Information	• CaseNoteQueryId • Name of service center or customer account number, if not a global query. • Routing • Message base GUID

4.4.8.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/CaseNoteQueryCiv" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/CaseNoteQueryCiv" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="CaseNoteQueryCiv" type="tns:CaseNoteQueryCivV1" />
<xs:complexType name="CaseNoteQueryCivV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<!--Requests is for all calls after this timestamp.-->
<xs:element minOccurs="1" maxOccurs="1" name="FromTimeStamp" type="xs:dateTime" />
<!--his uniquely identifies request-->
<xs:element minOccurs="0" maxOccurs="1" name="CaseNoteQueryId" type="xs:string" />
<!--Type of the query (global (for internal use only), service center, customer).
Follows numeric value of eCivQueryType.
public enum eCivQueryType
{
Global = 1,
OperatingCenter = 2,
ServicePoint = 3,
Customer = 4
```

```

    } -->
<xs:element minOccurs="1" maxOccurs="1" name="QueryType" type="xs:int" />
<!--Name of service center or customer, if not global query.-->
<xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="Routing" type="xs:string" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
</xs:schema>

```

4.4.8.3. eCivQueryType Schema Definition

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="eCivQueryType" type="eCivQueryType" />
<xs:simpleType name="eCivQueryType">
<xs:restriction base="xs:string">
<xs:enumeration value="Global" />
<xs:enumeration value="OperatingCenter" />
<xs:enumeration value="ServicePoint" />
<xs:enumeration value="Customer" />
</xs:restriction>
</xs:simpleType>
</xs:schema>

```

4.4.8.4. Sample XML Message

```

<?xml version="1.0"?>
<CaseNoteQueryCiv xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/CaseNoteQueryCiv">
<Sequence>59181</Sequence>
<Guid>d57bc994-e8eb-4f8e-bf58-c39ca7f5d498</Guid>
<FromTimeStamp>2017-11-24T00:44:16.2689299-08:00</FromTimeStamp>
<CaseNoteQueryId>db633561-8f23-472e-a377-d6031f7e5c98</CaseNoteQueryId>
<QueryType>2</QueryType>
<Name />
<Routing />

```

</CaseNoteQueryCiv>

4.4.9. Case Note Response

4.4.9.1. Characteristics

Message Type	XML
Frequency	As a response to Case Note Query.
Error Handling	None
Required Information	• UpToTimeStamp: Time stamp of the last case note in the system. This will be the From time stamp for the next query. • Message base sequence • ETR: Estimated time for restoration. For the first trouble call (when the customer is not part of any incident), the Initial estimated Time for Restoration (ITR) from the ETR matrix file is provided to the client (customer or Call Entry tool) based on the customer's city, service center, substation, and feeder. If the ETR matrix file is not available to the Call server, the default ITR (2.5 hours) is provided. • Has the case note been deleted? • Has the case note been updated? • Last Time Case note was updated • Case Note Number • System Code • Cause Code • Is Accessible • Suppress Notification • Suppress Reason • Start Time • ITR Disable • Use parent area setting? • Use Feeder Restoration Priority (FRP)? • Restoration Urgency

Optional Information

- QueryIDEcho: This is an echo of the original case note request.
 - Case notes
 - Routing echo
 - Message base GUID
 - Service center of the case note
 - Station
 - Feeder ID
 - Device ID
 - Device name
 - Area ID
 - Case note name
 - Trouble assigned to
 - Remarks
 - Who updated it last time
 - Case note message
 - Group type
 - Parent ID
 - Storm name
-

If the ADMS user (with the appropriate permissions) suppresses individual case note in a particular area, the case note query response will contain no results.

4.4.9.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/CaseNoteResponseCiv" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/CaseNoteResponseCiv" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CaseNoteResponseCiv" type="tns:CaseNoteResponseCivV1" />
  <xs:complexType name="CaseNoteResponseCivV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:MessageBaseV1">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="QueryIdEcho" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="UpToTimeStamp" type="xs:dateTime" />
          <xs:element minOccurs="0" maxOccurs="1" name="CaseNotes" type="tns:ArrayOfCaseNoteCivV1" />
          <xs:element minOccurs="0" maxOccurs="1" name="RoutingEcho" type="xs:string" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:NetMessageBase" />
    </xs:complexContent>
  </xs:complexType>
```

```

<xs:complexType name="NetMessageBase">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfCaseNoteCivV1">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="CaseNoteCivV1" nillable="true"
type="tns:CaseNoteCivV1" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="CaseNoteCivV1">
  <xs:attribute name="OpCenterCode" type="xs:string" />
  <xs:attribute name="Station" type="xs:string" />
  <xs:attribute name="FeederId" type="xs:string" />
  <xs:attribute name="DeviceId" type="xs:string" />
  <xs:attribute name="DeviceName" type="xs:string" />
  <xs:attribute name="AreaId" type="xs:string" />
  <xs:attribute name="Name" type="xs:string" />
  <xs:attribute name="TroubleAssignedTo" type="xs:string" />
  <xs:attribute name="ETR" type="xs:dateTime" use="required" />
  <xs:attribute name="IsDeleted" type="xs:boolean" use="required" />
  <xs:attribute name="IsUpdated" type="xs:boolean" use="required" />
  <xs:attribute name="OverrideCaseNote" type="xs:boolean" use="required" />
  <xs:attribute name="LastUpdateTime" type="xs:dateTime" use="required" />
  <xs:attribute name="Remarks" type="xs:string" />
  <xs:attribute name="LastUpdateByUserId" type="xs:string" />
  <xs:attribute name="CaseNoteNumber" type="xs:int" use="required" />
  <xs:attribute name="CaseNoteMessage" type="xs:string" />
  <xs:attribute name="SystemCode" type="xs:int" use="required" />
  <xs:attribute name="CauseCode" type="xs:int" use="required" />
  <xs:attribute name="IsAccessible" type="xs:boolean" use="required" />
  <xs:attribute name="GroupType" type="xs:string" />
  <xs:attribute name="ParentId" type="xs:string" />
  <xs:attribute name="SuppressNotification" type="xs:boolean" use="required" />
  <xs:attribute name="SuppressReason" type="xs:int" use="required" />
  <xs:attribute name="SystemMode" type="xs:int" use="required" />
  <xs:attribute name="StormName" type="xs:string" />
  <xs:attribute name="StartTime" type="xs:dateTime" use="required" />
  <xs:attribute name="ITRDisable" type="xs:boolean" use="required" />
  <xs:attribute name="UseParent" type="xs:boolean" use="required" />
  <xs:attribute name="UseFRP" type="xs:boolean" use="required" />
  <xs:attribute name="RestorationUrgency" type="xs:boolean" use="required" />
</xs:complexType>
</xs:schema>

```

4.4.9.3. Sample XML Message

```

<CaseNoteResponseCiv xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/CaseNoteResponseCiv">

```

```

<Sequence>23807</Sequence>
<Guid>a648ab6e-caaa-45e4-96d5-c035b325013d</Guid>
<QueryIdEcho>63a7ca61-69d5-432a-95b9-a01efa39ace9</QueryIdEcho>
<UpToTimeStamp>2019-02-06T05:43:06.7570605-08:00</UpToTimeStamp>
<CaseNotes>
  <CaseNoteCivV1 OpCenterCode="WINGED FOOT F7862" Station="Winged Foot" FeederId="" DeviceId=""
DeviceName="" AreaId="" Name="" TroubleAssignedTo="" ETR="0001-01-01T00:00:00" IsDeleted="true"
IsUpdated="false" OverrideCaseNote="false" LastUpdateTime="0001-01-01T00:00:00" Remarks=""
LastUpdateByUserId="" CaseNoteNumber="0" CaseNoteMessage="No Area Note Assigned." SystemCode="0"
CauseCode="0" IsAccessible="false" GroupType="FEEDER" ParentId="WINGED FOOT"
SuppressNotification="false" SuppressReason="0" StormName="" StartTime="0001-01-01T00:00:00"
ITRDisable="false" />
</CaseNotes>
<RoutingEcho />
</CaseNoteResponseCiv>

```

[1] The maximum integer value is stored in the Call Server configuration and is used for validation of trouble call messages.

5. Advanced Metering Infrastructure Interfaces

! Important

It is recommended to use the Call Server HTTP interface instead of the legacy TCP/IP interface for communicating with external systems.

5.1. Overview

The Advanced Metering Infrastructure (AMI) (and in particular, Automated Meter Reading (AMR) system) can provide information on whether individual customers are impacted by an outage. This information is used by ADMS in addition to phone calls and other sources to determine the existence and extent of outages.

The ADMS system interfaces with AMI through the Channel Adapter and the Enterprise Service Bus.

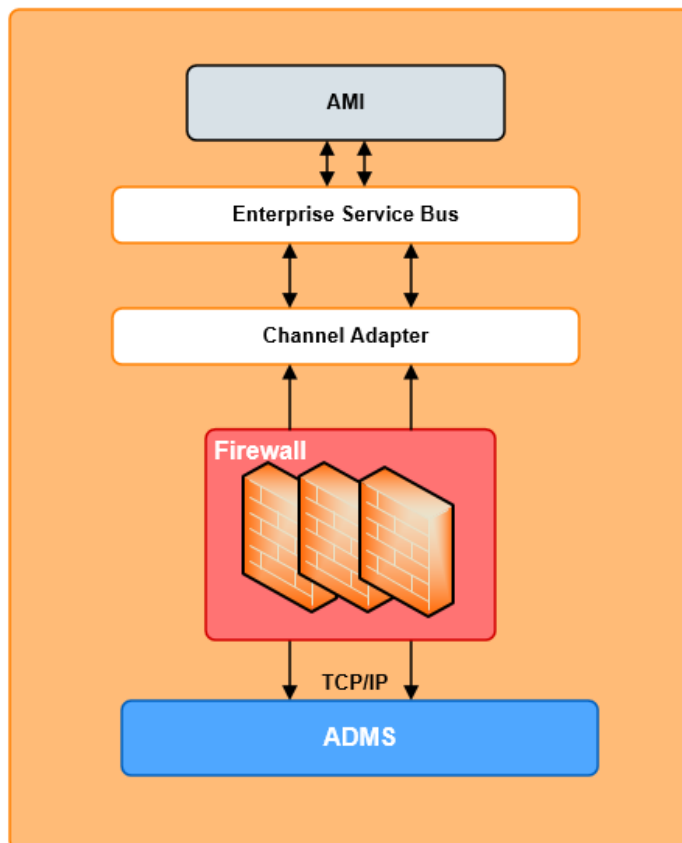


Figure 24. AMI Interface

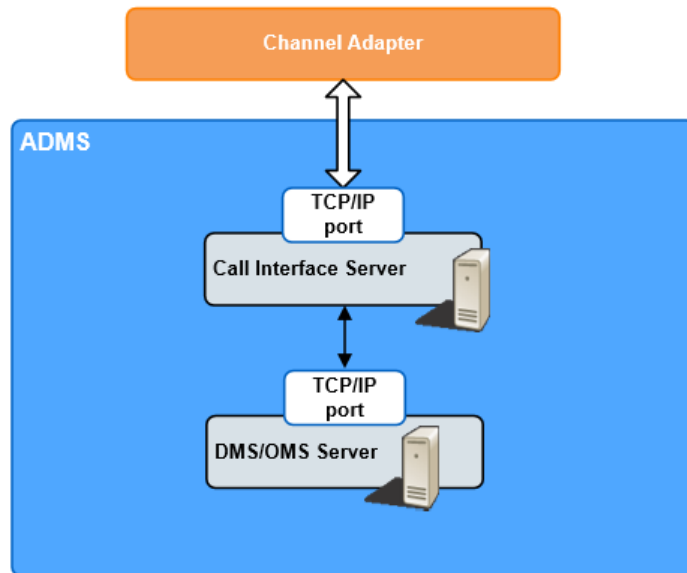


Figure 25. ADMS Components

5.2. Interface Characteristics

The characteristics of the interface are the following:

Interface Type	XML Messaging
Protocol	TCP/IP

5.3. Message Flow

The interface to the Call server covers these functions relating to AMI:

- Meter Event: The AMI system can report power outage and power restoration messages. These meter outage reports are similar to outage calls except that they refer to a meter rather than a customer account, and there are no comments associated with the message since they are automatically generated.
- Meter Pings: The ADMS system can ping a meter for its power state.
- Meter Voltage Event/Voltage Pings: Used to get voltage and brownout messages from a meter.

5.3.1. General Message Flow

The General message flow is the same as the interfaces with CIS.

5.3.2. AMI Message Flow

The ADMS system will query AMI for meter statuses (power out or power on). Technically, there are not separate power-on and power-off messages; they are derived from the AMI outage report. AMI will return any last gasp outage messages it has received since the last query, and any new power on messages it has received.

It is assumed that meters are on unless reported otherwise, that is, it is not necessary for AMI to report that all customers are on initially. It is possible for an analyst to request (through the Call server) that the link be reinitialized. In this case, all current AMI messages should be reported.

The ADMS system will query for the outage status of meters on an individual basis (a "ping") in some circumstances. These may need to be adjusted based on the throughput capability of the infrastructure. Meters will be pinged for these cases initially:

- To confirm an individual outage (the customer calls in, but the ping will verify if it is a customer problem or a real outage).
- To detect whether a single outage is really a transformer outage (only one meter reported an outage, but there are two other meters on the transformer).

Additional pings may be added to the system, to better define the outage extent, and in lieu of callbacks.

In the system initialization event, the adapter can specify whether it supports AMI or not, and also whether it supports meter pinging or not.

Some AMI systems that use pilot wire technology require that the OMS reports changes in connectivity (it is required that AMI communicate through the proper feeder and on the proper phase). This capability is not being implemented at this time.

The AMI message flow is illustrated in the below image.

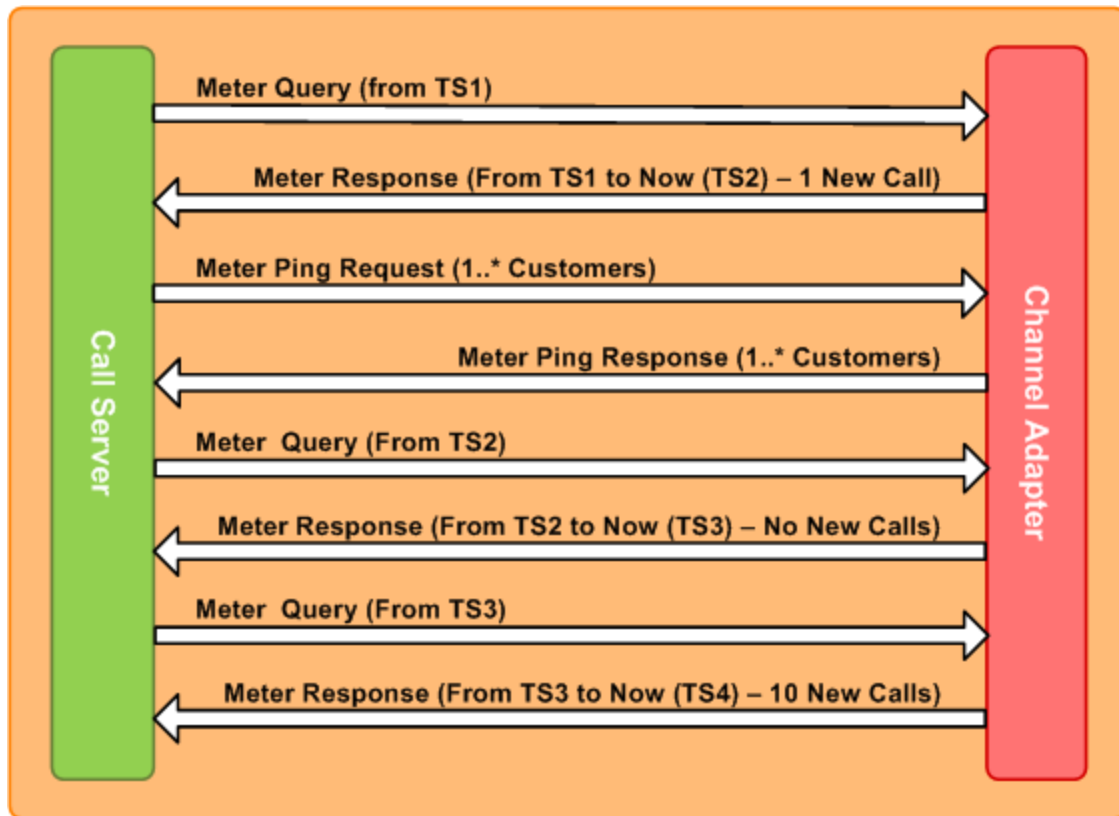


Figure 26. AMI Message Flow

5.4. Message Details

5.4.1. List of Messages

The list of supported messages is as follows:

- Hello
- Hello Response
- Positive/Negative Acknowledgement
- Meter Event Query
- Meter Event Response
- Meter Ping Request
- Meter Ping Response
- Meter Voltage Event Query
- Meter Voltage Event Response
- Meter Voltage Ping Request

- Meter Voltage Ping Response

The following table describes the flow of the messages between AMI and ADMS:

Table 2. Message Flow Between AMI and ADMS

Message	From AMI	To AMI
Hello	X	X
Positive/Negative Acknowledgement	X	X
Meter Event Query		X
Meter Event Response	X	
Meter Ping Request		X
Meter Ping Response	X	
Meter Voltage Event Query		X
Meter Voltage Event Response	X	
Meter Voltage Ping Request		X
Meter Voltage Ping Response	X	

5.4.2. Hello

Characteristics, schema definition, and sample messages for the "Hello" message are provided in [Hello](#).

5.4.3. Hello Response

Characteristics, schema definition, and sample messages for the "Hello Response" message are provided in [Hello Response](#).

5.4.4. Positive/Negative Acknowledgement

This has been described in [Positive/Negative Acknowledgement Messages](#).

5.4.5. Meter Event Query

5.4.5.1. Characteristics

Message Type	XML
Frequency	Configurable time period
Error Handling	

Required Information	• FromTimeStamp • Message base sequence
Optional Information	• Message base GUID

5.4.5.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/MeterEventQueryAmi" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/MeterEventQueryAmi" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="MeterEventQueryAmi" type="tns:MeterEventQueryAmiV1" />
<xs:complexType name="MeterEventQueryAmiV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<!--Requests is for all calls after this timestamp.-->
<xs:element minOccurs="1" maxOccurs="1" name="FromTimeStamp" type="xs:dateTime" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
```

5.4.5.3. Sample XML Message

```
<?xml version="1.0"?>
<MeterEventQueryAmi xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/MeterEventQueryAmi">
<Sequence>59167</Sequence>
<Guid>8c28cdbf-34e9-477d-975a-3a20528f46a4</Guid>
<FromTimeStamp>0001-01-01T00:00:00</FromTimeStamp>
</MeterEventQueryAmi>
```

5.4.6. Meter Event Response

5.4.6.1. Characteristics

Message Type	XML
Frequency	As a response to MeterEventQuery request.
Error Handling	NACK message will be sent to the sender.
Required Information	• UpToTimeStamp • Message base sequence • The meter status is Out? • The meter status is In? • Trouble code • Time reported • Query time stamp
Optional Information	• Message base Guid • MeterId • Meter responses • Trouble Code

5.4.6.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/MeterEventResponseAmi" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/MeterEventResponseAmi"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="MeterEventResponseAmi" type="tns:MeterEventResponseAmiV1" />
<xs:complexType name="MeterEventResponseAmiV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="1" name="MeterResponses" type="tns:ArrayOfMeterResponseAmiV1" />
<xs:element minOccurs="1" maxOccurs="1" name="UpToTimeStamp" type="xs:dateTime" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
```

```

</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfMeterResponseAmiV1">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="MeterResponseAmiV1" nillable="true"
type="tns:MeterResponseAmiV1" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="MeterResponseAmiV1">
<!--Meter number.-->
<xs:attribute name="MeterId" type="xs:string" />
<!--Is Meter status out?-->
<xs:attribute name="IsOut" type="xs:boolean" use="required" />
<!-- Is meter status restored?-->
<xs:attribute name="IsIn" type="xs:boolean" use="required" />
<xs:attribute name="TroubleCode" type="xs:string" />
<!--TimeReported-->
<xs:attribute name="TimeReported" type="xs:dateTime" use="required" />
<!--Time stamp in channel adapter database.-->
<xs:attribute name="QueryTimeStamp" type="xs:dateTime" use="required" />
</xs:complexType>
</xs:schema>

```

5.4.6.3. Sample XML Message

```

<?xml version="1.0"?>
<MeterEventResponseAmi xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://www.esca.com/MeterEventResponseAmi">
<Sequence>59165</Sequence>
<Guid>bb13c272-edc0-4fef-bcbc-152d47017a85</Guid>
<MeterResponses />
<UpToTimeStamp>0001-01-01T00:00:00</UpToTimeStamp>
</MeterEventResponseAmi>

```

5.4.7. Meter Ping Request

MeterPing request sent by ADMS with the list of meter IDs and the configurable timeout in minutes, the response coming back will have the possible status of:

- On
- Undetermined
- NotTried
- Communication error
- Invalid Meter

The ADMS system will retry the ping request for the list of meters coming back with the response of

NotTried. The number of retries will be configurable in ADMS.

5.4.7.1. Characteristics

Message Type	XML
Frequency	On Operator's request
Error Handling	NACK message is expected
Required Information	• TimeOut • Message base sequence
Optional Information	• Ping Request ID • CompanyId • Meter IDs • Routing • Message base Guid • Meter ID type • Value

5.4.7.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/MeterPingRequestAmi" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/MeterPingRequestAmi" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="MeterPingRequestAmi" type="tns:MeterPingRequestAmiV1" />
<xs:complexType name="MeterPingRequestAmiV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<!--To uniquely identify the meter ping request. (Might include the transformer number.)-->
<xs:element minOccurs="0" maxOccurs="1" name="PingRequestId" type="xs:string" />
<!-- Meter ping after TimeOut minutes and
the response of the result will be sent.
Result might come in the form of in, out, not tried
in case of not tried meter ids, ADMS will send ping
query again.-->
<xs:element minOccurs="1" maxOccurs="1" name="TimeOut" type="xs:int" />
<!--CompanyId that the first meter belongs to -->
<xs:element minOccurs="0" maxOccurs="1" name="CompanyId" type="xs:string" />
<!--List of meter numbers to ping.-->
<xs:element minOccurs="0" maxOccurs="1" name="MeterIds" type="tns:ArrayOfMeterIdType" />
<!--Internal use only-->
<xs:element minOccurs="0" maxOccurs="1" name="Routing" type="xs:string" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
```

```

<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfMeterIdType">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="MeterIdType" type="tns:MeterIdType" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="MeterIdType">
<xs:attribute name="value" type="xs:string" />
</xs:complexType>
</xs:schema>

```

5.4.7.3. Sample XML Message

```

<?xml version="1.0"?>
<MeterPingRequestAmi xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/MeterPingRequestAmi">
<Sequence>59182</Sequence>
<Guid>0b7e7ecd-156f-4d9d-b181-1c393af32f3a</Guid>
<PingRequestId>de4481d4-c4b1-4022-a42c-f03fbb86dd0</PingRequestId>
<TimeOut>180</TimeOut>
<CompanyId>COMPANY</CompanyId>
<MeterIds>
<MeterIdType value="395864UI" />
</MeterIds>
<Routing />
</MeterPingRequestAmi>

```

5.4.8. Meter Ping Response

5.4.8.1. Characteristics

Message Type	XML
Frequency	After the TimeOut delay requested in MeterPingRequest
Error Handling	NACK message will be sent

Required Information	• Message base sequence • Status • Last communication time • VOLT A • VOLT B • VOLT C
Optional Information	• Ping request ID • Meter statuses • Routing echo • Message base Guid • Meter ID

5.4.8.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/MeterPingResponseAmi" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/MeterPingResponseAmi"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="MeterPingResponseAmi" type="tns:MeterPingResponseAmiV1" />
<xs:complexType name="MeterPingResponseAmiV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<!--RequestID copied from the ping request-->
<xs:element minOccurs="0" maxOccurs="1" name="PingRequestId" type="xs:string" />
<!--List of meter statuses-->
<xs:element minOccurs="0" maxOccurs="1" name="MeterStatuses" type="tns:ArrayOfMeterStatusAmiV1" />
<!--Internal use only-->
<xs:element minOccurs="0" maxOccurs="1" name="RoutingEcho" type="xs:string" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfMeterStatusAmiV1">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="MeterStatusAmiV1" nillable="true"
type="tns:MeterStatusAmiV1" />

```

```

</xs:sequence>
</xs:complexType>
<xs:complexType name="MeterStatusAmiV1">
<!--Meter number-->
<xs:attribute name="MeterId" type="xs:string" />
<!--Result of the ping status. Follows numeric value of enum public enum eAmiQueryResult
{
On = 1,
Undetermined = 2,
NotTried = 3,
CommunicationError = 4,
InvalidMeter = 5
} -->
<xs:attribute name="Status" type="xs:int" use="required" />
<!--Time of last communication.-->
<xs:attribute name="LastCommunicationTime" type="xs:dateTime" use="required" />
<xs:attribute name="VoltA" type="xs:float" use="required" />
<xs:attribute name="VoltB" type="xs:float" use="required" />
<xs:attribute name="VoltC" type="xs:float" use="required" />
</xs:complexType>
</xs:schema>

```

5.4.8.3. Enum eAmiQueryResult Schema Definition

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="eAmiQueryResult" type="eAmiQueryResult" />
<xs:simpleType name="eAmiQueryResult">
<xs:restriction base="xs:string">
<xs:enumeration value="On" />
<xs:enumeration value="Undetermined" />
<xs:enumeration value="NotTried" />
<xs:enumeration value="CommunicationError" />
<xs:enumeration value="InvalidMeter" /> <-Not in USE
<xs:enumeration value="Out" />
<xs:enumeration value="Expired" />
</xs:restriction>
</xs:simpleType>
</xs:schema>

```

5.4.8.4. Sample XML Message

```

<?xml version="1.0"?>
<MeterPingResponseAmi xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://www.esca.com/MeterPingResponseAmi">
<Sequence>59182</Sequence>
<Guid>6b8da4f2-ea0d-48da-81ff-e603870e2992</Guid>
<PingRequestId>de4481d4-c4b1-4022-a42c-f03fbdb86dd0</PingRequestId>

```

```

<MeterStatuses>
<MeterStatusAmiV1 MeterId="395864UI" Status="1" LastCommunicationTime="2017-11-24T00:45:17.7340427-
08:00" VoltA="118.6885" VoltB="NaN" VoltC="NaN" />
</MeterStatuses>
<RoutingEcho />
</MeterPingResponseAmi>

```

5.4.9. Meter Voltage Event Query (DROP)

This query will be used to get the brown out message for DROP (Distribution Regulation Options program).

5.4.9.1. Characteristics

Message Type	XML
Frequency	Configurable time period
Error Handling	
Required Information	• Message base sequence • From Time Stamp
Optional Information	• Message base GUID

5.4.9.2. Schema Definition

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/MeterVoltEventQueryAmi" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/MeterVoltEventQueryAmi"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="MeterVoltEventQueryAmi" type="tns:MeterVoltEventQueryAmiV1" />
<xs:complexType name="MeterVoltEventQueryAmiV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="FromTimeStamp" type="xs:dateTime" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />

```

```
</xs:sequence>
</xs:complexType>
</xs:schema>
```

5.4.9.3. Sample XML Message

```
<?xml version="1.0"?>
<MeterVoltEventQueryAmi xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://www.esca.com/MeterVoltEventQueryAmi">
<Sequence>59174</Sequence>
<Guid>e9757abe-61b0-467b-a0c2-75a290b0a3f1</Guid>
<FromTimeStamp>0001-01-01T00:00:00</FromTimeStamp>
</MeterVoltEventQueryAmi>
```

5.4.10. Meter Voltage Event Response

This is in response to the query for meter voltage event. The ADMS system expects that the brown-out message and meter-out message will be mutually exclusive. That is, ADMS will not receive a meter-out message when a brown-out message is received.

5.4.10.1. Characteristics

Message Type	XML
Frequency	As a response to Meter Voltage Event Query request.
Error Handling	NACK message will be sent to the sender.

Required Information	• UpToTimeStamp • Message base sequence • Event type: Nominal, Below Threshold, or Above Threshold. • Volt A • Volt B • Volt C • Amp A • Amp B • Amp C • Watt A • Watt B • Watt C • VAR A • VAR B • VAR C • Time reported • Query time stamp
Optional Information	• MeterResponses • Meter ID • Message base GUID

5.4.10.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/MeterVoltEventResponseAmi" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/MeterVoltEventResponseAmi"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="MeterVoltEventResponseAmi" type="tns:MeterVoltEventResponseAmiV1" />
<xs:complexType name="MeterVoltEventResponseAmiV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="1" name="MeterResponses" type="tns:ArrayOfMeterVoltResponseAmi"
/>
<xs:element minOccurs="1" maxOccurs="1" name="UpToTimeStamp" type="xs:dateTime" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
```

```

<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfMeterVoltResponseAmi">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="MeterVoltResponseAmi" nillable="true"
type="tns:MeterVoltResponseAmi" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="MeterVoltResponseAmi">
<!--Meter number.-->
<xs:attribute name="MeterId" type="xs:string" />
<xs:attribute name="EventType" type="xs:int" use="required" />
<xs:attribute name="VoltA" type="xs:float" use="required" />
<xs:attribute name="VoltB" type="xs:float" use="required" />
<xs:attribute name="VoltC" type="xs:float" use="required" />
<xs:attribute name="AmpA" type="xs:float" use="required" />
<xs:attribute name="AmpB" type="xs:float" use="required" />
<xs:attribute name="AmpC" type="xs:float" use="required" />
<xs:attribute name="WattA" type="xs:float" use="required" />
<xs:attribute name="WattB" type="xs:float" use="required" />
<xs:attribute name="WattC" type="xs:float" use="required" />
<xs:attribute name="VarA" type="xs:float" use="required" />
<xs:attribute name="VarB" type="xs:float" use="required" />
<xs:attribute name="VarC" type="xs:float" use="required" />
<xs:attribute name="TimeReported" type="xs:dateTime" use="required" />
<xs:attribute name="QueryTimeStamp" type="xs:dateTime" use="required" />
</xs:complexType>
</xs:schema>

```

5.4.10.3. Sample XML Message

```

<?xml version="1.0"?>
<MeterVoltEventResponseAmi xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xmlns="http://www.esca.com/MeterVoltEventResponseAmi">
<Sequence>59163</Sequence>
<Guid>001c3336-f468-453a-b7ca-61a432ba26b3</Guid>
<MeterResponses />
<UpToTimeStamp>0001-01-01T00:00:00</UpToTimeStamp>
</MeterVoltEventResponseAmi>

```

5.4.11. Meter Voltage Ping Request

MeterVoltagePing request sent by ADMS with the list of meter IDs and the configurable timeout in minutes, the response coming back will have the possible status of:

- Good
- NotTried
- Communication error
- Invalid Meter

The ADMS system will retry the ping request for the list of meters coming back with the response of NotTried. The number of retries will be configurable in ADMS.

5.4.11.1. Characteristics

Message Type	XML
Frequency	On Operator's request
Error Handling	NACK message is expected
Required Information	• TimeOut • Message base sequence
Optional Information	• Ping request ID • Company ID • Meter IDs • Routing • Message base Guid • Meter ID Type • Value

5.4.11.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/MeterVoltPingReqAmi" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/MeterVoltPingReqAmi" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="MeterVoltPingReqAmi" type="tns:MeterVoltPingReqAmiV1" />
<xs:complexType name="MeterVoltPingReqAmiV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:MessageBaseV1">
<xs:sequence>
<!--To uniquely identify the meter ping request.-->
<xs:element minOccurs="0" maxOccurs="1" name="PingRequestId" type="xs:string" />
<!--The company ID of the company for which the ping request will be sent. This will be extracted from
the customer file-->
<xs:element minOccurs="0" maxOccurs="1" name="CompanyId" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="TimeOut" type="xs:int" />
<!--List of meter numbers to ping.-->
<xs:element minOccurs="0" maxOccurs="1" name="MeterIds" type="tns:ArrayOfMeterIdType" />
<xs:element minOccurs="0" maxOccurs="1" name="Routing" type="xs:string" />
</xs:sequence>
</xs:extension>
```

```

</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
<xs:complexContent mixed="false">
<xs:extension base="tns:NetMessageBase" />
</xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfMeterIdType">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="MeterIdType" type="tns:MeterIdType" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="MeterIdType">
<xs:attribute name="value" type="xs:string" />
</xs:complexType>
</xs:schema>

```

5.4.11.3. Sample XML Message

```

<?xml version="1.0"?>
<MeterVoltPingReqAmi xmlns:xsd="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/MeterVoltPingReqAmi">
<Sequence>59693</Sequence>
<Guid>ba3b6ce9-ff9b-4804-8e7b-9429faf6ddf7</Guid>
<PingRequestId />
<CompanyId />
<TimeOut>0</TimeOut>
<MeterIds>
<MeterIdType value="259265MG" />
</MeterIds>
<Routing />
</MeterVoltPingReqAmi>

```

5.4.12. Meter Voltage Ping Response

5.4.12.1. Characteristics

Message Type	XML
Frequency	After the TimeOut delay requested in MeterVoltPingRequest
Error Handling	NACK message will be sent

Required Information	• Result • Message base sequence • Phase A voltage • Phase B voltage • Phase C voltage • Phase A VAR • Phase B VAR • Phase C VAR • Phase A Watt • Phase B Watt • Phase C Watt • Phase A Amp • Phase B Amp • Phase C Amp • Last Communication Time
Optional Information	• Ping Request ID • Meter Volt Stats • Routing Echo • Message base GUID • Meter ID

5.4.12.2. Schema Definition

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/MeterVoltPingResAmi" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/MeterVoltPingResAmi" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="MeterVoltPingResAmi" type="tns:MeterVoltPingResAmiV1" />
  <xs:complexType name="MeterVoltPingResAmiV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:MessageBaseV1">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="PingRequestId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="MeterVoltStats"
type="tns:ArrayOfMeterVoltStatusAmi" />
          <xs:element minOccurs="1" maxOccurs="1" name="Result" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="RoutingEcho" type="xs:string" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:NetMessageBase" />
    </xs:complexContent>
  </xs:complexType>
```

```

<xs:complexType name="NetMessageBase">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfMeterVoltStatusAmi">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="MeterVoltStatusAmi" nillable="true"
type="tns:MeterVoltStatusAmi" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="MeterVoltStatusAmi">
  <xs:attribute name="MeterId" type="xs:string" />
  <xs:attribute name="VoltA" type="xs:float" use="required" />
  <xs:attribute name="VoltB" type="xs:float" use="required" />
  <xs:attribute name="VoltC" type="xs:float" use="required" />
  <xs:attribute name="VarA" type="xs:float" use="required" />
  <xs:attribute name="VarB" type="xs:float" use="required" />
  <xs:attribute name="VarC" type="xs:float" use="required" />
  <xs:attribute name="WattA" type="xs:float" use="required" />
  <xs:attribute name="WattB" type="xs:float" use="required" />
  <xs:attribute name="WattC" type="xs:float" use="required" />
  <xs:attribute name="AmpA" type="xs:float" use="required" />
  <xs:attribute name="AmpB" type="xs:float" use="required" />
  <xs:attribute name="AmpC" type="xs:float" use="required" />
  <xs:attribute name="LastCommunicationTime" type="xs:dateTime" use="required" />
</xs:complexType>
</xs:schema>

```

5.4.12.3. Enum eVoltAmiQueryResult Schema Definition

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:element name="eVoltAmiQueryResult" type="eVoltAmiQueryResult" />
<xs:simpleType name="eVoltAmiQueryResult">
<xs:restriction base="xs:string">
<xs:enumeration value="Good" />
<xs:enumeration value="NotTried" />
<xs:enumeration value="CommunicationError" />
<xs:enumeration value="InvalidMeter" /><-Not in USE
</xs:restriction>
</xs:simpleType>
</xs:schema>

```

5.4.12.4. Sample XML Message

```

<?xml version="1.0"?>
<MeterVoltPingResAmi xmlns:xsd="http://www.w3.org/2001/XMLSchema"

```

```

xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance" xmlns="http://www.esca.com/MeterVoltPingResAmi">
<Sequence>59693</Sequence>
<Guid>63566b3a-14fe-4172-8401-62388a35db4d</Guid>
<PingRequestId />
<MeterVoltStats>
<MeterVoltStatusAmi MeterId="259265MG" VoltA="NaN" VoltB="NaN" VoltC="116.8963" VarA="NaN" VarB="NaN"
VarC="0.97519" WattA="NaN" WattB="NaN" WattC="1.855156" LastCommunicationTime="2017-11-
24T01:08:24.9575405-08:00" />
</MeterVoltStats>
<Result>0</Result>
<RoutingEcho />
</MeterVoltPingResAmi>

```

5.4.13. Meter Disconnect Request

5.4.13.1. Characteristics

Message Type	XML
Frequency	On operator's request
Error Handling	NACK message is expected
Required Information	• Message base sequence
Optional Information	• RequestId • MeterId • Routing • Message base GUID

5.4.13.2. Schema Definition

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/MeterDisconnectRequestAmi" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/MeterDisconnectRequestAmi"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="MeterDisconnectRequestAmi" type="tns:MeterDisconnectRequestAmiV1" />
  <xs:complexType name="MeterDisconnectRequestAmiV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:AmiOperationRequestV1" />
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="AmiOperationRequestV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:MessageBaseV1">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="RequestId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="MeterId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Routing" type="xs:string" />

```

```

        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:NetMessageBase" />
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="NetMessageBase">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

5.4.14. Meter Reconnect Request

5.4.14.1. Characteristics

Message Type	XML
Frequency	On operator's request
Error Handling	NACK message is expected
Required Information	• Message base sequence
Optional Information	• RequestId • MeterId • Routing • Message base GUID

5.4.14.2. Schema Definition

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema xmlns:tns="http://www.esca.com/MeterReconnectRequestAmi" elementFormDefault="qualified"
targetNamespace="http://www.esca.com/MeterReconnectRequestAmi"
xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="MeterReconnectRequestAmi" type="tns:MeterReconnectRequestAmiV1" />
  <xs:complexType name="MeterReconnectRequestAmiV1">
    <xs:complexContent mixed="false">
      <xs:extension base="tns:AmiOperationRequestV1">
        <xs:sequence>
          <xs:element minOccurs="1" maxOccurs="1" name="RequiresUserIntervention" type="xs:boolean" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>

```

```

</xs:complexType>
<xs:complexType name="AmiOperationRequestV1">
  <xs:complexContent mixed="false">
    <xs:extension base="tns:MessageBaseV1">
      <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="RequestId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="MeterId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Routing" type="xs:string" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV1">
  <xs:complexContent mixed="false">
    <xs:extension base="tns:NetMessageBase" />
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="NetMessageBase">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="Guid" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

6. Crew Server Web Service Version 1

Important

The Crew Server web service version 1 is deprecated. It is recommended that you use the Crew Server web interface version 2 (described in [Crew Server Web Interface](#)) for communicating with external systems, such as Workforce Management System (WFM).

Crew Server web service version 1 can post notifications, send data to external interfaces, allow external applications to update incident damage assessment location, crew status, and incident status, and also update vehicle locations.

To enable Crew Server web service version 1, configure `<EnableCrewWebServices>` and `<WebServiceBaseUrl>` in the `CrewServerConfig.xml` file.

The Crew Server web service version 1 base URI is: `http://<YourServerIP>:<Port>/crew` (For example, `http://localhost:8802/crew`). The service name is `CrewServerService`.

6.1. Service Types

The following tables list details about the service calls that are exposed at the root level. For more details, refer to `http://localhost:8802/crew/services/help`.

6.1.1. AssessmentStatus

Request to update the damage assessment location.

Message Type	HTTP POST
URI	<code>http://localhost:8802/crew/services/AssessmentStatus</code>
Source	WFM
Body	JSON/XML
Type	<code>AssessmentStatus</code>
Frequency	On demand

XML request schema:

```
<xs:schema xmlns:tns="http://com.alstom.dms.rest" elementFormDefault="qualified"
targetNamespace="http://com.alstom.dms.rest" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="AssessmentStatus">
    <xs:sequence>
      <xs:element minOccurs="0" name="Id" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="OrderNumber" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="AssessmentLocations" nillable="true"
type="tns:ArrayOfAssessmentLocation" />
    
```

```

    </xs:sequence>
  </xs:complexType>
  <xs:element name="AssessmentStatus" nillable="true" type="tns:AssessmentStatus" />
  <xs:complexType name="ArrayOfAssessmentLocation">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="AssessmentLocation" nillable="true"
type="tns:AssessmentLocation" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfAssessmentLocation" nillable="true" type="tns:ArrayOfAssessmentLocation" />
  <xs:complexType name="AssessmentLocation">
    <xs:sequence>
      <xs:element minOccurs="0" name="Id" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Latitude" type="xs:double" />
      <xs:element minOccurs="0" name="Longitude" type="xs:double" />
      <xs:element minOccurs="0" name="XCoordinate" type="xs:double" />
      <xs:element minOccurs="0" name="YCoordinate" type="xs:double" />
      <xs:element minOccurs="0" name="Status" type="xs:int" />
      <xs:element minOccurs="0" name="Radius" type="xs:int" />
      <xs:element minOccurs="0" name="RepairHours" type="xs:int" />
      <xs:element minOccurs="0" name="IsTruckAccessible" type="xs:boolean" />
      <xs:element minOccurs="0" name="TreeTrimmingRequired" type="xs:int" />
      <xs:element minOccurs="0" name="Comments" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="NumOfPolesDown" type="xs:int" />
      <xs:element minOccurs="0" name="NumOfTransformers" type="xs:int" />
      <xs:element minOccurs="0" name="ConductorSpan" type="xs:int" />
      <xs:element minOccurs="0" name="NumOfServicesDown" type="xs:int" />
      <xs:element minOccurs="0" name="NumOfCrossArms" type="xs:int" />
      <xs:element minOccurs="0" name="NumOfDevicesDown" type="xs:int" />
      <xs:element minOccurs="0" name="NumOfTreeTrimSpans" type="xs:int" />
      <xs:element minOccurs="0" name="ServiceCenter" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="AdditionalData" nillable="true" type="tns:ArrayOfPropertyValue"
/>
    </xs:sequence>
  </xs:complexType>
  <xs:element name="AssessmentLocation" nillable="true" type="tns:AssessmentLocation" />
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="tns:PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfPropertyValue" nillable="true" type="tns:ArrayOfPropertyValue" />
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" name="PropertyName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Value" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="PropertyValue" nillable="true" type="tns:PropertyValue" />
</xs:schema>

```

6.1.2. CanClearIncident

Request to check whether an incident can be cleared, according to the incident clearance rules.

Message Type	HTTP GET
URI	http://localhost:8802/crew/services/CanClearIncident?Id={ID}
Source	WFM
Body (response)	JSON/XML
Type (response)	RequestCanClearResponse
Frequency	On demand

XML response schema:

```
<xs:schema xmlns:tns="http://com.alstom.dms.rest" elementFormDefault="qualified"
targetNamespace="http://com.alstom.dms.rest" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
  <xs:import namespace="http://schemas.datacontract.org/2004/07/NetUtil" />
  <xs:complexType name="RequestCanClearResponse">
    <xs:sequence>
      <xs:element minOccurs="0" name="out" nillable="true" type="tns:CanClearOut" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="RequestCanClearResponse" nillable="true" type="tns:RequestCanClearResponse" />
  <xs:complexType name="CanClearOut">
    <xs:sequence>
      <xs:element minOccurs="0" name="ValidationMessages" nillable="true"
xmlns:q1="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q1:ArrayOfstring" />
      <xs:element minOccurs="0" name="ValidationStatus"
xmlns:q2="http://schemas.datacontract.org/2004/07/NetUtil" type="q2:ValidationResult" />
      <xs:element minOccurs="0" name="Errors" nillable="true" type="tns:ArrayOfResponseError" />
      <xs:element minOccurs="0" name="Acknowledgement" nillable="true" type="tns:ResponseAct" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="CanClearOut" nillable="true" type="tns:CanClearOut" />
  <xs:complexType name="ArrayOfResponseError">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="ResponseError" nillable="true"
type="tns:ResponseError" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfResponseError" nillable="true" type="tns:ArrayOfResponseError" />
  <xs:complexType name="ResponseError">
    <xs:sequence>
      <xs:element minOccurs="0" name="errorMessage" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="creationTime" nillable="true" type="xs:dateTime" />
      <xs:element minOccurs="0" name="systemID" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="ErrorObject" nillable="true" type="tns:ErrorObject" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ResponseError" nillable="true" type="tns:ResponseError" />
</xs:schema>
```

```

<xs:complexType name="ErrorObject">
  <xs:sequence>
    <xs:element minOccurs="0" name="InstanceIDs" nillable="true" type="tns:InstanceIDs" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ErrorObject" nillable="true" type="tns:ErrorObject" />
<xs:complexType name="InstanceIDs">
  <xs:sequence>
    <xs:element minOccurs="0" name="identifierName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="identifierValue" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="InstanceIDs" nillable="true" type="tns:InstanceIDs" />
<xs:complexType name="ResponseAct">
  <xs:sequence>
    <xs:element minOccurs="0" name="outcome" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="creationTime" nillable="true" type="xs:dateTime" />
    <xs:element minOccurs="0" name="description" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ResponseAct" nillable="true" type="tns:ResponseAct" />
</xs:schema>

```

6.1.3. CrewStatus

Request to update the crew status.

Message Type	HTTP POST
URI	http://localhost:8802/crew/services/CrewStatus
Source	WFM
Body	JSON/XML
Type	CrewStatus
Frequency	On demand

Note

In ADMS 3.13, JobStatus, MobileCommunicationStatus, ShiftStatus, and AvailabilityStatus fields were added to provide detailed crew status information. The legacy Status field is still supported for backward compatibility. However, either the Status or JobStatus value can be used. If both the Status and JobStatus values are provided, the Status value is used and the JobStatus value is ignored.

XML request schema:

```

<xs:schema xmlns:tns="http://com.alstom.dms.rest" elementFormDefault="qualified"
targetNamespace="http://com.alstom.dms.rest" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="CrewStatus">
    <xs:sequence>

```

```

<xs:element minOccurs="0" name="CrewId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Status" nillable="true" type="xs:int" />
<xs:element minOccurs="0" name="IncidentId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="OtherInformation" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Comments" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="IncidentName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Vehicle" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ErrorNumber" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ErrorCode" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ARMSTimeStamp" type="xs:dateTime" />
<xs:element minOccurs="0" name="ETA" type="xs:dateTime" />
<xs:element minOccurs="0" name="WorkStatus" nillable="true" type="xs:int" />
<xs:element minOccurs="0" name="MobileCommunicationStatus" nillable="true" type="xs:int" />
<xs:element minOccurs="0" name="ShiftStatus" nillable="true" type="xs:int" />
<xs:element minOccurs="0" name="AvailabilityStatus" nillable="true" type="xs:int" />
</xs:sequence>
</xs:complexType>
<xs:element name="CrewStatus" nillable="true" type="tns:CrewStatus" />
</xs:schema>

```

6.1.4. DeviceStatus

Request to update the device state.

Message Type	HTTP POST
URI	http://localhost:8802/crew/services/DeviceStatus
Source	WFM
Body	JSON/XML
Type	DeviceStateUpdate
Frequency	On demand

XML request schema:

```

<xs:schema xmlns:tns="http://com.alstom.dms.rest" elementFormDefault="qualified"
targetNamespace="http://com.alstom.dms.rest" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:complexType name="DeviceStateUpdate">
<xs:sequence>
<xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
<xs:element minOccurs="1" name="DeviceId" type="xs:string" />
<xs:element minOccurs="1" name="Station" type="xs:string" />
<xs:element minOccurs="1" name="Feeder" type="xs:string" />
<xs:element minOccurs="1" name="DeviceState" type="xs:int" />
<xs:element minOccurs="1" name="Phases" type="xs:int" />
</xs:sequence>
</xs:complexType>
<xs:element name="DeviceStateUpdate" nillable="true" type="tns:DeviceStateUpdate" />
</xs:schema>

```

6.1.5. IncidentStatus

Request to update the incident status.

Message Type	HTTP POST
URI	http://localhost:8802/crew/services/IncidentStatus
Source	WFM
Body	JSON/XML
Type	IncidentStatus
Frequency	On demand

XML request schema:

```
<xs:schema xmlns:tns="http://com.alstom.dms.rest" elementFormDefault="qualified"
targetNamespace="http://com.alstom.dms.rest" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:complexType name="IncidentStatus">
<xs:sequence>
<xs:element minOccurs="0" name="IncidentId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Status" type="xs:int" />
<xs:element minOccurs="0" name="CrewId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ErrorNumber" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ErrorCode" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Comment" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ActualRestorationTime" type="xs:dateTime" />
<xs:element minOccurs="0" name="ActualRestorationTimeSpecified" type="xs:boolean" />
<xs:element minOccurs="0" name="Cause" type="xs:int" />
<xs:element minOccurs="0" name="CauseSpecified" type="xs:boolean" />
<xs:element minOccurs="0" name="OutageType" type="xs:int" />
<xs:element minOccurs="0" name="OutageTypeSpecified" type="xs:boolean" />
<xs:element minOccurs="0" name="PhasesOut" type="xs:int" />
<xs:element minOccurs="0" name="PhasesOutSpecified" type="xs:boolean" />
<xs:element minOccurs="0" name="FailureType" type="xs:int" />
<xs:element minOccurs="0" name="ActionTaken" type="xs:int" />
<xs:element minOccurs="0" name="ActionTakenSpecified" type="xs:boolean" />
<xs:element minOccurs="0" name="CorrCustCount" type="xs:int" />
<xs:element minOccurs="0" name="CorrCustCountSpecified" type="xs:boolean" />
<xs:element minOccurs="0" name="WfmEtr" type="xs:dateTime" />
<xs:element minOccurs="0" name="WfmEtrSpecified" type="xs:boolean" />
<xs:element minOccurs="0" name="NoOfPolesDown" type="xs:int" />
<xs:element minOccurs="0" name="NoOfPolesDownSpecified" type="xs:boolean" />
<xs:element minOccurs="0" name="NoOfTransformers" type="xs:int" />
<xs:element minOccurs="0" name="NoOfTransformersSpecified" type="xs:boolean" />
<xs:element minOccurs="0" name="ConductorSpan" type="xs:int" />
<xs:element minOccurs="0" name="ConductorSpanSpecified" type="xs:boolean" />
<xs:element minOccurs="0" name="NoServicesDown" type="xs:int" />
<xs:element minOccurs="0" name="NoServicesDownSpecified" type="xs:boolean" />
<xs:element minOccurs="0" name="Inaccessible" type="xs:boolean" />
<xs:element minOccurs="0" name="FollowupRequired" type="xs:boolean" />
<xs:element minOccurs="0" name="LineCrewNeeded" type="xs:boolean" />
<xs:element minOccurs="0" name="LineCrewNeededSpecified" type="xs:boolean" />
</xs:sequence>
</xs:complexType>
</xs:schema>
```

```

<xs:element minOccurs="0" name="TreeCrewNeeded" type="xs:boolean" />
<xs:element minOccurs="0" name="TreeCrewNeededSpecified" type="xs:boolean" />
<xs:element minOccurs="0" name="FirstResponderNeeded" type="xs:boolean" />
<xs:element minOccurs="0" name="PointOfFailure" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="NoOfDevicesDown" nillable="true" type="xs:int" />
<xs:element minOccurs="0" name="NoOfDevicesDownSpecified" type="xs:boolean" />
</xs:sequence>
</xs:complexType>
<xs:element name="IncidentStatus" nillable="true" type="tns:IncidentStatus" />
</xs:schema>

```

6.1.6. OrderStatus

Request to update the follow-up work order status.

Message Type	HTTP POST
URI	http://localhost:8802/crew/services/OrderStatus
Source	WFM
Body	JSON/XML
Type	OrderStatus
Frequency	On demand

XML request schema:

```

<xs:schema xmlns:tns="http://com.alstom.dms.rest" elementFormDefault="qualified"
targetNamespace="http://com.alstom.dms.rest" xmlns:xs="http://www.w3.org/2001/XMLSchema">
<xs:complexType name="OrderStatus">
<xs:sequence>
<xs:element minOccurs="0" name="Id" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Name" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Comment" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Aor" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="Organization" type="xs:int" />
<xs:element minOccurs="0" name="OrderType" type="xs:int" />
<xs:element minOccurs="0" name="Priority" type="xs:int" />
<xs:element minOccurs="0" name="Status" type="xs:int" />
<xs:element minOccurs="0" name="DeviceStation" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="CreatedBy" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="ModifiedBy" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="IncidentId" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="IncidentName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeleteOrder" type="xs:boolean" />
<xs:element minOccurs="0" name="DeviceType" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DeviceFeederName" nillable="true" type="xs:string" />
<xs:element minOccurs="0" name="DevicePhases" nillable="true" type="xs:string" />
</xs:sequence>
</xs:complexType>

```

```
<xs:element name="OrderStatus" nillable="true" type="tns:OrderStatus" />
</xs:schema>
```

6.1.7. PlannedOutageStatus

Request to update the planned outage status.

Message Type	HTTP POST
URI	http://localhost:8802/crew/services/PlannedOutageStatus
Source	WFM
Body	JSON/XML
Type	PlannedOutageStatus
Frequency	On demand

XML request schema:

```
<xs:schema xmlns:tns="http://com.alstom.dms.rest" elementFormDefault="qualified"
targetNamespace="http://com.alstom.dms.rest" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="PlannedOutageStatus">
    <xs:sequence>
      <xs:element minOccurs="0" name="PlannedOutageId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Status" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="PrimaryStartTime" type="xs:dateTime" />
      <xs:element minOccurs="0" name="PrimaryEndTime" type="xs:dateTime" />
      <xs:element minOccurs="0" name="WorkOrderId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="AdditionalData" nillable="true" type="tns:ArrayOfPropertyValue"
    />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="PlannedOutageStatus" nillable="true" type="tns:PlannedOutageStatus" />
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="tns:PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfPropertyValue" nillable="true" type="tns:ArrayOfPropertyValue" />
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" name="PropertyName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Value" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="PropertyValue" nillable="true" type="tns:PropertyValue" />
</xs:schema>
```

6.1.8. VehicleLocation

Request to update a vehicle location.

Message Type	HTTP POST
URI	http://localhost:8802/crew/services/VehicleLocation
Source	WFM
Body	JSON/XML
Type	VehicleLocation
Frequency	On demand

XML request schema:

```
<xs:schema xmlns:tns="http://com.alstom.dms.rest" elementFormDefault="qualified"
targetNamespace="http://com.alstom.dms.rest" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="VehicleLocation">
    <xs:sequence>
      <xs:element minOccurs="0" name="SystemId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="VehicleLocations" nillable="true" type="tns:ArrayOfVehicleLocationV1" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="VehicleLocation" nillable="true" type="tns:VehicleLocation" />
  <xs:complexType name="ArrayOfVehicleLocationV1">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="VehicleLocationV1" nillable="true" type="tns:VehicleLocationV1" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfVehicleLocationV1" nillable="true" type="tns:ArrayOfVehicleLocationV1" />
  <xs:complexType name="VehicleLocationV1">
    <xs:sequence>
      <xs:element minOccurs="0" name="VehicleId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="CrewId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Latitude" type="xs:double" />
      <xs:element minOccurs="0" name="Longitude" type="xs:double" />
      <xs:element minOccurs="0" name="X" type="xs:double" />
      <xs:element minOccurs="0" name="Y" type="xs:double" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="VehicleLocationV1" nillable="true" type="tns:VehicleLocationV1" />
</xs:schema>
```

Note

Currently, the Crew server does not query for vehicle locations when it runs as a web service. The Crew server forwards vehicle location messages to the Net server.

6.1.9. Damages

Request to add, update, or delete a damage object.

Message Type	HTTP POST
URI	http://localhost:8802/crew/services/Damages
Source	WFM
Body	JSON/XML
Type	AddDamages
	UpdateDamages
	DeleteDamages
Frequency	On demand

XML request schema (AddDamages):

```
<xs:schema xmlns:tns="http://com.alstom.dms.rest" elementFormDefault="qualified"
targetNamespace="http://com.alstom.dms.rest" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="AddDamages">
    <xs:sequence>
      <xs:element minOccurs="0" name="Damages" nillable="true" type="tns:ArrayOfAddDamage" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="AddDamages" nillable="true" type="tns:AddDamages" />
  <xs:complexType name="ArrayOfAddDamage">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="AddDamage" nillable="true"
type="tns:AddDamage" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="ArrayOfAddDamage" nillable="true" type="tns:ArrayOfAddDamage" />
  <xs:complexType name="AddDamage">
    <xs:sequence>
      <xs:element minOccurs="0" name="DamageId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="DamageAssessmentLocationId" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="AssetType" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="AssetSubType" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Quantity" type="xs:int" />
      <xs:element minOccurs="0" name="Description" nillable="true" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="AddDamage" nillable="true" type="tns:AddDamage" />
</xs:schema>
```

XML request schema (UpdateDamages):

```
<xs:schema xmlns:tns="http://com.alstom.dms.rest" elementFormDefault="qualified"
targetNamespace="http://com.alstom.dms.rest" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:complexType name="UpdateDamages">
    <xs:sequence>
```

```

    <xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Damages" nillable="true" type="tns:ArrayOfUpdateDamage" />
  </xs:sequence>
</xs:complexType>
<xs:element name="UpdateDamages" nillable="true" type="tns:UpdateDamages" />
<xs:complexType name="ArrayOfUpdateDamage">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="UpdateDamage" nillable="true"
type="tns:UpdateDamage" />
  </xs:sequence>
</xs:complexType>
<xs:element name="ArrayOfUpdateDamage" nillable="true" type="tns:ArrayOfUpdateDamage" />
<xs:complexType name="UpdateDamage">
  <xs:sequence>
    <xs:element minOccurs="0" name="DamageId" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="AssetType" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="AssetSubType" nillable="true" type="xs:string" />
    <xs:element minOccurs="0" name="Quantity" type="xs:int" />
    <xs:element minOccurs="0" name="Description" nillable="true" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:element name="UpdateDamage" nillable="true" type="tns:UpdateDamage" />
</xs:schema>

```

XML request schema (DeleteDamages):

```

<xs:schema xmlns:tns="http://com.alstom.dms.rest" elementFormDefault="qualified"
targetNamespace="http://com.alstom.dms.rest" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:import namespace="http://schemas.microsoft.com/2003/10/Serialization/Arrays" />
  <xs:complexType name="DeleteDamages">
    <xs:sequence>
      <xs:element minOccurs="0" name="UserName" nillable="true" type="xs:string" />
      <xs:element minOccurs="0" name="Damages" nillable="true"
xmlns:q1="http://schemas.microsoft.com/2003/10/Serialization/Arrays" type="q1:ArrayOfstring" />
    </xs:sequence>
  </xs:complexType>
  <xs:element name="DeleteDamages" nillable="true" type="tns:DeleteDamages" />
</xs:schema>

```

6.2. Notification Types

An event notification mechanism on the Crew server web service can send notifications about crew and incident updates to external applications. These events are sent via REST calls to the registered URLs.

Note

For the applications to receive notifications, configure `<NotificationBaseUrl>` and `<NotificationTypes>` in the `CrewServerConfig.xml` file.

The following notification events are supported by the Crew server web service:

- Incident update notification
- Crew update notification
- Follow-up work order update notification
- Crew hello message notification

6.2.1. IncidentUpdateNotification

Sends notifications about incident updates.

Message Type	HTTP POST
URI	http://localhost:9090/dms/event/Crew/IncidentUpdateNotification
Source	Crew server
Body	JSON/XML
Type	INC-UPDATE
Frequency	On demand

6.2.2. CrewUpdateNotification

Sends notifications about crew updates.

Message Type	HTTP POST
URI	http://localhost:9090/dms/event/Crew/CrewUpdateNotification
Source	Crew server
Body	JSON/XML
Type	CREW-UPDATE
Frequency	On demand

6.2.3. OrderUpdateNotification

Sends notifications about follow-up work order updates.

Message Type	HTTP POST
URI	http://localhost:9090/dms/event/Crew/OrderUpdateNotification
Source	Crew server
Body	JSON/XML
Type	ORDER-UPDATE
Frequency	On demand

6.2.4. HelloNotification

Sends notifications about crew hello messages.

Message Type	HTTP POST
URI	http://localhost:9090/dms/event/Crew/HelloNotification
Source	Crew server
Body	JSON/XML
Type	HELLO
Frequency	On demand

7. Crew Server Web Interface

The ADMS Crew server supports communicating with a Workforce Management System (WFM) via HTTP instead of legacy TCP. When HTTP communication is enabled, interface messages are exchanged via web services. The web service is loaded by the Crew server using the Plugin API.

To enable the Crew server web service (WFM web API services), set `PluginDefinition> Enabled` to `True` in the Crew server configuration file (`CrewServerConfig.xml`).

The WFM, or an Enterprise Service Bus (ESB), provides a web services portal for all messages to or from the ADMS. Although this chapter refers to messages sent to or from WFM, it is implied that the messages are sent to or from the WFM web services portal.

The Crew server web service interface uses version 2 message schemas. The original message schemas used by the TCP/IP interface are referred to as version1.

i Note

The external WFM system can send messages with only the fields they want to change. Most of the remaining fields can be omitted (fields with `nillable="true"`). Any omitted field will default to null or blank and will not change the incident or crew data in the OMS system.

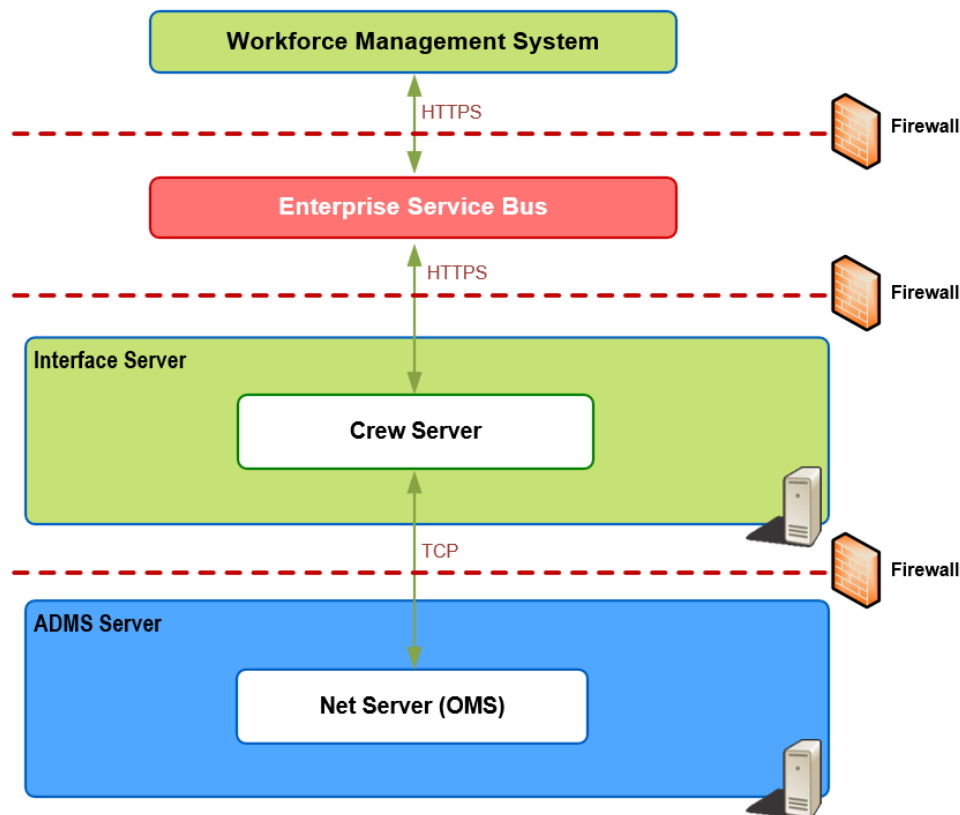


Figure 27. ADMS-WFM Data Flow

The use of web services means that ADMS and WFM do not maintain a constant connection. Each

request is a new connection. To restrict connections originating from less secure zones, all message exchanges are initiated by ADMS, which sends query requests to WFM, and WFM responds with queued messages. The endpoints are hosted in the ESB.

For each message query response that ADMS receives from WFM, ADMS sends a response with the result of the requested action to WFM. For example, if WFM sends a request to confirm open a device, ADMS may send an error message if the device could not be open for some reason (for example, this is a non-protective device).

ADMS posts a response to each request asynchronously, in a subsequent HTTPS message. Each response references the associated request message.

Note

ADMS web API projects (Crew server, Call server, Relay server, and AMI server web services) use ADMS Open API server and Plugin API. The Open API server provides startup, logging, security, and schema generation functions. The Plugin API provides loading, persistence, notification, query, message queue, and configuration functions.

7.1. Security

All messages among an ADMS web API application and an external application (for example, WFM) can be exchanged using the HTTPS secure communications protocol.

To provide additional security, WFM can require an authorization header for all requests from ADMS. The authorization header, combined with WFM web service configuration parameters, is used by WFM to verify that requests come from a trusted client (ADMS).

Requests from WFM to ADMS do not need to include an authorization header. WFM sends messages to ADMS only in response to message requests from ADMS. HTTPS ensures that the response from WFM (which contains requests for ADMS) is not tampered with during transmission.

ADMS web API application servers also support using SSL certificates for the HTTPS mode.

7.2. Service Types

The Crew server web interface is configured in the CrewServerConfig.xml configuration file. Web services and web service clients are defined as separate plugins. The default base URI configuration applies to all plugins.

The default base URI configuration is as follows:

```
<UriScheme>http</UriScheme>
<UriHost>localhost</UriHost>
<UriPort>8443</UriPort>
<UriPath>api</UriPath>
```

The default base URI configuration results in the following base URI:

```
http://localhost:8443/api/
```

The base URI composite configuration is used for each plugin except the plugins with the BaseUri parameter, which explicitly defines the base URI.

Each Crew server web interface URL has the following format:

```
<UriScheme>://<UriHost>:<UriPort>/<UriPath>/<BasePath>/v<Version>/<Endpoint>
```

Or:

```
<BaseUri>/<Endpoint>
```

For example:

```
http://localhost:8445/api/adms/crew/model/v2/Status
```

For web services, you can configure uri scheme, uri host, uri port, uri path, base path, and version values. For web service clients, you can additionally configure endpoints.

The Crew server web interface (and other plugin web interfaces) support two service types:

- **WebService:** This is an interface to a base class that implements a web service using the Microsoft Owin technology. All plugin definitions with the WebService type are services that are hosted by ADMS. This means that an external WFM system can communicate to this endpoint.

The following example shows a WebService plugin definition:

```
<PluginDefinition>
  <Enabled>true</Enabled>
  <Type>WebService</Type>
  <Name>CrewModelingWebService</Name>
  <Config>
    <!--BaseUri>http://localhost:8445/api/ext/wms/v2</BaseUri-->
    <BasePath>adms/crew/model</BasePath>
    <Name>CrewModelingWebService</Name>
    <Version>2</Version>
    <Description>Implements Crew Modeling Web Service</Description>
  ...
```

In this example, the BaseUri parameter is not defined (commented out). Therefore the composed URI configuration is used: http://localhost:8443/api/adms/crew/model/v2

- **WebServiceClient:** This type is used to communicate with existing endpoints in the external WFM system. WebServiceClient uses the HTTPClient object to send-request-to and receive-response-from external WFM system endpoints. The following example shows a WebServiceClient type plugin definition:

```

<PluginDefinition>
  <Enabled>true</Enabled>
  <Type>WebServiceClient</Type>
  <Name>CrewServerWmsWebClient</Name>
  <Config>
    <Name>CrewServerWmsWebClient</Name>
    <BaseUri>http://localhost:8445/api/ext/wms/v2</BaseUri>
    <MessagesStatusPostUri>Status</MessagesStatusPostUri>
  </Config>
</PluginDefinition>

```

In this example, the BaseUri parameter explicitly defines the base URI and the MessagesStatusPostUri defines the Status endpoint. Therefore, the status URI will be translated to: <http://localhost:8445/api/ext/wms/v2/Status>

You can use separate URLs for each endpoint. The following example shows the configuration.

```

<PluginDefinition>
  <Type>WebServiceClient</Type>
  <Name>AmiInterfaceWebServiceClient</Name>
  <Enabled>true</Enabled>
  <Config>
    <Name>AmiInterfaceWebServiceClient</Name>
    <BaseUri>http://localhost:8445/api/ext/ami/v2</BaseUri>
    <MessagesUri>http://localhost:8446/api/ext/ami/v2/AmiMessages</MessagesUri>
    <StatusUri>http://localhost:8447/api/ext/ami/v2/Status</StatusUri>
    <PingUri>Ping</PingUri>
  </Config>
</PluginDefinition>

```

Before sending any request through HTTPClient, WebServiceClient checks the endpoint. If the endpoint starts with "http" then it will not consider the BaseUri parameter. The following translated URI's are for above configuration.

Message URI - <http://localhost:8446/api/ext/ami/v2/AmiMessages> (same as configured)

Status URI - <http://localhost:8447/api/ext/ami/v2/Status> (same as configured)

Ping URI - <http://localhost:8445/api/ext/ami/v2/Ping> (base URI + ping URI as ping URI not contains any url)

In this way, PluginDefination supports multiple URLs for processing different endpoints. For Message, it is using 8446 port, for Status using 8447 port, and for Ping using 8445 port(base URI port).

Note

By convention, all services hosted by ADMS have /adms/ in the base path. Similarly, all services hosted by external systems (for example, WFM), have /ext/ in the base path.

7.2.1. Crew Server Web Services Hosted by ADMS

The following are default Crew Server web interface URLs hosted by ADMS. The Data Type section provides a reference to a section that shows the data type format that is accepted by ADMS (message schema).

- **/api/adms/crew/v2/**: This is the root of the Crew Server web service.

- AssignmentQuery

Hosted by	ADMS
Description	WFM uses this URL to send AssignmentQuery messages to the Crew server.
Default URI	http://localhost:8443/api/adms/crew/v2/AssignmentQuery
HTTP Operation	POST
Data Type	See AssignmentQuery .

- UpdateCrewStatus

Hosted by	ADMS
Description	WFM uses this URL to send crew updates to the Crew server.
Default URI	http://localhost:8443/api/adms/crew/v2/UpdateCrewStatus
HTTP Operation	POST
Data Type	See UpdateCrewStatus .

- UpdateDeviceState

Hosted by	ADMS
Description	WFM uses this URL to send device state updates to the Crew server.
Default URI	http://localhost:8443/api/adms/crew/v2/UpdateDeviceState
HTTP Operation	POST
Data Type	See UpdateDeviceState .

- UpdateIncident

Hosted by	ADMS
Description	WFM uses this URL to send incident updates to the Crew server.
Default URI	http://localhost:8443/api/adms/crew/v2/UpdateIncident
HTTP Operation	POST
Data Type	See UpdateIncident .

- UpdateVehicleLocation

Hosted by	ADMS
Description	WFM uses this URL to send vehicle location updates to the Crew server.
Default URI	http://localhost:8443/api/adms/crew/v2/UpdateVehicleLocation
HTTP Operation	POST
Data Type	An array of type VehicleLocationV2. See UpdateVehicleLocation .

- OrderStatus

Hosted by	ADMS
Description	WFM uses this URL to send follow-up work order updates to the Crew server.
Default URI	http://localhost:8443/api/adms/crew/v2/OrderStatus
HTTP Operation	POST
Data Type	See OrderStatus .

- PlannedOutageStatus

Hosted by	ADMS
Description	WFM uses this URL to send planned outage updates to the Crew server.
Default URI	http://localhost:8443/api/adms/crew/v2/PlannedOutageStatus
HTTP Operation	POST
Data Type	See PlannedOutageStatus .

- AssessmentStatus

Hosted by	ADMS
Description	WFM uses this URL to send damage assessment location updates to the Crew server.
Default URI	http://localhost:8443/api/adms/crew/v2/AssessmentStatus
HTTP Operation	POST
Data Type	See AssessmentStatus .

- Damages

Hosted by	ADMS
Description	WFM uses this URL to send messages to add, update, or delete damages to the Crew server.
Default URI	http://localhost:8443/api/adms/crew/v2/Damages
HTTP Operation	POST PUT DELETE
Data Type	See Damages .

- CanClearIncident

Hosted by	ADMS
Description	WFM uses this URL to check whether an incident can be cleared according to incident clearance rules.
Default URI	http://localhost:8443/api/adms/crew/v2/CanClearIncident?incidentId='incident ID'
HTTP Operation	GET
Data Type (response)	See CanClearIncident .

- Status (GET)

Hosted by	ADMS
------------------	------

Description	WFM uses this URL to get the status of the Crew server web service.
Default URI	http://localhost:8443/api/adms/crew/v2/Status
HTTP Operation	GET
Data Type	See Status (Crew Server Web Service) .

- Status (POST)

Hosted by	ADMS
Description	WFM uses this URL to send the message processing status.
Default URI	http://localhost:8443/api/adms/crew/v2/Status
HTTP Operation	POST
Data Type	See Status (Crew Server Web Service) (POST) .

- **/api/adms/crew/model/v2/:** This is the root of the Crew server crew modeling web service.

- addCrew

Hosted by	ADMS
Description	WFM uses this URL to add crews to the ADMS crew model.
Default URI	http://localhost:8443/api/adms/crew/model/v2/addCrew
HTTP Operation	POST
Data Type	An array of type AddCrew. See addCrew .

- deleteCrew

Hosted by	ADMS
Description	WFM uses this URL to delete crews from the ADMS crew model.
Default URI	http://localhost:8443/api/adms/crew/model/v2/deleteCrew
HTTP Operation	POST
Data Type	An array of type DeleteCrew. See deleteCrew .

- updateCrew

Hosted by	ADMS
Description	WFM uses this URL to update crew data in the ADMS crew model.
Default URI	http://localhost:8443/api/adms/crew/model/v2/updateCrew
HTTP Operation	POST
Data Type	An array of type UpdateCrew. See updateCrew .

- addEmployee

Hosted by	ADMS
Description	WFM uses this URL to add employees to the ADMS crew model.

Default URI	http://localhost:8443/api/adms/crew/model/v2/addEmployee
HTTP Operation	POST
Data Type	An array of type AddEmployee. See addEmployee .

- deleteEmployee

Hosted by	ADMS
Description	WFM uses this URL to delete employees from the ADMS crew model.
Default URI	http://localhost:8443/api/adms/crew/model/v2/deleteEmployee
HTTP Operation	POST
Data Type	An array of type DeleteEmployee. See delete-employee[deleteEmployee] .

- updateEmployee

Hosted by	ADMS
Description	WFM uses this URL to update employee data in the ADMS crew model.
Default URI	http://localhost:8443/api/adms/crew/model/v2/updateEmployee
HTTP Operation	POST
Data Type	An array of type UpdateEmployee. See updateEmployee .

- addVehicle

Hosted by	ADMS
Description	WFM uses this URL to add vehicles to the ADMS crew model
Default URI	http://localhost:8443/api/adms/crew/model/v2/addVehicle
HTTP Operation	POST
Data Type	An array of type AddVehicle. See add-vehicle[addVehicle] .

- deleteVehicle

Hosted by	ADMS
Description	WFM uses this URL to delete vehicles from the ADMS crew model.
Default URI	http://localhost:8443/api/adms/crew/model/v2/deleteVehicle
HTTP Operation	POST
Data Type	An array of type DeleteVehicle. See delete-vehicle[deleteVehicle] .

- updateVehicle

Hosted by	ADMS
Description	WFM uses this URL to update vehicle data in the ADMS crew model.
Default URI	http://localhost:8443/api/adms/crew/model/v2/updateVehicle
HTTP Operation	POST

Data Type	An array of type UpdateVehicle. See update-vehicle[updateVehicle].
------------------	--

- assignCrewEmployees

Hosted by	ADMS
Description	WFM uses this URL to assign employees to or unassign employees from the crew.
Default URI	http://localhost:8443/api/adms/crew/model/v2/assignCrewEmployees
HTTP Operation	POST
Data Type	See assignCrewEmployees .

- assignCrewVehicles

Hosted by	ADMS
Description	WFM uses this URL to assign vehicles to or unassign vehicles from the crew.
Default URI	http://localhost:8443/api/adms/crew/model/v2/assignCrewVehicles
HTTP Operation	POST
Data Type	See assignCrewVehicles .

- Status

Hosted by	ADMS
Description	WFM uses this URL to send the message processing status.
Default URI	http://localhost:8443/api/adms/crew/model/v2/Status
HTTP Operation	POST
Data Type	See Status (Crew Server Crew Modeling Web Service) .

- **/api/adms/crew/logger/v2:** This is the root of the Crew server OmsLogger web service.

- Log (future functionality)

Hosted by	ADMS
Description	WFM uses this URL to get HTTP response messages.
Default URI	http://localhost:8443/api/adms/crew/logger/v2/Log?from={from}
HTTP Operation	GET
Data Type	HttpResponseMessage

7.2.2. Crew Server Web Interfaces Hosted by WFM System

The following are default Crew Server web interface URLs that are hosted by WFM. The Data Type section provides a reference to a section which shows the data type format that is accepted by WFM (message schema).

- **/api/ext/wms/v2/:** This is the root of the Crew server web service client.

- ModelingStatus

Hosted by	WFM system
Description	Crew server uses this URL to send crew modeling status updates to the crew modeling web service client.
Default URI	http://localhost:8445/api/ext/wms/v2/ModelingStatus
HTTP Operation	POST
Data Type	See ModelingStatus .

- WMSMessages

Hosted by	WFM system
Description	Crew server uses this URL to return all pending crew status, incident status, vehicle location, assignment query, device state update, and assessment status messages from WFM to ADMS.
Default URI	http://localhost:8445/api/ext/wms/v2/WMSMessages
HTTP Operation	GET
Data Type	See WMSMessages .

- IncidentUpdate

Hosted by	WFM system
Description	Crew server uses this URL to send incident updates to WFM.
Default URI	http://localhost:8445/api/ext/wms/v2/IncidentUpdate
HTTP Operation	POST
Data Type	See IncidentUpdate .

- CrewUpdate

Hosted by	WFM system
Description	Crew server uses this URL to send crew updates to WFM.
Default URI	http://localhost:8445/api/ext/wms/v2/CrewUpdate
HTTP Operation	POST
Data Type	See CrewUpdate .

- OrderUpdate

Hosted by	WFM system
Description	Crew server uses this URL to send follow-up work order updates to WFM.
Default URI	http://localhost:8445/api/ext/wms/v2/OrderUpdate
HTTP Operation	POST
Data Type	See OrderUpdate .

- PlannedOutageUpdate

Hosted by	WFM system
Description	Crew server uses this URL to send planned outage updates to WFM.
Default URI	http://localhost:8445/api/ext/wms/v2/PlannedOutageUpdate
HTTP Operation	POST
Data Type	See PlannedOutageUpdate .

- SynchronizeCrews

Hosted by	WFM system
Description	Crew server uses this URL to send crew modeling updates to Mobility Gateway, which in turn notifies Mobile Enterprise.
Default URI	http://localhost:8445/api/ext/wms/v2/SynchronizeCrews
HTTP Operation	POST
Data Type	See SynchronizeCrews .

- SwitchingOrder

Hosted by	WFM system
Description	Crew server uses this URL to send switching order updates to WFM (Future functionality).
Default URI	Future functionality
HTTP Operation	Future functionality
Data Type	Future functionality

- Status

Hosted by	WFM system
Description	Crew server uses this URL to send message processing status to WFM.
Default URI	http://localhost:8445/api/ext/wms/v2/Status
HTTP Operation	POST
Data Type	See Status (Crew Server) .

- **/api/ext/wms/v2/:** This is the root of the Crew server modeling web service client.

- WMSModelMessages

Hosted by	WFM system
Description	Crew server uses this URL to return all pending crew modeling messages from WFM to ADMS.
Default URI	http://localhost:8445/api/ext/wms/v2/WMSModelMessages
HTTP Operation	GET
Data Type	See WMSModelMessages .

- ModelMessageStatus

Hosted by	WFM system
Description	Crew server uses this URL to send modeling message processing status to WFM.
Default URI	http://localhost:8445/api/ext/wms/v2/ModelMessageStatus
HTTP Operation	POST
Data Type	See ModelMessageStatus .

7.2.3. AssignmentQuery

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/v2/AssignmentQuery
Source	WFM
Body (Response)	JSON/XML
Type (Response)	AssignmentQueryV2

WFM uses this request to send AssignmentQuery messages to the Crew server.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="AssignmentQueryV2" nillable="true" type="AssignmentQueryV2" />
  <xs:complexType name="AssignmentQueryV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="CustomerResults" nillable="true" type="xs:int"
        />
      </xs:sequence>
    </xs:extension>
  </xs:complexType>
</xs:schema>
<xs:complexType name="MessageBaseV2">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
  </xs:sequence>
</xs:complexType>
```

```

type="PropertyValue" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.4. UpdateCrewStatus

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/v2/UpdateCrewStatus
Source	WFM
Body	JSON/XML
Type	CrewStatusV2
Frequency	On demand

WFM uses this URL to send crew updates to the Crew server.

To trigger the Crew Server to send the IncidentUpdateV2 message with the complete list of customer calls to the WFM, send a CrewStatusV2 message with the "201" status. For example:

```

{
  "CrewId": "FI1940",
  "Status": 201,
  "IncidentId": "1870000121"
}

```

Note

In the CrewStatusV2 message, the only required field is the "CrewId"; all other fields are optional.

Note

In ADMS 3.13, JobStatus, MobileCommunicationStatus, ShiftStatus, and AvailabilityStatus fields were added to provide detailed crew status information. The legacy Status field is still supported for backward compatibility. However, either the Status or WorkStatus value can be used. If both the Status and WorkStatus values are provided, the Status value is used and the WorkStatus value is ignored.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CrewStatusV2" nillable="true" type="CrewStatusV2" />
  <xs:complexType name="CrewStatusV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="WorkOrderId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Status" nillable="true" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="StatusName" nillable="true"
type="eWfmCrewStatus" />
          <xs:element minOccurs="1" maxOccurs="1" name="WorkStatus" nillable="true" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="MobileCommunicationStatus" nillable="true"
type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="ShiftStatus" nillable="true" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="AvailabilityStatus" nillable="true"
type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="SwitchingOrderId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="OtherInformation" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Comments" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Vehicle" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="ErrorNumber" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="ErrorCode" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewArea" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="ETA" type="xs:dateTime" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

```

</xs:complexType>
<xs:simpleType name="eWfmCrewStatus">
  <xs:restriction base="xs:string">
    <xs:enumeration value="NotSelected" />
    <xs:enumeration value="Unavailable" />
    <xs:enumeration value="SignOff" />
    <xs:enumeration value="SignOn" />
    <xs:enumeration value="OnBreak" />
    <xs:enumeration value="StandBy" />
    <xs:enumeration value="Dispatched" />
    <xs:enumeration value="EnRoute" />
    <xs:enumeration value="OnSite" />
    <xs:enumeration value="WaitingOnEquipment" />
    <xs:enumeration value="Released" />
    <xs:enumeration value="Unassigned" />
    <xs:enumeration value="Available" />
    <xs:enumeration value="OutOfCoverage" />
    <xs:enumeration value="Acknowledged" />
    <xs:enumeration value="Suspended" />
    <xs:enumeration value="PendingDispatch" />
    <xs:enumeration value="Assigned" />
  </xs:restriction>
</xs:simpleType>
</xs:schema>

```

7.2.5. UpdateDeviceState

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/v2/UpdateDeviceState
Source	WFM
Body	JSON/XML
Type	DeviceStateUpdateV2
Frequency	On demand

WFM uses this URL to send device state status updates to the Crew server.

FoundOpen Messages

WFM can send DeviceStateUpdateV2 messages with the DeviceState value set to 3 (FoundOpen) to notify the ADMS system of devices which were found open in the field. This option is available for non-SCADA switches, fuses, distribution transformers, and reclosers.

When a FoundOpen message is received, the system opens the device and creates a confirmed incident with the Found Open bit set for a device. When a device is already part of an incident, a manually confirmed incident is created and downstream customers are split to this incident.



Note

A new incident is not created if the device is open and already has the associated incident.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="DeviceStateUpdateV2" nillable="true" type="DeviceStateUpdateV2" />
  <xs:complexType name="DeviceStateUpdateV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="1" maxOccurs="1" name="DeviceId" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="DeviceState" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="DeviceStateName" nillable="true"
type="eConfirmDeviceState" />
          <xs:element minOccurs="1" maxOccurs="1" name="Phases" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="PhasesName" nillable="true" type="ePhaseOut"
/>
          <xs:element minOccurs="0" maxOccurs="1" name="UserName" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Station" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Feeder" type="xs:string" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="eConfirmDeviceState">
    <xs:restriction base="xs:string">
      <xs:enumeration value="ConfirmOpen" />
      <xs:enumeration value="ConfirmClosed" />
      <xs:enumeration value="FoundOpen" />
    </xs:restriction>
  </xs:simpleType>
</xs:schema>
```

```

</xs:simpleType>
<xs:simpleType name="ePhaseOut">
  <xs:restriction base="xs:string">
    <xs:enumeration value="ABC" />
    <xs:enumeration value="AB" />
    <xs:enumeration value="BC" />
    <xs:enumeration value="AC" />
    <xs:enumeration value="A" />
    <xs:enumeration value="B" />
    <xs:enumeration value="C" />
    <xs:enumeration value="None" />
  </xs:restriction>
</xs:simpleType>
</xs:schema>

```

7.2.6. UpdateIncident

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/v2/UpdateIncident
Source	WFM
Body	JSON/XML
Type	IncidentStatusV2
Frequency	On demand

WFM uses this URL to send incident updates to the Crew server.

Note

Messages with StatusCode set to 103 (Complete), 3 (Dispatched), 4 (EnRoute), 5 (OnSite), or 10 (Evaluating) require that CrewId is set to the crew ID that is assigned to the incident.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="IncidentStatusV2" nillable="true" type="IncidentStatusV2" />
  <xs:complexType name="IncidentStatusV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="1" maxOccurs="1" name="StatusCode" nillable="true" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="StatusName" nillable="true"
type="eOrderStatus" />
          <xs:element minOccurs="1" maxOccurs="1" name="CrewId" nillable="true" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="ErrorNumber" type="xs:string" />

```

```

        <xs:element minOccurs="0" maxOccurs="1" name="ErrorCode" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Comment" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="ActualRestorationTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="CauseGroup" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="Cause" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="OutageType" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="OutageTypeName" nillable="true"
type="eOutageType" />
        <xs:element minOccurs="1" maxOccurs="1" name="PhasesOut" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="PhasesOutName" nillable="true"
type="ePhaseOut" />
        <xs:element minOccurs="1" maxOccurs="1" name="FailureType" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="FailureMode" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="FailedEquipment" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="PowerQualityType" nillable="true"
type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="Manufacturer" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="ActionTaken" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="CorrCustCount" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="WfmEtr" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="NoOfPolesDown" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="NoOfTransformers" nillable="true"
type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="NoOfCrossArms" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="NoOfTreeTrimSpans" nillable="true"
type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="ConductorSpan" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="NoServicesDown" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="Inaccessible" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="FollowupRequired" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="FollowupType" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="LineCrewNeeded" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="SystemAffected" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="TreeCrewNeeded" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="FirstResponderNeeded" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="EvaluatorNeeded" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="0" maxOccurs="1" name="PointOfFailure" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="Weather" type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="WorkOrderId" type="xs:string" />
</xs:sequence>

```

```

    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="eOrderStatus">
  <xs:restriction base="xs:string">
    <xs:enumeration value="NotSelected" />
    <xs:enumeration value="NoTroubleFound" />
    <xs:enumeration value="DamageAssessment" />
    <xs:enumeration value="CancelOrder" />
    <xs:enumeration value="Complete" />
    <xs:enumeration value="Patrolled" />
    <xs:enumeration value="PendingAssigned" />
    <xs:enumeration value="BackToAssigned" />
    <xs:enumeration value="ConfirmedOpen" />
    <xs:enumeration value="Cleared" />
    <xs:enumeration value="Restored" />
    <xs:enumeration value="CancelIncident" />
  </xs:restriction>
</xs:simpleType>
<xs:simpleType name="eOutageType">
  <xs:restriction base="xs:string">
    <xs:enumeration value="NotSelected" />
    <xs:enumeration value="None" />
    <xs:enumeration value="CustomerTrouble" />
    <xs:enumeration value="ExternalSource" />
    <xs:enumeration value="Primary" />
    <xs:enumeration value="Secondary" />
    <xs:enumeration value="Service" />
    <xs:enumeration value="Substation" />
    <xs:enumeration value="Transformer" />
    <xs:enumeration value="Transmission" />
  </xs:restriction>

```

```

</xs:simpleType>
<xs:simpleType name="ePhaseOut">
  <xs:restriction base="xs:string">
    <xs:enumeration value="ABC" />
    <xs:enumeration value="AB" />
    <xs:enumeration value="BC" />
    <xs:enumeration value="AC" />
    <xs:enumeration value="A" />
    <xs:enumeration value="B" />
    <xs:enumeration value="C" />
    <xs:enumeration value="None" />
  </xs:restriction>
</xs:simpleType>
</xs:schema>

```

7.2.7. UpdateVehicleLocation

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/v2/UpdateVehicleLocation
Source	WFM
Body	JSON/XML
Type	VehicleLocationV2
Frequency	On demand

WFM uses this URL to send vehicle location updates to the Crew server.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="VehicleLocationV2" nillable="true" type="VehicleLocationV2" />
  <xs:complexType name="VehicleLocationV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="VehicleId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Latitude" type="xs:double" />
          <xs:element minOccurs="1" maxOccurs="1" name="Longitude" type="xs:double" />
          <xs:element minOccurs="1" maxOccurs="1" name="X" type="xs:double" />
          <xs:element minOccurs="1" maxOccurs="1" name="Y" type="xs:double" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>

```

```

    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.8. OrderStatus

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/v2/OrderStatus
Source	WFM
Body	JSON/XML
Type	OrderStatusV2
Frequency	On demand

WFM uses this URL to send follow-up work order updates to the Crew server.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="OrderStatusV2" nillable="true" type="OrderStatusV2" />
  <xs:complexType name="OrderStatusV2">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Comment" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="DeviceName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Aor" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="Organization" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="OrderType" type="xs:int" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

```

<xs:element minOccurs="1" maxOccurs="1" name="Priority" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceStation" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CreatedBy" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="ModifiedBy" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="IncidentName" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="DeleteOrder" type="xs:boolean" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceType" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceFeederName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DevicePhases" type="xs:string" />
</xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.9. PlannedOutageStatus

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/v2/PlannedOutageStatus
Source	WFM
Body	JSON/XML
Type	PlannedOutageStatusV2
Frequency	On demand

WFM uses this URL to send planned outage updates to the Crew server.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="PlannedOutageStatusV2" nillable="true" type="PlannedOutageStatusV2" />
  <xs:complexType name="PlannedOutageStatusV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="PrimaryStartTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="PrimaryEndTime" type="xs:dateTime" />
          <xs:element minOccurs="0" maxOccurs="1" name="WorkOrderId" type="xs:string" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

```

<xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
<xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
<xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.10. AssessmentStatus

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/v2/AssessmentStatus
Source	WFM
Body	JSON/XML
Type	AssessmentStatusV2
Frequency	On demand

WFM uses this URL to send damage assessment location updates to the Crew server.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="AssessmentStatusV2" nillable="true" type="AssessmentStatusV2" />
  <xs:complexType name="AssessmentStatusV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="OrderNumber" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="AssessmentLocations"
type="ArrayOfAssessmentLocation" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>

```

```

    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfAssessmentLocation">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="AssessmentLocation" nillable="true"
type="AssessmentLocation" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="AssessmentLocation">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="Latitude" nillable="true" type="xs:double" />
      <xs:element minOccurs="1" maxOccurs="1" name="Longitude" nillable="true" type="xs:double" />
      <xs:element minOccurs="1" maxOccurs="1" name="XCoordinate" nillable="true" type="xs:double" />
      <xs:element minOccurs="1" maxOccurs="1" name="YCoordinate" nillable="true" type="xs:double" />
      <xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="Radius" nillable="true" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="RepairHours" nillable="true" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="IsTruckAccessible" nillable="true"
type="xs:boolean" />
      <xs:element minOccurs="1" maxOccurs="1" name="TreeTrimmingRequired" nillable="true"
type="xs:int" />
      <xs:element minOccurs="0" maxOccurs="1" name="Comments" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="NumOfDevicesDown" nillable="true" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="NumOfPolesDown" nillable="true" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="NumOfTransformers" nillable="true" type="xs:int"
/>
      <xs:element minOccurs="1" maxOccurs="1" name="ConductorSpan" nillable="true" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="NumOfServicesDown" nillable="true" type="xs:int"
/>
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValueV1" />
    </xs:sequence>
  </xs:complexType>

```

```

</xs:complexType>
<xs:complexType name="ArrayOfPropertyValueV1">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValueV1" nillable="true"
type="PropertyValueV1" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValueV1">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.11. Damages

Characteristics

Message Type	HTTP POST PUT DELETE
URI	api/adms/crew/v2/Damages
Source	WFM
Body	JSON/XML
Type	AddDamagesV2
	UpdateDamagesV2
	DeleteDamagesV2
Frequency	On demand

WFM uses this URL to send messages to add, update, or delete damages to the Crew server.

Message Schema (AddDamagesV2)

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="AddDamagesV2" nillable="true" type="AddDamagesV2" />
  <xs:complexType name="AddDamagesV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Damages" type="ArrayOfAddDamage" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

```

    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfAddDamage">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="AddDamage" nillable="true"
type="AddDamage" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="AddDamage">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="DamageId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="DamageAssessmentLocationId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="AssetType" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="AssetSubType" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="Quantity" type="xs:int" />
    <xs:element minOccurs="0" maxOccurs="1" name="Description" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

Message Schema (UpdateDamagesV2)

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="UpdateDamagesV2" nillable="true" type="UpdateDamagesV2" />
  <xs:complexType name="UpdateDamagesV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="UserName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Damages" type="ArrayOfUpdateDamage" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>

```

```

    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfUpdateDamage">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="UpdateDamage" nillable="true"
type="UpdateDamage" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="UpdateDamage">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="DamageId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="AssetType" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="AssetSubType" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="Quantity" type="xs:int" />
    <xs:element minOccurs="0" maxOccurs="1" name="Description" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

Message Schema (DeleteDamagesV2)

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="DeleteDamagesV2" nillable="true" type="DeleteDamagesV2" />
  <xs:complexType name="DeleteDamagesV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="UserName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Damages" type="ArrayOfString" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>

```

```

    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfString">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="string" nillable="true" type="xs:string"
/>
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.12. CanClearIncident

Characteristics

Message Type	HTTP GET
URI	api/adms/crew/v2/CanClearIncident?incidentId='incident ID'
Source	WFM
Body	JSON/XML
Type (response)	CanClearIncidentResponseV2
Frequency	On demand

WFM uses this URL to check whether an incident can be cleared according to incident clearance rules.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CanClearIncidentResponseV2" nillable="true" type="CanClearIncidentResponseV2" />
  <xs:complexType name="CanClearIncidentResponseV2">
    <xs:complexContent mixed="false">

```

```

    <xs:extension base="MessageBaseV2">
      <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="ValidationMessages" type="ArrayOfString" />
        <xs:element minOccurs="1" maxOccurs="1" name="ValidationStatus" type="ValidationResult" />
        <xs:element minOccurs="0" maxOccurs="1" name="Errors" type="ArrayOfError" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfString">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="string" nillable="true" type="xs:string"
/>
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="ValidationResult">
  <xs:restriction base="xs:string">
    <xs:enumeration value="Error" />
    <xs:enumeration value="Warning" />
    <xs:enumeration value="Success" />
    <xs:enumeration value="Failed" />
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="ArrayOfError">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="Error" nillable="true" type="Error" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="Error">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="Message" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />

```

```
</xs:sequence>
</xs:complexType>
</xs:schema>
```

7.2.13. Status (Crew Server Web Service) (GET)

Characteristics

Message Type	HTTP GET
URI	api/adms/crew/v2/Status
Source	WFM
Body (Response)	JSON/XML
Type (Response)	Status
Frequency	On demand

WFM uses this URL to get the status of the Crew server web service.

For the Status schema, see [Status](#).

7.2.14. Status (Crew Server Web Service) (POST)

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/v2/Status
Source	WFM
Body	JSON/XML
Type (Response)	MessageProcessingUpdate
Frequency	On demand

WFM uses this URL to send the message processing status.

For the MessageProcessingUpdate schema, see [Status \(Call Server\)](#).

7.2.15. addCrew

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/model/v2/addCrew
Source	WFM

Body	JSON/XML
Type	AddCrew (array)
Frequency	On demand

WFM uses this URL to add crews to the ADMS crew model.

Important

The HasMDT field must be set to True for the crew to enable integration with the Crew server web interface and the Workforce Management System.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="AddCrew" nillable="true" type="AddCrew" />
  <xs:complexType name="AddCrew">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="DivisionTypeCode" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="ServiceCenter" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewDivision" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewArea" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Headquarters" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Dept" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsEvaluator" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="DefaultLocationX" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="DefaultLocationY" type="xs:float" />
          <xs:element minOccurs="0" maxOccurs="1" name="EmployeeIdsString" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="PrimaryEmployee" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="FirstName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="LastName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="PhoneNumber" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="RadioNumber" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewTypeId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewTypeName" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="CrewStatusCode" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="HasMDT" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsTemporary" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsForeign" type="xs:boolean" />
          <xs:element minOccurs="0" maxOccurs="1" name="Company" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsContractor" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsAdhoc" type="xs:boolean" />
          <xs:element minOccurs="0" maxOccurs="1" name="CreatedBy" type="xs:string" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>
```

```

<xs:complexType name="MessageBaseV2">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.16. deleteCrew

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/model/v2/deleteCrew
Source	WFM
Body	JSON/XML
Type	DeleteCrew (array)
Frequency	On demand

WFM uses this URL to delete crews from the ADMS crew model.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="DeleteCrew" nillable="true" type="DeleteCrew" />
  <xs:complexType name="DeleteCrew">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>

```

```

</xs:complexType>
<xs:complexType name="MessageBaseV2">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.17. updateCrew

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/model/v2/updateCrew
Source	WFM
Body	JSON/XML
Type	UpdateCrew (array)
Frequency	On demand

WFM uses this URL to update crew data in the ADMS crew model.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="UpdateCrew" nillable="true" type="UpdateCrew" />
  <xs:complexType name="UpdateCrew">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Updates" type="ArrayOfPropertyValue" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>

```

```

    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.18. addEmployee

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/model/v2/addEmployee
Source	WFM
Body	JSON/XML
Type	AddEmployee (array)
Frequency	On demand

WFM uses this URL to add employees to the ADMS crew model.

Important

Set the MDTFlag field to True and populate the AutomaticDispatchId field for primary employees of crews that are equipped with a Mobile Data Terminal (MDT) to enable integration with the Crew server web interface and the Workforce Management System.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
```

```

<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="AddEmployee" nillable="true" type="AddEmployee" />
  <xs:complexType name="AddEmployee">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="EmployeeId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="FirstName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="LastName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="AutomaticDispatchId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="RadioNumber" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="DivisionTypeCode" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="OpCenter" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewDivision" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="DefaultCrewArea" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewArea" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewAreaId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Dept" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="DeptId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="EmployeeTypeId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="EmployeeTypeName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="ShoeSize" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsContractor" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="CapabilityCode" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="CanSwitch" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsAvailable" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="EmployeeStatusCode" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="MDTFlag" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsForeign" type="xs:boolean" />
          <xs:element minOccurs="0" maxOccurs="1" name="SupervisorId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="SupervisorName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="DefaultLeadId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CurrentLeadId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="PhoneNumber" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="SecondaryPhoneNumber" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsAdhoc" type="xs:boolean" />
          <xs:element minOccurs="0" maxOccurs="1" name="CreatedBy" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="EMPHQ" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="MembersCount" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="VehicleIds" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>

```

```

</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.19. deleteEmployee

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/model/v2/deleteEmployee
Source	WFM
Body	JSON/XML
Type	DeleteEmployee (array)
Frequency	On demand

WFM uses this URL to delete employees from the ADMS crew model.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="DeleteEmployee" nillable="true" type="DeleteEmployee" />
  <xs:complexType name="DeleteEmployee">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="EmployeeId" type="xs:string" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>

```

```

    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

7.2.20. updateEmployee

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/model/v2/updateEmployee
Source	WFM
Body	JSON/XML
Type	UpdateEmployee (array)
Frequency	On demand

WFM uses this URL to update employee data in the ADMS crew model.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="UpdateEmployee" nillable="true" type="UpdateEmployee" />
  <xs:complexType name="UpdateEmployee">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="EmployeeId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Updates" type="ArrayOfPropertyValue" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

```

        <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.21. addVehicle

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/model/v2/addVehicle
Source	WFM
Body	JSON/XML
Type	AddVehicle (array)
Frequency	On demand

WFM uses this URL to add vehicles to the ADMS crew model.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
    <xs:element name="AddVehicle" nillable="true" type="AddVehicle" />
    <xs:complexType name="AddVehicle">
        <xs:complexContent mixed="false">
            <xs:extension base="MessageBaseV2">
                <xs:sequence>
                    <xs:element minOccurs="0" maxOccurs="1" name="VehicleId" type="xs:string" />
                    <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
                    <xs:element minOccurs="0" maxOccurs="1" name="VehicleTypeId" type="xs:string" />
                    <xs:element minOccurs="0" maxOccurs="1" name="EmployeeId" type="xs:string" />
                    <xs:element minOccurs="0" maxOccurs="1" name="RadioNumber" type="xs:string" />
                    <xs:element minOccurs="1" maxOccurs="1" name="IsAdhoc" type="xs:boolean" />
                    <xs:element minOccurs="0" maxOccurs="1" name="CreatedBy" type="xs:string" />
                </xs:sequence>
            </xs:extension>
        </xs:complexContent>
    </xs:complexType>
</xs:schema>

```

```

</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.22. deleteVehicle

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/model/v2/deleteVehicle
Source	WFM
Body	JSON/XML
Type	DeleteVehicle (array)
Frequency	On demand

WFM uses this URL to delete vehicles from the ADMS crew model.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="DeleteVehicle" nillable="true" type="DeleteVehicle" />
  <xs:complexType name="DeleteVehicle">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="VehicleId" type="xs:string" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>

```

```

    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.23. updateVehicle

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/model/v2/updateVehicle
Source	WFM
Body	JSON/XML
Type	UpdateVehicle (array)
Frequency	On demand

WFM uses this URL to update vehicle data in the ADMS crew model.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="UpdateVehicle" nillable="true" type="UpdateVehicle" />
  <xs:complexType name="UpdateVehicle">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="VehicleId" type="xs:string" />

```

```

        <xs:element minOccurs="0" maxOccurs="1" name="Updates" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.24. assignCrewEmployees

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/model/v2/assignCrewEmployees
Source	WFM
Body	JSON/XML
Type	AssignEmployee
Frequency	On demand

WFM uses this URL to assign or unassign employees from the crew.

Request Example

```

{
  "CrewId": "FI1507",
  "Action": "Assign",
  "EmployeeIds": [
    {
      "PropertyName": "EmployeeId",

```

```

    "Value": "FI1903"
  }
]
}

```

Supported Action values are Assign or Unassign (case insensitive). PropertyName must contain employee ID, other values are ignored

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="AssignEmployee" nillable="true" type="AssignEmployee" />
  <xs:complexType name="AssignEmployee">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Action" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="EmployeeIds" type="ArrayOfPropertyValue" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

7.2.25. assignCrewVehicles

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/model/v2/assignCrewVehicles
Source	WFM
Body	JSON/XML
Type	AssignVehicle
Frequency	On demand

WFM uses this URL to update vehicle data in the ADMS crew model.

Request Example

```
{
  "CrewId": "FI1507",
  "Action": "Assign",
  "EmployeeId": "FI1903",
  "VehicleIds": [
    {
      "PropertyName": "VehicleId",
      "Value": "VH20001"
    },
    {
      "PropertyName": "VehicleId",
      "Value": "VHXXXX"
    }
  ]
}
```

Supported Action values are Assign or Unassign (case insensitive). PropertyName must contain vehicle ID, other values are ignored

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="AssignVehicle" nillable="true" type="AssignVehicle" />
  <xs:complexType name="AssignVehicle">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Action" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="EmployeeId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="VehicleIds" type="ArrayOfPropertyValue" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

```

<xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
<xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
<xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.26. Status (Crew Server Crew Modeling Web Service)

Characteristics

Message Type	HTTP POST
URI	api/adms/crew/model/v2/Status
Source	WFM
Body	JSON/XML
Type	MessageProcessingUpdate
Frequency	On demand

WFM uses this URL to send the message processing status.

For the MessageProcessingUpdate schema, see [Status \(Call Server\)](#).

7.2.27. ModelingStatus

Characteristics

Message Type	HTTP POST
URI	api/ext/wms/v2/ModelingStatus
Source	Crew server
Body	JSON/XML
Type	MessageProcessingUpdate

Frequency	On demand
------------------	-----------

Crew server uses this URL to send crew modeling status updates to the crew modeling web service client. The URL is specified by the `ResponseBaseUri` and `ResponseStatusMessagesUri` parameters in the `CrewServerConfig.xml` file.

For the `MessageProcessingUpdate` schema, see [Status](#).

7.2.28. WMSMessages

Characteristics

Message Type	HTTP GET
URI	<code>api/ext/wms/v2/WMSMessages?from={from}</code>
Source	Crew server
Body (Response)	JSON/XML
Type (Response)	<code>CrewMessageResponseV2</code>
Frequency	Use the <code>MessageGetPollingInterval</code> parameter to configure the time period for automatically sending these requests
From	Date and time of the latest message from WFM

The Crew server uses this request to get a collection of messages that were added to the message queue since the time specified in the associated request. The Crew server periodically sends these messages to the WFM to retrieve new messages. WFM also returns the timestamp of the latest returned message. The Crew server uses this "up-to" timestamp as the "from" parameter in the next request.

Note

In the `IncidentStatusV2` message, the only required field is "IncidentId"; all other fields are optional, including "StatusCode". The only exception is the "CrewId" field, which is required if the `StatusCode` value is `BackToAssigned` (106), `EnRoute` (4), `OnSite` (5), or `Evaluating` (10). In this case, NetServer validates that the crew ID is valid, and this crew is assigned to the reported incident.

Enhanced Error Handling

ADMS supports enhanced error handling for `CrewMessageResponseV2` messages from WFM when ADMS encounters de-serializing or parsing errors in individual `IncidentStatusV2`, `CrewStatusV2`, `AssignmentQueryV2`, `VehicleLocationV2`, `DeviceStatusUpdateV2`, `PlannedOutageStatusV2`, or `AssessmentStatusV2` messages received as part of a `CrewMessageResponseV2` message. ADMS reports errors in a collection of the `Error` type messages with message ID and error details back to WFM. For example:

```
{
```

```

"Body": {
  "RequestId": "",
  "MessagesReceived": 0,
  "MessagesProcessed": 0,
  "Errors": [
    { "Message": "IncidentStatusMessages: Error converting value \"BackToAssigned1\" to type
      'System.Nullable`1[NetUtil.eOrderStatus]'. ", "MessageId": "8063214.1557941597277509363" }
  ]
}

```

CrewMessageResponseV2 Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CrewMessageResponseV2" nillable="true" type="CrewMessageResponseV2" />
  <xs:complexType name="CrewMessageResponseV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="1" maxOccurs="1" name="MaxResults" nillable="true" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="UpTo" type="xs:dateTime" />
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentStatusMessages"
            type="ArrayOfIncidentStatusV2" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewStatusMessages" type="ArrayOfCrewStatusV2" />
          <xs:element minOccurs="0" maxOccurs="1" name="AssignmentQueryMessages"
            type="ArrayOfAssignmentQueryV2" />
          <xs:element minOccurs="0" maxOccurs="1" name="VehicleLocationMessages"
            type="ArrayOfVehicleLocationV2" />
          <xs:element minOccurs="0" maxOccurs="1" name="DeviceStateUpdateMessages"
            type="ArrayOfDeviceStateUpdateV2" />
          <xs:element minOccurs="0" maxOccurs="1" name="AssessmentStatusMessages"
            type="ArrayOfAssessmentStatusV2" />
          <xs:element minOccurs="0" maxOccurs="1" name="PlannedOutageStatusMessages"
            type="ArrayOfPlannedOutageStatusV2" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
        type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>

```

```

<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PlannedOutageStatusV2">
  <xs:complexContent mixed="false">
    <xs:extension base="MessageBaseV2">
      <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="PrimaryStartTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="PrimaryEndTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="WorkOrderId" type="xs:string" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="AssessmentStatusV2">
  <xs:complexContent mixed="false">
    <xs:extension base="MessageBaseV2">
      <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="OrderNumber" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="AssessmentLocations"
type="ArrayOfAssessmentLocation" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="ArrayOfAssessmentLocation">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="AssessmentLocation" nillable="true"
type="AssessmentLocation" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="AssessmentLocation">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="Latitude" nillable="true" type="xs:double" />
    <xs:element minOccurs="1" maxOccurs="1" name="Longitude" nillable="true" type="xs:double" />
    <xs:element minOccurs="1" maxOccurs="1" name="XCoordinate" nillable="true" type="xs:double" />
    <xs:element minOccurs="1" maxOccurs="1" name="YCoordinate" nillable="true" type="xs:double" />
    <xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
    <xs:element minOccurs="1" maxOccurs="1" name="Radius" nillable="true" type="xs:int" />
    <xs:element minOccurs="1" maxOccurs="1" name="RepairHours" nillable="true" type="xs:int" />
    <xs:element minOccurs="1" maxOccurs="1" name="IsTruckAccessible" nillable="true"
type="xs:boolean" />
    <xs:element minOccurs="1" maxOccurs="1" name="TreeTrimmingRequired" nillable="true"
type="xs:int" />
    <xs:element minOccurs="0" maxOccurs="1" name="Comments" type="xs:string" />
  </xs:sequence>
</xs:complexType>

```

```

        <xs:element minOccurs="1" maxOccurs="1" name="NumOfDevicesDown" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="NumOfPolesDown" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="NumOfTransformers" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="ConductorSpan" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="NumOfServicesDown" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValueV1" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValueV1">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValueV1" nillable="true"
type="PropertyValueV1" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValueV1">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="DeviceStateUpdateV2">
    <xs:complexContent mixed="false">
        <xs:extension base="MessageBaseV2">
            <xs:sequence>
                <xs:element minOccurs="1" maxOccurs="1" name="DeviceId" type="xs:string" />
                <xs:element minOccurs="1" maxOccurs="1" name="DeviceState" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="DeviceStateName" nillable="true"
type="eConfirmDeviceState" />
                <xs:element minOccurs="1" maxOccurs="1" name="Phases" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="PhasesName" nillable="true" type="ePhaseOut"
/>
                <xs:element minOccurs="0" maxOccurs="1" name="UserName" type="xs:string" />
                <xs:element minOccurs="1" maxOccurs="1" name="Station" type="xs:string" />
                <xs:element minOccurs="1" maxOccurs="1" name="Feeder" type="xs:string" />
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:simpleType name="eConfirmDeviceState">
    <xs:restriction base="xs:string">
        <xs:enumeration value="ConfirmOpen" />
        <xs:enumeration value="ConfirmClosed" />
        <xs:enumeration value="FoundOpen" />
    </xs:restriction>
</xs:simpleType>
<xs:simpleType name="ePhaseOut">
    <xs:restriction base="xs:string">
        <xs:enumeration value="ABC" />
        <xs:enumeration value="AB" />
        <xs:enumeration value="BC" />
        <xs:enumeration value="AC" />
    </xs:restriction>
</xs:simpleType>

```

```

        <xs:enumeration value="A" />
        <xs:enumeration value="B" />
        <xs:enumeration value="C" />
        <xs:enumeration value="None" />
    </xs:restriction>
</xs:simpleType>
<xs:complexType name="VehicleLocationV2">
    <xs:complexContent mixed="false">
        <xs:extension base="MessageBaseV2">
            <xs:sequence>
                <xs:element minOccurs="0" maxOccurs="1" name="VehicleId" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
                <xs:element minOccurs="1" maxOccurs="1" name="Latitude" type="xs:double" />
                <xs:element minOccurs="1" maxOccurs="1" name="Longitude" type="xs:double" />
                <xs:element minOccurs="1" maxOccurs="1" name="X" type="xs:double" />
                <xs:element minOccurs="1" maxOccurs="1" name="Y" type="xs:double" />
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:complexType name="AssignmentQueryV2">
    <xs:complexContent mixed="false">
        <xs:extension base="MessageBaseV2">
            <xs:sequence>
                <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
                <xs:element minOccurs="1" maxOccurs="1" name="CustomerResults" nillable="true" type="xs:int"
/>
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:complexType name="CrewStatusV2">
    <xs:complexContent mixed="false">
        <xs:extension base="MessageBaseV2">
            <xs:sequence>
                <xs:element minOccurs="0" maxOccurs="1" name="WorkOrderId" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
                <xs:element minOccurs="1" maxOccurs="1" name="Status" nillable="true" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="StatusName" nillable="true"
type="eWfmCrewStatus" />
                <xs:element minOccurs="1" maxOccurs="1" name="WorkStatus" nillable="true" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="MobileCommunicationStatus" nillable="true"
type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="ShiftStatus" nillable="true" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="AvailabilityStatus" nillable="true"
type="xs:int" />
                <xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="SwitchingOrderId" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="OtherInformation" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="Comments" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="IncidentName" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="Vehicle" type="xs:string" />

```

```

        <xs:element minOccurs="0" maxOccurs="1" name="ErrorNumber" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="ErrorCode" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CrewArea" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="ETA" type="xs:dateTime" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:simpleType name="eWfmCrewStatus">
    <xs:restriction base="xs:string">
        <xs:enumeration value="NotSelected" />
        <xs:enumeration value="Unavailable" />
        <xs:enumeration value="SignOff" />
        <xs:enumeration value="SignOn" />
        <xs:enumeration value="OnBreak" />
        <xs:enumeration value="StandBy" />
        <xs:enumeration value="Dispatched" />
        <xs:enumeration value="EnRoute" />
        <xs:enumeration value="OnSite" />
        <xs:enumeration value="WaitingOnEquipment" />
        <xs:enumeration value="Released" />
        <xs:enumeration value="Unassigned" />
        <xs:enumeration value="Available" />
        <xs:enumeration value="OutOfCoverage" />
        <xs:enumeration value="Acknowledged" />
        <xs:enumeration value="Suspended" />
        <xs:enumeration value="PendingDispatch" />
        <xs:enumeration value="Assigned" />
    </xs:restriction>
</xs:simpleType>
<xs:complexType name="IncidentStatusV2">
    <xs:complexContent mixed="false">
        <xs:extension base="MessageBaseV2">
            <xs:sequence>
                <xs:element minOccurs="1" maxOccurs="1" name="StatusCode" nillable="true" type="xs:int" />
                <xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
                <xs:element minOccurs="1" maxOccurs="1" name="StatusName" nillable="true"
type="eOrderStatus" />
                <xs:element minOccurs="1" maxOccurs="1" name="CrewId" nillable="true" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="ErrorNumber" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="ErrorCode" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="Comment" type="xs:string" />
                <xs:element minOccurs="1" maxOccurs="1" name="ActualRestorationTime" type="xs:dateTime" />
                <xs:element minOccurs="1" maxOccurs="1" name="CauseGroup" nillable="true" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="Cause" nillable="true" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="OutageType" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="OutageTypeName" nillable="true"
type="eOutageType" />
                <xs:element minOccurs="1" maxOccurs="1" name="PhasesOut" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="PhasesOutName" nillable="true"
type="ePhaseOut" />
                <xs:element minOccurs="1" maxOccurs="1" name="FailureType" nillable="true" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="FailureMode" nillable="true" type="xs:int" />
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>

```

```

        <xs:element minOccurs="1" maxOccurs="1" name="FailedEquipment" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="PowerQualityType" nillable="true"
type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="Manufacturer" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="ActionTaken" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="CorrCustCount" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="WfmEtr" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="NoOfPolesDown" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="NoOfTransformers" nillable="true"
type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="NoOfCrossArms" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="NoOfTreeTrimSpans" nillable="true"
type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="ConductorSpan" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="NoServicesDown" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="Inaccessible" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="FollowupRequired" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="FollowupType" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="LineCrewNeeded" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="SystemAffected" nillable="true" type="xs:int"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="TreeCrewNeeded" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="FirstResponderNeeded" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="EvaluatorNeeded" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="0" maxOccurs="1" name="PointOfFailure" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="Weather" type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="WorkOrderId" type="xs:string" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:simpleType name="eOrderStatus">
    <xs:restriction base="xs:string">
        <xs:enumeration value="NotSelected" />
        <xs:enumeration value="NoTroubleFound" />
        <xs:enumeration value="DamageAssessment" />
        <xs:enumeration value="CancelOrder" />
        <xs:enumeration value="Complete" />
        <xs:enumeration value="Patrolled" />
        <xs:enumeration value="PendingAssigned" />
        <xs:enumeration value="BackToAssigned" />
    </xs:restriction>
</xs:simpleType>

```

```

        <xs:enumeration value="ConfirmedOpen" />
        <xs:enumeration value="Cleared" />
        <xs:enumeration value="Restored" />
        <xs:enumeration value="CancelIncident" />
    </xs:restriction>
</xs:simpleType>
<xs:simpleType name="eOutageType">
    <xs:restriction base="xs:string">
        <xs:enumeration value="NotSelected" />
        <xs:enumeration value="None" />
        <xs:enumeration value="CustomerTrouble" />
        <xs:enumeration value="ExternalSource" />
        <xs:enumeration value="Primary" />
        <xs:enumeration value="Secondary" />
        <xs:enumeration value="Service" />
        <xs:enumeration value="Substation" />
        <xs:enumeration value="Transformer" />
        <xs:enumeration value="Transmission" />
    </xs:restriction>
</xs:simpleType>
<xs:complexType name="ArrayOfIncidentStatusV2">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="IncidentStatusV2" nillable="true"
type="IncidentStatusV2" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfCrewStatusV2">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="CrewStatusV2" nillable="true"
type="CrewStatusV2" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfAssignmentQueryV2">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="AssignmentQueryV2" nillable="true"
type="AssignmentQueryV2" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfVehicleLocationV2">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="VehicleLocationV2" nillable="true"
type="VehicleLocationV2" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfDeviceStateUpdateV2">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="DeviceStateUpdateV2" nillable="true"
type="DeviceStateUpdateV2" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfAssessmentStatusV2">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="AssessmentStatusV2" nillable="true"

```

```

type="AssessmentStatusV2" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPlannedOutageStatusV2">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PlannedOutageStatusV2" nillable="true"
type="PlannedOutageStatusV2" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.29. IncidentUpdate

Characteristics

Message Type	HTTP POST
URI	api/ext/wms/v2/IncidentUpdate
Source	Crew server
Body	JSON/XML
Type	IncidentUpdateV2
Frequency	On demand

The Crew server uses this URL to notify WFM of created or updated incidents in the system. The notification message includes all incident properties, not just the updated properties.

Not all incident updates trigger an update notification. The WFM_UPDATE_FIELDS parameter in the NetServerOmsConfig.txt configuration file defines the fields whose update triggers an IncidentUpdateWfm message in the system.

Note

Selecting Line Crew, Tree Crew, or Evaluator in the Order Worksheet or Update Incident UI automatically sets the UpdateTypeName to DamageAssessment. Selecting First Responder does not set UpdateTypeName to DamageAssessment, because First Responders do not perform damage assessment.

Note

When an incident with an MDT crew assigned by WFM rolls up into a new upstream incident, the new upstream incident name is set to the name of the merged downstream incident. For this incident, the Crew server sends an IncidentUpdateV2 message with the UpdateType field set to code 5 (DeviceReprediction) to the WFM.

The IncidentUpdateV2 schema includes two CallType fields:

- The CallType field under IncidentUpdateV2 shows the call type for non-outage incidents with a

call type (set by the operator or extracted from a customer call). This field value is "null" for outage incidents and non-outage incidents without a call type. (Null values are not sent in JSON messages because JSON only formats non-null values.)

- The CallType field under CustomerV2 shows the call type for each customer associated with an incident. This field only appears for outage incidents.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="IncidentUpdateV2" nillable="true" type="IncidentUpdateV2" />
  <xs:complexType name="IncidentUpdateV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="1" maxOccurs="1" name="CaseNoteCode" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="Comments" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CompanyId" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="ConductorSpan" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="CrewAssignedCount" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewType" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewTypeIds" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CrewTypeNames" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="CritCust" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="CritCust2" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="CritCust3" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="CritCust4" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="CritCust5" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="CustomerList" type="ArrayOfCustomerV2" />
          <xs:element minOccurs="1" maxOccurs="1" name="DeviceStatus" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="DeviceType" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="DevId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="DevName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="DevSize" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="DrivingDirections" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="ETR" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="FailedEquipment" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="FdrName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentName" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="IncidentType" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsAssisted" nillable="true" type="xs:boolean" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>
```

```

<xs:element minOccurs="1" maxOccurs="1" name="NoOfTreeTrimSpans" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="NoServicesDown" type="xs:int" />
<xs:element minOccurs="0" maxOccurs="1" name="OpCenter" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="OperatedFdrName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="OperatedSubstationName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="PointOfFailure" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="PredCust" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="TotalPredictedCustomerCount" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Priority" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Ranking" type="xs:int" />
<xs:element minOccurs="0" maxOccurs="1" name="Regions" type="ArrayOfRegionLevel" />
<xs:element minOccurs="1" maxOccurs="1" name="StartTime" type="xs:dateTime" />
<xs:element minOccurs="0" maxOccurs="1" name="SubArea" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="SubstationName" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="SystemAffected" type="xs:int" />
<xs:element minOccurs="0" maxOccurs="1" name="TrblPredPhase" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="UpdateTypeCode" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="UpdateTypeName" nillable="true"
type="eUpdateType" />
<xs:element minOccurs="0" maxOccurs="1" name="WorkOrderId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CrewComments" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="WeatherCode" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="ReasonGroup" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="Reason" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="RestorationTime" type="xs:dateTime" />
<xs:element minOccurs="1" maxOccurs="1" name="CallType" nillable="true" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="HazardousCustomer" type="xs:boolean" />
<xs:element minOccurs="1" maxOccurs="1" name="HasAnUpstreamReclosingDisabledDevice"
type="xs:boolean" />
<xs:element minOccurs="1" maxOccurs="1" name="PredictedFaultLocationX" nillable="true"
type="xs:double" />
<xs:element minOccurs="1" maxOccurs="1" name="PredictedFaultLocationY" nillable="true"
type="xs:double" />
<xs:element minOccurs="1" maxOccurs="1" name="AreaSettingCode" nillable="true" type="xs:int"
/>
<xs:element minOccurs="0" maxOccurs="1" name="AreaSettingText" type="xs:string" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
<xs:sequence>
<xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
<xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
<xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
<xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
<xs:sequence>
<xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />

```

```

    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfCustomerV2">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="CustomerV2" nillable="true"
type="CustomerV2" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="CustomerV2">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="Address" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="CallType" type="xs:int" />
      <xs:element minOccurs="0" maxOccurs="1" name="City" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="Critical" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="Critical2" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="Critical3" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="Critical4" type="xs:int" />
      <xs:element minOccurs="1" maxOccurs="1" name="Critical5" type="xs:int" />
      <xs:element minOccurs="0" maxOccurs="1" name="HazardousCondition" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="Latitude" type="xs:double" />
      <xs:element minOccurs="1" maxOccurs="1" name="Longitude" type="xs:double" />
      <xs:element minOccurs="0" maxOccurs="1" name="MeterNumber" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="PhoneNumber" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="PhoneCall" type="xs:boolean" />
      <xs:element minOccurs="1" maxOccurs="1" name="Priority" type="xs:int" />
      <xs:element minOccurs="0" maxOccurs="1" name="Remarks" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="State" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="SwitchName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="TfName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="TroubleCode" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="WantsCallback" type="xs:boolean" />
      <xs:element minOccurs="0" maxOccurs="1" name="Zipcode" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="AccountNumber" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="DrivingDirections" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="CustomerKey" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="CallReferenceId" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PointD">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="X" type="xs:double" />
      <xs:element minOccurs="1" maxOccurs="1" name="Y" type="xs:double" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfRegionLevel">
    <xs:sequence>

```

```

    <xs:element minOccurs="0" maxOccurs="unbounded" name="RegionLevel" nillable="true"
type="RegionLevel" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="RegionLevel">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="RegionType" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="RegionName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="RegionID" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:simpleType name="eUpdateType">
  <xs:restriction base="xs:string">
    <xs:enumeration value="None" />
    <xs:enumeration value="DispatchCrew" />
    <xs:enumeration value="AssignWithAssist" />
    <xs:enumeration value="Unassign" />
    <xs:enumeration value="CancelIncident" />
    <xs:enumeration value="DeviceReprediction" />
    <xs:enumeration value="DamageAssessment" />
    <xs:enumeration value="ClearIncident" />
    <xs:enumeration value="AssignCrew" />
  </xs:restriction>
</xs:simpleType>
</xs:schema>

```

7.2.30. CrewUpdate

Characteristics

Message Type	HTTP POST
URI	api/ext/wms/v2/CrewUpdate
Source	Crew server
Body	JSON/XML
Type	CrewUpdateV2
Frequency	On demand

The Crew server uses this URL to notify WFM of crew updates that occurred in the system.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CrewUpdateV2" nillable="true" type="CrewUpdateV2" />
  <xs:complexType name="CrewUpdateV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />

```

```

        <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="UpdateType" nillable="true" type="eUpdateType"
/>

        <xs:element minOccurs="1" maxOccurs="1" name="UpdateTypeCode" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="CrewType" nillable="true" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsAssisted" nillable="true" type="xs:boolean"
/>

        <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:simpleType name="eUpdateType">
    <xs:restriction base="xs:string">
        <xs:enumeration value="None" />
        <xs:enumeration value="DispatchCrew" />
        <xs:enumeration value="AssignWithAssist" />
        <xs:enumeration value="Unassign" />
        <xs:enumeration value="CancelIncident" />
        <xs:enumeration value="DeviceReprediction" />
        <xs:enumeration value="DamageAssessment" />
        <xs:enumeration value="ClearIncident" />
        <xs:enumeration value="AssignCrew" />
    </xs:restriction>
</xs:simpleType>
</xs:schema>

```

7.2.31. OrderUpdate

Characteristics

Message Type	HTTP POST
URI	api/ext/wms/v2/OrderUpdate
Source	Crew server
Body	JSON/XML
Type	OrderUpdateV2
Frequency	On demand

The Crew server uses this URL to notify WFM of follow-up work updates that occurred in the system.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="OrderUpdateV2" nillable="true" type="OrderUpdateV2" />
  <xs:complexType name="OrderUpdateV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Comment" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="DeviceName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="DeviceType" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="DeviceFeederName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="DevicePhases" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Aor" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Organization" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="OrderType" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="Priority" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="DeviceStation" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CreatedBy" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="ModifiedBy" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="IncidentName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CustomerAddress" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Created" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="Modified" type="xs:dateTime" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

```

</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.32. PlannedOutageUpdate

Characteristics

Message Type	HTTP POST
URI	api/ext/wms/v2/PlannedOutageUpdate
Source	Crew server
Body	JSON/XML
Type	PlannedOutageUpdateV2
Frequency	On demand

The Crew server uses this URL to notify WFM of planned outage updates that occurred in the system.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="PlannedOutageUpdateV2" nillable="true" type="PlannedOutageUpdateV2" />
  <xs:complexType name="PlannedOutageUpdateV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="AOR" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Status" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="CreatedTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="PrimaryStartTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="PrimaryEndTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="AlternateStartTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="AlternateEndTime" type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="UseAlternateDate" type="xs:boolean" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>

```

```

        <xs:element minOccurs="0" maxOccurs="1" name="EventType" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CustomerAccountsString" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CustomerAccountsNotifyString" type="xs:string"
/>
        <xs:element minOccurs="0" maxOccurs="1" name="AddedCustomersString" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="AddedCustomersNotifyString" type="xs:string"
/>
        <xs:element minOccurs="0" maxOccurs="1" name="RemovedCustomersString" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="RemovedCustomersNotifyString" type="xs:string"
/>
        <xs:element minOccurs="0" maxOccurs="1" name="Station" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="RepresentativeName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="RepresentativeNumber" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="RepresentativeEmail" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="RepresentativeAddress" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="OutageMessage" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CreatorComments" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="Modified" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="FeederName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="WorkOrderId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="PlannedOutageType" type="xs:int" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.33. SynchronizeCrews

Characteristics

Message Type	HTTP POST
URI	api/ext/wms/v2/SynchronizeCrews
Source	Crew server
Body	JSON/XML
Type	SynchCrewV2
Frequency	On demand

The Crew server uses this URL to notify Mobility Gateway of crew modeling updates that occurred in the system. Mobility Gateway will notify Mobile Enterprise of these updates.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="MobileCrew" nillable="true" type="MobileCrew" />
  <xs:complexType name="MobileCrew">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="BaseLocation" type="GeoLocation" />
      <xs:element minOccurs="0" maxOccurs="1" name="ContactInfo" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="CrewMembers" type="ArrayOfCrewMember" />
      <xs:element minOccurs="0" maxOccurs="1" name="Vehicles" type="ArrayOfVehicle" />
      <xs:element minOccurs="1" maxOccurs="1" name="IsMobileDispatchable" nillable="true"
type="xs:boolean" />
      <xs:element minOccurs="1" maxOccurs="1" name="IsTemporary" nillable="true" type="xs:boolean" />
      <xs:element minOccurs="0" maxOccurs="1" name="Type" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Radio" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="VehicleId" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="OperatingArea" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="OpCenter" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="OperatingAreaId" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Company" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Dept" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="CrewDivision" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="DefaultCrewArea" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="AutomaticDispatchId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="IsContractor" nillable="true" type="xs:boolean" />
      <xs:element minOccurs="1" maxOccurs="1" name="IsEvaluator" nillable="true" type="xs:boolean" />
      <xs:element minOccurs="1" maxOccurs="1" name="ActiveEmployeesCount" nillable="true"
type="xs:int" />
      <xs:element minOccurs="0" maxOccurs="1" name="CustomFields" type="ArrayOfCustomField" />
      <xs:element minOccurs="0" maxOccurs="1" name="TempArea" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="PrimaryEmployeeName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="PrimaryFirstName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="PrimaryLastName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="EmployeeIds" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:complexType name="GeoLocation">
```

```

<xs:sequence>
  <xs:element minOccurs="0" maxOccurs="1" name="ProjectionType" type="xs:string" />
  <xs:element minOccurs="1" maxOccurs="1" name="X" nillable="true" type="xs:double" />
  <xs:element minOccurs="1" maxOccurs="1" name="Y" nillable="true" type="xs:double" />
  <xs:element minOccurs="1" maxOccurs="1" name="Zone" nillable="true" type="xs:int" />
</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfCrewMember">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="CrewMember" nillable="true"
type="CrewMember" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="CrewMember">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="IsLeader" nillable="true" type="xs:boolean" />
    <xs:element minOccurs="0" maxOccurs="1" name="Employee" type="CrewMemberEmployee" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="CrewMemberEmployee">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="EmployeeId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="FirstName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="LastName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="RadioNumber" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="IsEvaluator" nillable="true" type="xs:boolean" />
    <xs:element minOccurs="1" maxOccurs="1" name="IsPrimary" nillable="true" type="xs:boolean" />
    <xs:element minOccurs="1" maxOccurs="1" name="MdtFlag" nillable="true" type="xs:boolean" />
    <xs:element minOccurs="0" maxOccurs="1" name="PhoneNumber" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="EmployeeTypeName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="DefaultLeadName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="CrewArea" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="CrewDivision" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="DefaultCrewArea" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="AutomaticDispatchId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="IsContractor" nillable="true" type="xs:boolean" />
    <xs:element minOccurs="1" maxOccurs="1" name="IsForeign" nillable="true" type="xs:boolean" />
    <xs:element minOccurs="0" maxOccurs="1" name="Company" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="TempCrewName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="EmployeeTypeId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="CapabilityCode" nillable="true" type="xs:int" />
    <xs:element minOccurs="0" maxOccurs="1" name="CrewAreaId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="SupervisorId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="DepartmentId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="SecondaryPhoneNumber" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="VehicleIds" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="UserId" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfVehicle">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="Vehicle" nillable="true" type="Vehicle" />
  </xs:sequence>
</xs:complexType>

```

```

    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="Vehicle">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="Id" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Type" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="HasMobileDataTerminal" nillable="true"
type="xs:boolean" />
      <xs:element minOccurs="1" maxOccurs="1" name="HasGPS" nillable="true" type="xs:boolean" />
      <xs:element minOccurs="0" maxOccurs="1" name="EmployeeId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="ReportedTime" nillable="true" type="xs:dateTime"
/>
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfCustomField">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="CustomField" nillable="true"
type="CustomField" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="CustomField">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

7.2.34. Status (Crew Server)

Characteristics

Message Type	HTTP POST
URI	api/ext/wms/v2/Status
Source	Crew server
Body	JSON/XML
Type	MessageProcessingUpdate
Frequency	Triggered by CrewMessageResponseV2

The Crew server uses this URL to provide a list of errors to the external application after receiving a batch of messages. If no errors are encountered, the Status message contains only the number of messages processed.

For the MessageProcessingUpdate schema, see [Status](#).

7.2.35. WMSModelMessages

Characteristics

Message Type	HTTP GET
URI	api/ext/wms/v2/WMSModelMessages?from={from}
Source	Crew server
Body (Response)	JSON/XML
Type (Response)	CrewModelingMessageResponseV2
Frequency	Use the MessageGetPollingInterval parameter to configure the time period for automatically sending these requests
From	Date and time of the latest message from WFM

The Crew server uses this request to get a collection of model update messages that were added to the message queue since the time specified in the associated request. The Crew server periodically sends these messages to the WFM to retrieve new messages. WFM also returns the timestamp of the latest returned message. The Crew server uses this "up-to" timestamp as the "from" parameter in the next request.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CrewModelingMessageResponseV2" nillable="true"
type="CrewModelingMessageResponseV2" />
  <xs:complexType name="CrewModelingMessageResponseV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="1" maxOccurs="1" name="MaxResults" type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="UpTo" type="xs:dateTime" />
          <xs:element minOccurs="0" maxOccurs="1" name="AddEmployeeMessages" type="ArrayOfAddEmployee"
/>
          <xs:element minOccurs="0" maxOccurs="1" name="UpdateEmployeeMessages"
type="ArrayOfUpdateEmployee" />
          <xs:element minOccurs="0" maxOccurs="1" name="DeleteEmployeeMessages"
type="ArrayOfDeleteEmployee" />
          <xs:element minOccurs="0" maxOccurs="1" name="AddVehicleMessages" type="ArrayOfAddVehicle"
/>
          <xs:element minOccurs="0" maxOccurs="1" name="UpdateVehicleMessages"
type="ArrayOfUpdateVehicle" />
          <xs:element minOccurs="0" maxOccurs="1" name="DeleteVehicleMessages"
type="ArrayOfDeleteVehicle" />
          <xs:element minOccurs="0" maxOccurs="1" name="AddCrewMessages" type="ArrayOfAddCrew" />
          <xs:element minOccurs="0" maxOccurs="1" name="UpdateCrewMessages" type="ArrayOfUpdateCrew"
/>
          <xs:element minOccurs="0" maxOccurs="1" name="DeleteCrewMessages" type="ArrayOfDeleteCrew"
/>
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>
```

```

    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="DeleteCrew">
  <xs:complexContent mixed="false">
    <xs:extension base="MessageBaseV2">
      <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="UpdateCrew">
  <xs:complexContent mixed="false">
    <xs:extension base="MessageBaseV2">
      <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Updates" type="ArrayOfPropertyValue" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="AddCrew">
  <xs:complexContent mixed="false">
    <xs:extension base="MessageBaseV2">
      <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="DivisionTypeCode" type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="ServiceCenter" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CrewDivision" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CrewArea" type="xs:string" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>

```

```

        <xs:element minOccurs="0" maxOccurs="1" name="Headquarters" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Dept" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsEvaluator" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="DefaultLocationX" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="DefaultLocationY" type="xs:float" />
        <xs:element minOccurs="0" maxOccurs="1" name="EmployeeIdsString" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="PrimaryEmployee" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="FirstName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="LastName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="PhoneNumber" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="RadioNumber" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CrewTypeId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CrewTypeName" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="CrewStatusCode" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="HasMDT" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsTemporary" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsForeign" type="xs:boolean" />
        <xs:element minOccurs="0" maxOccurs="1" name="Company" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsContractor" type="xs:boolean" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="DeleteVehicle">
    <xs:complexContent mixed="false">
        <xs:extension base="MessageBaseV2">
            <xs:sequence>
                <xs:element minOccurs="0" maxOccurs="1" name="VehicleId" type="xs:string" />
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:complexType name="UpdateVehicle">
    <xs:complexContent mixed="false">
        <xs:extension base="MessageBaseV2">
            <xs:sequence>
                <xs:element minOccurs="0" maxOccurs="1" name="VehicleId" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="Updates" type="ArrayOfPropertyValue" />
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:complexType name="AddVehicle">
    <xs:complexContent mixed="false">
        <xs:extension base="MessageBaseV2">
            <xs:sequence>
                <xs:element minOccurs="0" maxOccurs="1" name="VehicleId" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="VehicleTypeId" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="EmployeeId" type="xs:string" />
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>

```

```

</xs:complexType>
<xs:complexType name="DeleteEmployee">
  <xs:complexContent mixed="false">
    <xs:extension base="MessageBaseV2">
      <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="EmployeeId" type="xs:string" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="UpdateEmployee">
  <xs:complexContent mixed="false">
    <xs:extension base="MessageBaseV2">
      <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="EmployeeId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Updates" type="ArrayOfPropertyValue" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="AddEmployee">
  <xs:complexContent mixed="false">
    <xs:extension base="MessageBaseV2">
      <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="EmployeeId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="FirstName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="LastName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="AutomaticDispatchId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="RadioNumber" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="DivisionTypeCode" type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="OpCenter" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CrewDivision" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="DefaultCrewArea" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CrewArea" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CrewAreaId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Dept" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="DeptId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="EmployeeTypeId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="EmployeeTypeName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="ShoeSize" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsContractor" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="CapabilityCode" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="CanSwitch" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsAvailable" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="EmployeeStatusCode" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="MDTFlag" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsForeign" type="xs:boolean" />
        <xs:element minOccurs="0" maxOccurs="1" name="SupervisorId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="SupervisorName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="DefaultLeadId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CurrentLeadId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="PhoneNumber" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="SecondaryPhoneNumber" type="xs:string" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>

```

```

        <xs:element minOccurs="0" maxOccurs="1" name="EMPHQ" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="MembersCount" type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="VehicleIds" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CrewId" type="xs:string" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="ArrayOfAddEmployee">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="AddEmployee" nillable="true"
type="AddEmployee" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfUpdateEmployee">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="UpdateEmployee" nillable="true"
type="UpdateEmployee" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfDeleteEmployee">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="DeleteEmployee" nillable="true"
type="DeleteEmployee" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfAddVehicle">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="AddVehicle" nillable="true"
type="AddVehicle" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfUpdateVehicle">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="UpdateVehicle" nillable="true"
type="UpdateVehicle" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfDeleteVehicle">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="DeleteVehicle" nillable="true"
type="DeleteVehicle" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfAddCrew">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="AddCrew" nillable="true" type="AddCrew" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfUpdateCrew">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="UpdateCrew" nillable="true"
type="UpdateCrew" />
    </xs:sequence>
</xs:complexType>

```

```

</xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfDeleteCrew">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="DeleteCrew" nillable="true"
type="DeleteCrew" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

7.2.36. ModelMessageStatus

Characteristics

Message Type	HTTP POST
URI	api/ext/wms/v2/ModelMessageStatus
Source	Crew server
Body	JSON/XML
Type	MessageProcessingUpdate
Frequency	Triggered by CrewModelingMessageResponseV2

Crew server uses this URL to provide a list of errors to the external application after receiving a batch of messages. If no errors are encountered, the ModelMessageStatus message contains only the number of messages processed.

For the MessageProcessingUpdate schema, see [Status](#).

8. Call Server Web Interface

ADMS Call server supports communicating with external interfaces (CIS, IVR, and AMI interfaces) via HTTP instead of legacy TCP. When HTTP communication is enabled, interface messages are exchanged via web services.

To enable the Call server web services, set `PluginDefinition> Enabled` to `True` in the Call server configuration file (`CallServerConfig.xml`). The web service is loaded by the Call server using the Plugin API, which is a framework that hosts web service servers and clients.

The Call server web interface supports exchanging call, case note, and customer data between ADMS and any external interfaces.

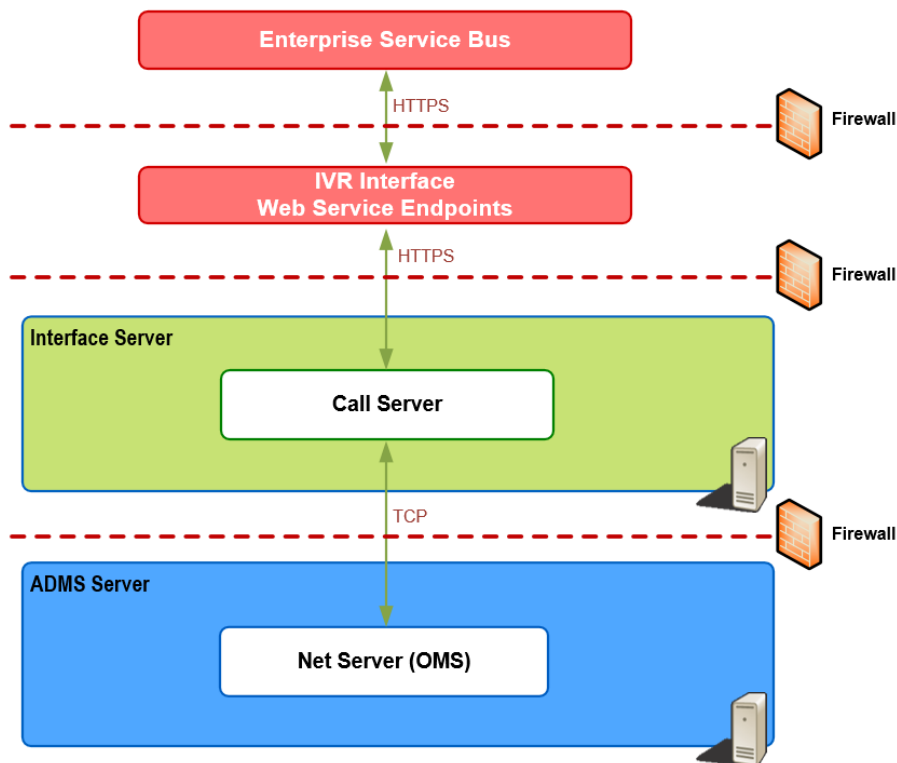


Figure 28. ADMS-IVR Data Flow

For more details about web service security, see [Security](#).

8.1. Service Types

The Call server web interface is configured in the `CallServerConfig.xml` configuration file. For web services, you can configure host, port, and root values. For web service clients, you can also configure endpoints.

Each Call server web interface URL has the following format:

```
<Host:Port><Root><Endpoint>
```

For example:

`http://localhost:8443/api/adms/call/v2/Calls`

The following are default Call server web interface URLs:

- **/api/adms/call/v2/**: This is the root of the Call server web service.
 - **Calls**: IVR uses this URL to get calls entered by operators from ADMS (GET).
 - **Calls**: IVR uses this URL to send calls to ADMS (POST).
 - **CaseNotes**: IVR uses this URL to get casenote data from ADMS (GET).
 - **Customers**: IVR uses this URL to get customer model data from ADMS (GET).
 - **AreaSettings**: IVR uses this URL to get or send area settings data to ADMS (GET, POST).
 - **Status**: IVR uses this URL to get the status of the Call server web service (GET).
- **/api/ext/call/v2/**: This is the root of the Call server web service client.
 - **CallMessages**: The Call server uses this URL to return all pending messages from IVR (GET). This request does the following:
 - Returns pending call messages from IVR.
 - Returns pending case note query messages from IVR.
 - Returns pending customer data query messages from IVR.
 - **CallNotification**: The Call server uses this URL to send calls entered by operators to IVR (POST).
 - **Callbacks**: The Call server uses this URL to send callback requests to IVR (POST).
 - **CaseNoteResponse**: The Call server uses this URL to send casenote responses to IVR (POST) (Future functionality).
 - **CaseNotes**: The Call server uses this URL to send casenote data to IVR (POST).
 - **CustomerMove**: The Call server uses this URL to send a customer move notification to a notification system, such as ANPS (POST).
 - **CustomerResponse**: The Call server uses this URL to send customer model data to IVR (POST).
 - **Status**: The Call server uses this URL to provide a message processing status to IVR (POST).
- **/api/adms/call/logger/v2**: This is the root of the Call server OmsLogger web service.

8.1.1. Calls (GET)

Characteristics

Message Type	HTTP GET
URI	<code>api/adms/call/v2/Calls?from={from}</code>
Source	IVR

Body	JSON/XML
Type (Response)	CallQueryResponseV2
Frequency	On demand
From	Date and time

IVR uses this URL to get calls entered by operators from ADMS.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CallQueryResponseV2" nillable="true" type="CallQueryResponseV2" />
  <xs:complexType name="CallQueryResponseV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="1" maxOccurs="1" name="UpTo" nillable="true" type="xs:dateTime" />
          <xs:element minOccurs="0" maxOccurs="1" name="Calls" type="ArrayOfCallV2" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="CallV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="MeterNumber" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Address" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsForwardedCall" type="xs:boolean" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>
```

```

<xs:element minOccurs="0" maxOccurs="1" name="ReferenceId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="AccountId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CustomerKey" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="IsCancelledCallbackRequest" type="xs:boolean"
/>

<xs:element minOccurs="1" maxOccurs="1" name="CallTime" type="xs:dateTime" />
<xs:element minOccurs="0" maxOccurs="1" name="Region_1" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="Region_2" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CallbackNumber" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="Comments" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="CallType" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Condition_1" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Condition_2" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Description" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="CallDuration" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="PriorityCode" type="xs:int" />
<xs:element minOccurs="0" maxOccurs="1" name="FeederId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="IsCancelledCall" type="xs:boolean" />
<xs:element minOccurs="1" maxOccurs="1" name="WantsCallback" type="xs:boolean" />
<xs:element minOccurs="1" maxOccurs="1" name="OriginalCallTime" type="xs:dateTime" />
<xs:element minOccurs="0" maxOccurs="1" name="LatestCallbackTime" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="X" type="xs:double" />
<xs:element minOccurs="1" maxOccurs="1" name="Y" type="xs:double" />
<xs:element minOccurs="1" maxOccurs="1" name="Latitude" type="xs:float" />
<xs:element minOccurs="1" maxOccurs="1" name="Longitude" type="xs:float" />
<xs:element minOccurs="1" maxOccurs="1" name="IsFromCallback" type="xs:boolean" />
<xs:element minOccurs="0" maxOccurs="1" name="CallbackName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DrivingDirections" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="AgentId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="Source" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="TroubleCode" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="Unassociated" type="xs:boolean" />
<xs:element minOccurs="0" maxOccurs="1" name="PhoneNumber" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="UserName" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="PowerIsOn" type="xs:boolean" />
<xs:element minOccurs="1" maxOccurs="1" name="KeepNonOutageSeparate" type="xs:boolean" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="ArrayOfCallV2">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="CallV2" nillable="true" type="CallV2" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

8.1.2. Calls (POST)

Characteristics

Message Type	HTTP POST
URI	api/adms/call/v2/Calls
Source	IVR
Body	JSON/XML
Type	CallV2
Frequency	On demand

IVR uses this URL to send calls to ADMS.

You can send a call for a customer using CustomerKey ##or AccountId. Using the CustomerKey is recommended, so that a call for a customer with multiple meters is associated with the correct meter. When ADMS receives a call from IVR using POST /Calls, ADMS primarily uses CustomerKey to associate the call with a customer. If CustomerKey is not specified, ADMS uses AccountId to associate the call with a customer.

Note

This URL is supported but not certified, because the performance was not verified. Using the "api/ext/call/v2/CallMessages" message is recommended.

LatestCallbackTime Field Validation

The LatestCallbackTime field in the CallV2 schema is validated based on the WantsCallback field:

- When WantsCallback is set to true, the LatestCallbackTime field is required and must be provided in one of the following formats:
 - hh:mm:ss (for example, "14:30:00")
 - hh:mm (for example, "14:30")
- When WantsCallback is set to false, validation of the LatestCallbackTime field is skipped.

Note

Ensure that WantsCallback is set to true when providing a LatestCallbackTime value. If WantsCallback is false, the field is not validated and may result in integration issues.

Request Example

http://localhost:8443/api/adms/call/v2/Calls

```
[
{
  "CustomerKey": "555360577555360577555360577",
  "CallType": 1,
  "CallTime": "2021-07-27T12:52:00.0000000+02:00"
}
]
```

Or:

```
[
{
"AccountId": "5553605775",
"CallType": 1,
"CallTime": "2021-07-27T12:52:00.0000000+02:00"
}
]
```

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CallV2" nillable="true" type="CallV2" />
  <xs:complexType name="CallV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="MeterNumber" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Address" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsForwardedCall" type="xs:boolean" />
          <xs:element minOccurs="0" maxOccurs="1" name="ReferenceId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="AccountId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CustomerKey" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsCancelledCallbackRequest" type="xs:boolean" />
        </xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="CallTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="Region_1" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Region_2" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="CallbackNumber" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Comments" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="CallType" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="Condition_1" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="Condition_2" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="Description" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="CallDuration" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="PriorityCode" type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="FeederId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsCancelledCall" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="WantsCallback" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="OriginalCallTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="LatestCallbackTime" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="X" type="xs:double" />
        <xs:element minOccurs="1" maxOccurs="1" name="Y" type="xs:double" />
        <xs:element minOccurs="1" maxOccurs="1" name="Latitude" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="Longitude" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsFromCallback" type="xs:boolean" />
        <xs:element minOccurs="0" maxOccurs="1" name="CallbackName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="DrivingDirections" type="xs:string" />
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>
```

```

        <xs:element minOccurs="0" maxOccurs="1" name="AgentId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Source" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="TroubleCode" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="Unassociated" type="xs:boolean" />
        <xs:element minOccurs="0" maxOccurs="1" name="PhoneNumber" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="UserName" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="PowerIsOn" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="KeepNonOutageSeparate" type="xs:boolean" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
</xs:schema>

```

8.1.3. CaseNotes

Characteristics 1

Message Type	HTTP GET
URI	api/adms/call/v2/CaseNotes?queryType={queryType}&name={EntityName_or_CustomerKey_or_AccountNumber}&from={from}
Source	IVR
Body (Response)	JSON/XML
Type (Response)	CaseNoteResponseV2
Frequency	On demand

queryType	1 = Global 2 = ServiceCenter 3 = ServicePoint 4 = Customer
name	Area or service point name
from	Date and time

IVR uses this URL to get casenote data for a management area hierarchy entity or a customer from ADMS.

Characteristics 2

Message Type	HTTP GET
URI	api/adms/call/v2/CaseNotes
Source	IVR
Body	JSON/XML
Type (Response)	CaseNoteResponseV2
Frequency	On demand

IVR uses this URL to get casenote data for all management area hierarchy entities from ADMS.

Request Example

http://localhost:8443/api/adms/call/v2/CaseNotes?queryType=4&name=555254932555254932555254932

Response Example (in JSON format)

```
{
  "QueryIdEcho": "",
  "UpTo": "0001-01-01T00:00:00.0010000",
  "CaseNotes": [
    {
      "OpCenterCode": "WSTC",
      "FeederId": "F5865",
      "DeviceId": "SP_10543614",
      "DeviceName": "6-7707-4823-0-9",
      "AreaId": "GE",
      "Name": "555254932555254932555254932",
      "TroubleAssignedTo": "",
      "ETR": "2021-07-27T23:59:00.0000000-07:00",
      "IsDeleted": false,
      "IsUpdated": false,
      "LastUpdateTime": "2021-07-27T02:11:24.0313604-07:00",
      "Remarks": "",
      "LastUpdateByUserId": "ADMS",
      "CaseNoteNumber": 301,
      "CaseNoteMessage": "Outage Recorded",
      "SystemCode": 0,
    }
  ]
}
```

```

        "CauseCode": 0,
        "IsAccessible": true,
        "SuppressNotification": false,
        "SuppressReason": 0,
        "SystemMode": 0,
        "StormName": "Test Area Setting",
        "StartTime": "0001-01-01T00:00:00.0000000",
        "ITRDisable": false,
        "OverrideCaseNote": true,
        "UseFRP": false,
        "RestorationUrgency": false
    }
],
"Sequence": 12510,
"MessageId": "cdb3f5ed-a3a4-4e4b-97bb-d77262ca790e",
"EventTime": "0001-01-01T00:00:00.0010000",
"MessageTime": "2021-07-27T02:13:04.0036207-07:00"
}

```

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CaseNoteResponseV2" nillable="true" type="CaseNoteResponseV2" />
  <xs:complexType name="CaseNoteResponseV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="QueryIdEcho" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="UpTo" nillable="true" type="xs:dateTime" />
          <xs:element minOccurs="0" maxOccurs="1" name="CaseNotes" type="ArrayOfCaseNoteV2" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />

```

```

        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfCaseNoteV2">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="CaseNoteV2" nillable="true"
type="CaseNoteV2" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="CaseNoteV2">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="OpCenterCode" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="FeederId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="DeviceName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="AreaId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="TroubleAssignedTo" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="ETR" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsDeleted" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsUpdated" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="LastUpdateTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="Remarks" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="LastUpdateByUserId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="CaseNoteNumber" type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="CaseNoteMessage" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="SystemCode" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="CauseCode" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsAccessible" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="SuppressNotification" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="SuppressReason" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="SystemMode" type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="StormName" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="StartTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="ITRDisable" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="OverrideCaseNote" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="UseFRP" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="RestorationUrgency" type="xs:boolean" />
    </xs:sequence>
</xs:complexType>
</xs:schema>

```

8.1.4. Customers

Characteristics

Message Type	HTTP GET
URI	api/adms/call/v2/Customers?select={select}&where={where}
Source	IVR
Body	JSON/XML

Type	CustomerQueryResponseV2
Type (Response)	MessageProcessingUpdate
Frequency	On demand
select	Customer properties to return separated by comma (",")
where	Query parameters separated by "%"

IVR uses this URL to get customer model data from ADMS.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CustomerQueryResponseV2" nillable="true" type="CustomerQueryResponseV2" />
  <xs:complexType name="CustomerQueryResponseV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="RequestId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Customers" type="ArrayOfCallCustomerV2" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfCallCustomerV2">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="CallCustomerV2" nillable="true"
type="CallCustomerV2" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="CallCustomerV2">
```

```

<xs:sequence>
  <xs:element minOccurs="1" maxOccurs="1" name="CustIdx" nillable="true" type="xs:int" />
  <xs:element minOccurs="0" maxOccurs="1" name="CustomerKey" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="ServicePoint" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="MeterNumber" type="xs:string" />
  <xs:element minOccurs="1" maxOccurs="1" name="AMROutageReport" nillable="true" type="xs:boolean"
/>
  <xs:element minOccurs="1" maxOccurs="1" name="AMRPowerOnReport" nillable="true"
type="xs:boolean" />
  <xs:element minOccurs="1" maxOccurs="1" name="PhoneCall" nillable="true" type="xs:boolean" />
  <xs:element minOccurs="1" maxOccurs="1" name="PriorityCall" nillable="true" type="xs:boolean" />
  <xs:element minOccurs="1" maxOccurs="1" name="X" nillable="true" type="xs:float" />
  <xs:element minOccurs="1" maxOccurs="1" name="Y" nillable="true" type="xs:float" />
  <xs:element minOccurs="1" maxOccurs="1" name="CallType" nillable="true" type="xs:int" />
  <xs:element minOccurs="1" maxOccurs="1" name="Condition_1" nillable="true" type="xs:int" />
  <xs:element minOccurs="1" maxOccurs="1" name="Condition_2" nillable="true" type="xs:int" />
  <xs:element minOccurs="1" maxOccurs="1" name="Description" nillable="true" type="xs:int" />
  <xs:element minOccurs="1" maxOccurs="1" name="PriorityCode" nillable="true" type="xs:int" />
  <xs:element minOccurs="0" maxOccurs="1" name="AccountNumber" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Address" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="AreaCode" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Comment" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="CriticalAccount" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="CriticalAccount2" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="CriticalAccount3" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="CriticalAccount4" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="CriticalAccount5" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="FirstName" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="LastName" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="CustomerName" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="PhonePrefix" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="PhoneSuffix" type="xs:string" />
  <xs:element minOccurs="1" maxOccurs="1" name="LastAMRReportTime" nillable="true"
type="xs:dateTime" />
  <xs:element minOccurs="1" maxOccurs="1" name="LastCallTime" nillable="true" type="xs:dateTime"
/>
  <xs:element minOccurs="1" maxOccurs="1" name="EnergizedTime" nillable="true" type="xs:dateTime"
/>
  <xs:element minOccurs="1" maxOccurs="1" name="Modified" nillable="true" type="xs:dateTime" />
  <xs:element minOccurs="0" maxOccurs="1" name="OperatingCenter" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Division" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Column1" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Column2" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Column3" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Column4" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Column5" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Column6" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Column7" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Column8" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Column9" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Column10" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Column11" type="xs:string" />
</xs:sequence>

```

```
</xs:complexType>
</xs:schema>
```

8.1.5. AreaSettings

Characteristics

Message Type	HTTP GET
	HTTP POST
URI	api/adms/call/v2/AreaSettings
Source	IVR
Body	JSON/XML
Type (Response)	AreaSettingsDataV2
Frequency	On demand

IVR uses this URL to get or send area settings data to ADMS.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="AreaSettingDataV2" nillable="true" type="AreaSettingDataV2" />
  <xs:complexType name="AreaSettingDataV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="AreaName" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="SystemMode" nillable="true" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="CaseNoteType" nillable="true" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="SuppressNotification" nillable="true"
type="xs:boolean" />
          <xs:element minOccurs="1" maxOccurs="1" name="ITRDisabled" nillable="true" type="xs:boolean"
/>
          <xs:element minOccurs="1" maxOccurs="1" name="SuppressReason" nillable="true" type="xs:int"
/>
          <xs:element minOccurs="0" maxOccurs="1" name="StormName" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="StartTime" nillable="true" type="xs:dateTime"
/>
          <xs:element minOccurs="1" maxOccurs="1" name="ETR" nillable="true" type="xs:dateTime" />
          <xs:element minOccurs="0" maxOccurs="1" name="CaseNoteRemark" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="ExcludeCMI" nillable="true" type="xs:boolean"
/>
          <xs:element minOccurs="1" maxOccurs="1" name="PSTDeltaKey" nillable="true" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="ExpirationTime" nillable="true"
type="xs:dateTime" />
          <xs:element minOccurs="1" maxOccurs="1" name="DisableAutoRestorationPings" nillable="true"
type="xs:boolean" />
          <xs:element minOccurs="0" maxOccurs="1" name="ReferenceId" type="xs:string" />

```

```

        <xs:element minOccurs="1" maxOccurs="1" name="UseFRP" nillable="true" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="RestorationUrgency" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="0" maxOccurs="1" name="InternalNote" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="CustomerFeedbackOption" nillable="true"
type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="CustomFields" type="ArrayOfPropertyValue" />
        <xs:element minOccurs="1" maxOccurs="1" name="DisableAutoCrewAssignment" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="DisableAutoCrewTransfer" nillable="true"
type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="AutoClearMomentaries" nillable="true"
type="xs:boolean" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
</xs:schema>

```

8.1.6. Status (Call Server Web Service)

Characteristics

Message Type	HTTP GET
URI	api/adms/call/v2/Status
Source	IVR
Body	JSON/XML
Type	Status

IVR uses this URL to get the status of the Call server web service.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="Status" nillable="true" type="Status" />
  <xs:complexType name="Status">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="ProcessName" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="StartTime" type="xs:dateTime" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

8.1.7. CallMessages

Characteristics

Message Type	HTTP GET
URI	api/ext/call/v2/CallMessages?from={from}
Source	Call server

Body (Response)	JSON/XML
Type (Response)	CallMessageResponseV2
Frequency	Use the CallMessagesInterval parameter to configure the time period for automatically sending these requests
From	Date and time of the latest message from IVR

The Call server uses this request to get all queued messages from IVR. IVR responds with CallMessageResponseV2, which contains the requested call data, queries for case note and customer data, and the "up-to" timestamp for future queries.

Call data may include CustomerKey, AccountId, or both values. When ADMS receives a call from IVR using GET /CallMessages, ADMS primarily uses CustomerKey to associate the call with a customer. If CustomerKey is not specified, ADMS uses AccountId to associate the call with a customer.

Message Schema (Response)

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CallMessageResponseV2" nillable="true" type="CallMessageResponseV2" />
  <xs:complexType name="CallMessageResponseV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Calls" type="ArrayOfCallV2" />
          <xs:element minOccurs="0" maxOccurs="1" name="CaseNoteQueries" type="ArrayOfCaseNoteQueryV2" />
          <xs:element minOccurs="0" maxOccurs="1" name="CustomerQueries" type="ArrayOfCustomerQueryV2" />
          <xs:element minOccurs="1" maxOccurs="1" name="UpTo" nillable="true" type="xs:dateTime" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true" type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

```

        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="CustomerQueryV2">
    <xs:complexContent mixed="false">
        <xs:extension base="MessageBaseV2">
            <xs:sequence>
                <xs:element minOccurs="0" maxOccurs="1" name="Where" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="Select" type="xs:string" />
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:complexType name="CaseNoteQueryV2">
    <xs:complexContent mixed="false">
        <xs:extension base="MessageBaseV2">
            <xs:sequence>
                <xs:element minOccurs="1" maxOccurs="1" name="FromTimeStamp" nillable="true"
type="xs:dateTime" />
                <xs:element minOccurs="1" maxOccurs="1" name="QueryTypeCode" type="xs:int" />
                <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>
<xs:complexType name="CallV2">
    <xs:complexContent mixed="false">
        <xs:extension base="MessageBaseV2">
            <xs:sequence>
                <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="MeterNumber" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="Address" type="xs:string" />
                <xs:element minOccurs="1" maxOccurs="1" name="IsForwardedCall" type="xs:boolean" />
                <xs:element minOccurs="0" maxOccurs="1" name="ReferenceId" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="AccountId" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="CustomerKey" type="xs:string" />
                <xs:element minOccurs="1" maxOccurs="1" name="IsCancelledCallbackRequest" type="xs:boolean"
/>
                <xs:element minOccurs="1" maxOccurs="1" name="CallTime" type="xs:dateTime" />
                <xs:element minOccurs="0" maxOccurs="1" name="Region_1" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="Region_2" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="CallbackNumber" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="Comments" type="xs:string" />
                <xs:element minOccurs="1" maxOccurs="1" name="CallType" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="Condition_1" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="Condition_2" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="Description" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="CallDuration" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="PriorityCode" type="xs:int" />
                <xs:element minOccurs="0" maxOccurs="1" name="FeederId" type="xs:string" />
                <xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
                <xs:element minOccurs="1" maxOccurs="1" name="IsCancelledCall" type="xs:boolean" />
                <xs:element minOccurs="1" maxOccurs="1" name="WantsCallback" type="xs:boolean" />
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>

```

```

    <xs:element minOccurs="1" maxOccurs="1" name="OriginalCallTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="LatestCallbackTime" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="X" type="xs:double" />
    <xs:element minOccurs="1" maxOccurs="1" name="Y" type="xs:double" />
    <xs:element minOccurs="1" maxOccurs="1" name="Latitude" type="xs:float" />
    <xs:element minOccurs="1" maxOccurs="1" name="Longitude" type="xs:float" />
    <xs:element minOccurs="1" maxOccurs="1" name="IsFromCallback" type="xs:boolean" />
    <xs:element minOccurs="0" maxOccurs="1" name="CallbackName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="DrivingDirections" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="AgentId" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Source" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="TroubleCode" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="Unassociated" type="xs:boolean" />
    <xs:element minOccurs="0" maxOccurs="1" name="PhoneNumber" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="UserName" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="PowerIsOn" type="xs:boolean" />
  </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="ArrayOfCallV2">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="CallV2" nillable="true" type="CallV2" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfCaseNoteQueryV2">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="CaseNoteQueryV2" nillable="true"
type="CaseNoteQueryV2" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfCustomerQueryV2">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="CustomerQueryV2" nillable="true"
type="CustomerQueryV2" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

8.1.8. CallNotification

Characteristics

Message Type	HTTP POST
URI	api/ext/call/v2/CallNotification
Source	Call server
Body	JSON/XML
Type	CallNotificationV2
Frequency	On Demand

The Call server uses this URL to send calls entered by operators to the IVR.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CallNotificationV2" nillable="true" type="CallNotificationV2" />
  <xs:complexType name="CallNotificationV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Calls" type="ArrayOfCallV2" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="CallV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="MeterNumber" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Address" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsForwardedCall" type="xs:boolean" />
          <xs:element minOccurs="0" maxOccurs="1" name="ReferenceId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="AccountId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CustomerKey" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="IsCancelledCallbackRequest" type="xs:boolean" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:element minOccurs="1" maxOccurs="1" name="CallTime" type="xs:dateTime" />
  <xs:element minOccurs="0" maxOccurs="1" name="Region_1" type="xs:string" />
  <xs:element minOccurs="0" maxOccurs="1" name="Region_2" type="xs:string" />
</xs:schema>
```

```

<xs:element minOccurs="0" maxOccurs="1" name="CallbackNumber" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="Comments" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="CallType" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Condition_1" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Condition_2" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="Description" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="CallDuration" type="xs:int" />
<xs:element minOccurs="1" maxOccurs="1" name="PriorityCode" type="xs:int" />
<xs:element minOccurs="0" maxOccurs="1" name="FeederId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="IsCancelledCall" type="xs:boolean" />
<xs:element minOccurs="1" maxOccurs="1" name="WantsCallback" type="xs:boolean" />
<xs:element minOccurs="1" maxOccurs="1" name="OriginalCallTime" type="xs:dateTime" />
<xs:element minOccurs="0" maxOccurs="1" name="LatestCallbackTime" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="X" type="xs:double" />
<xs:element minOccurs="1" maxOccurs="1" name="Y" type="xs:double" />
<xs:element minOccurs="1" maxOccurs="1" name="Latitude" type="xs:float" />
<xs:element minOccurs="1" maxOccurs="1" name="Longitude" type="xs:float" />
<xs:element minOccurs="1" maxOccurs="1" name="IsFromCallback" type="xs:boolean" />
<xs:element minOccurs="0" maxOccurs="1" name="CallbackName" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="DrivingDirections" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="AgentId" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="Source" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="TroubleCode" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="Unassociated" type="xs:boolean" />
<xs:element minOccurs="0" maxOccurs="1" name="PhoneNumber" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="UserName" type="xs:string" />
<xs:element minOccurs="1" maxOccurs="1" name="PowerIsOn" type="xs:boolean" />
</xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="ArrayOfCallV2">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="CallV2" nillable="true" type="CallV2" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

8.1.9. Callbacks

Characteristics

Message Type	HTTP POST
URI	api/ext/call/v2/Callbacks
Source	Call server
Body	JSON/XML
Type	CallbackRequestV2
Frequency	On demand

The Call server uses this URL to send callback requests to IVR.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CallbackRequestV2" nillable="true" type="CallbackRequestV2" />
  <xs:complexType name="CallbackRequestV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="CallbackRequests" type="ArrayOfCallbackV2" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfCallbackV2">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="CallbackV2" nillable="true"
type="CallbackV2" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="CallbackV2">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="ReferenceId" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="AccountId" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="OperatingCenterCode" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="CompanyCode" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="CallbackName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="CallbackPhoneNumber" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="IsCaseNoteChange" type="xs:boolean" />
      <xs:element minOccurs="0" maxOccurs="1" name="CaseNoteMessage" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

```

<xs:element minOccurs="1" maxOccurs="1" name="ETR" type="xs:dateTime" />
<xs:element minOccurs="1" maxOccurs="1" name="LastCallTime" type="xs:dateTime" />
<xs:element minOccurs="0" maxOccurs="1" name="TroubleCode" type="xs:string" />
<xs:element minOccurs="0" maxOccurs="1" name="CustomerKey" type="xs:string" />
</xs:sequence>
</xs:complexType>
</xs:schema>

```

8.1.10. CaseNoteResponse

Characteristics

Message Type	HTTP POST
URI	api/ext/call/v2/CaseNoteResponse
Source	Call server
Body	JSON/XML
Type	TBD
Frequency	On demand

The Call server uses this URL to send casenote responses to IVR (Future functionality).

8.1.11. CaseNotes

Characteristics

Message Type	HTTP POST
URI	api/ext/call/v2/CaseNotes
Source	Call server
Body	JSON/XML
Type	CaseNotesNotificationV2
Frequency	On demand

The Call server uses this URL to send casenote data to IVR.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CaseNotesNotificationV2" nillable="true" type="CaseNotesNotificationV2" />
  <xs:complexType name="CaseNotesNotificationV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="CaseNotes" type="ArrayOfCaseNoteV2" />

```

```

        </xs:sequence>
    </xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfCaseNoteV2">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="CaseNoteV2" nillable="true"
type="CaseNoteV2" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="CaseNoteV2">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="OpCenterCode" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="FeederId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="DeviceId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="DeviceName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="AreaId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Name" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="TroubleAssignedTo" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="ETR" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsDeleted" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsUpdated" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="LastUpdateTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="Remarks" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="LastUpdateByUserId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="CaseNoteNumber" type="xs:int" />
        <xs:element minOccurs="0" maxOccurs="1" name="CaseNoteMessage" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="SystemCode" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="CauseCode" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="IsAccessible" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="SuppressNotification" type="xs:boolean" />
        <xs:element minOccurs="1" maxOccurs="1" name="SuppressReason" type="xs:int" />
    </xs:sequence>
</xs:complexType>

```

```

    <xs:element minOccurs="1" maxOccurs="1" name="SystemMode" type="xs:int" />
    <xs:element minOccurs="0" maxOccurs="1" name="StormName" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="StartTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="ITRDisable" type="xs:boolean" />
    <xs:element minOccurs="1" maxOccurs="1" name="OverrideCaseNote" type="xs:boolean" />
    <xs:element minOccurs="1" maxOccurs="1" name="UseFRP" type="xs:boolean" />
    <xs:element minOccurs="1" maxOccurs="1" name="RestorationUrgency" type="xs:boolean" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

8.1.12. CustomerMove

Characteristics

Message Type	HTTP POST
URI	api/ext/call/v2/CustomerMove
Source	Call server
Body	JSON/XML
Type	CustomerMoveV2
Frequency	After a CustomerCorrectionCivV1 request

The Call server uses this URL to send a customer move notification to a notification system, such as ANPS.

Message Schema

```

<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="CustomerMoveV2" nillable="true" type="CustomerMoveV2" />
  <xs:complexType name="CustomerMoveV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="PremiseNumber" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="AccountNumber" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CustomerKey" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="MeterNumber" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="XfrmerName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="XfrmerId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="FdrName" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="CompanyId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="ServicePointID" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="ServiceDrop" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="XCoordinate" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="YCoordinate" type="xs:string" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>

```

```

</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

8.1.13. CustomerResponse

Characteristics

Message Type	HTTP POST
URI	api/ext/call/v2/CustomerResponse
Source	Call server
Body	JSON/XML
Type	CustomerQueryResponseV2
Frequency	On demand

The Call server uses this URL to send customer model data to IVR.

For the CustomerQueryResponseV2 schema, see [Customers](#).

8.1.14. Status (Call Server)

Characteristics

Message Type	HTTP POST
URI	api/ext/call/v2/Status

Source	Call server
Body	JSON/XML
Type	MessageProcessingUpdate
Frequency	Triggered by CallMessageResponseV2

The Call server uses this URL to provide a list of errors to IVR in the form of a MessageProcessingUpdate after receiving a batch of messages. If no errors are encountered, the MessageProcessingUpdate message contains only the number of messages processed.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="MessageProcessingUpdate" nillable="true" type="MessageProcessingUpdate" />
  <xs:complexType name="MessageProcessingUpdate">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="RequestId" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="MessagesReceived" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="MessagesProcessed" type="xs:int" />
          <xs:element minOccurs="0" maxOccurs="1" name="Errors" type="ArrayOfError" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfError">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="Error" nillable="true" type="Error" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

```
</xs:complexType>
<xs:complexType name="Error">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="Message" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>
```

9. AMI Server Web Interface

ADMS AMI server supports communicating with an external AMI interface via HTTP instead of legacy TCP. When HTTP communication is enabled, interface messages are exchanged via web services. The AMI web service is loaded by the AMI server using the Plugin API.

The AMI server web interface supports exchanging meter data between ADMS and an external AMI interface.

Note

The AMI server is essentially the Call server, which runs with the AmiServerConfig.xml configuration file.

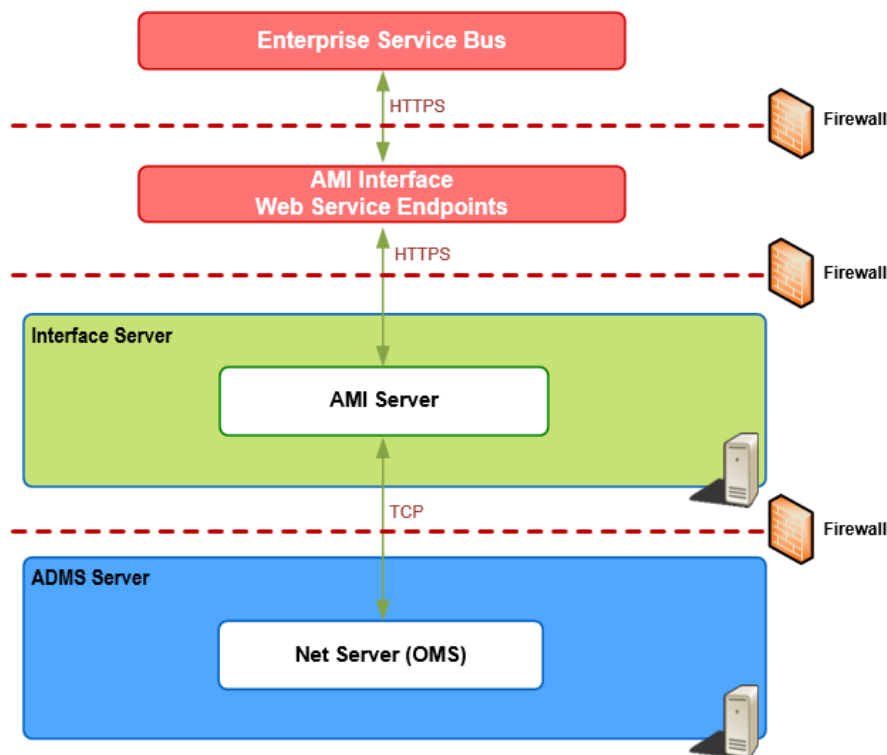


Figure 29. ADMS-IVR Data Flow

The external AMI interface periodically queries for power outage and power restoration messages from ADMS, calling the web service endpoint.

The external AMI interface posts messages entered by ADMS operators for Meter Ping and Meter Voltage Ping Requests to the web service endpoint. The external AMI interface writes response message to the ESB. The external AMI interface periodically queries the AMI endpoint for a Ping response and Voltage Ping Responses.

For more details about the web service security, see [Security](#).

9.1. Service Types

The AMI server web interface is configured in the `AmiServerConfig.xml` configuration file. For web services, you can configure host, port, and root values. For web service clients, you can also configure endpoints.

Each AMI server web interface URL has the following format:

`<Host:Port><Root><Endpoint>`

For example:

`http://localhost:8445/api/ext/ami/v2/AmiMessages`

The following are default AMI server web interface URLs:

- **/api/adms/ami/v2/**: This is the root of the AMI server web service.
 - **reading**: External applications use this message to send meter readings (voltage, power, and current) to the AMI server web service (POST).
 - **readings**: External applications use this message to send an array of meter readings (voltage, power, and current) to the AMI server web service (POST).
 - **event**: External applications use this message to send meter events (power on/off) to the AMI server web service (POST).
 - **event**: External applications use this message to send an array of meter events (power on/off) to the AMI server web service (POST).
 - **voltEvent**: External applications use this message to send meter voltage exception events (voltage sag/swell) to the AMI server web service (POST).
 - **voltEvents**: External applications use this message to send an array of meter voltage exception events (voltage sag/swell) to the AMI server web service (POST).
 - **ping**: External applications use this message to send meter pings to the AMI server web service (POST).
 - **Status**: External applications use this message to get the status of the AMI server web service (GET).
- **/api/ext/ami/v2/**: This is the root of the AMI server web service client.
 - **AmiMessages**: The AMI server uses this URL to return all pending messages from the external application (GET). This request does the following:
 - Returns pending MeterStatus messages
 - Returns pending MeterVoltStatus messages
 - Returns pending MeterEvent messages (power on/off)
 - Returns pending MeterVoltEvent messages (voltage sag/swell)
 - **Ping**: The AMI server uses this URL to send meter ping data to the external application (POST).
 - **VoltPing**: The AMI server uses this URL to send meter volt ping data to the external

application (POST).

- **Control/Disconnect:** The AMI server uses this URL to request disconnecting the specified meters (POST).
- **Control/Reconnect:** The AMI server uses this URL to request reconnecting the specified meters (POST).
- **Read:** The AMI server uses this URL to request data for the specified meters (POST). This endpoint is deprecated. Use Ping or VoltPing instead.
- **Status:** The AMI server uses this URL to send the message processing status to the external application (POST).

9.1.1. Reading

Characteristics

Message Type	HTTP POST
URI	api/adms/ami/v2/reading
Source	External application
Body (Response)	JSON/XML
Type (Response)	MeterVoltStatusV2
Frequency	On demand

External applications use this message to send meter readings (voltage, power, and current) to the AMI server web service.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="MeterVoltStatusV2" nillable="true" type="MeterVoltStatusV2" />
  <xs:complexType name="MeterVoltStatusV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="RequestId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="MeterId" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="VoltA" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="VoltB" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="VoltC" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="VarA" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="VarB" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="VarC" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="WattA" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="WattB" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="WattC" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="AmpA" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="AmpB" nillable="true" type="xs:float" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>
```

```

        <xs:element minOccurs="1" maxOccurs="1" name="AmpC" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="LastCommunicationTime" nillable="true"
type="xs:dateTime" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
</xs:schema>

```

9.1.2. Readings

Characteristics

Message Type	HTTP POST
URI	api/adms/ami/v2/readings
Source	External application
Body (Response)	JSON/XML
Type (Response)	MeterVoltStatusV2[]
Frequency	On demand

External applications use this message to send an array of meter readings to the AMI server web service.

For the MeterVoltStatusV2 schema, see [Reading](#).

9.1.3. Event

Characteristics

Message Type	HTTP POST
URI	api/adms/ami/v2/event
Source	External application
Body (Response)	JSON/XML
Type (Response)	MeterEventV2
Frequency	On demand

External applications use this message to send meter events (power on/off) to the AMI server web service.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="MeterEventV2" nillable="true" type="MeterEventV2" />
  <xs:complexType name="MeterEventV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="1" maxOccurs="1" name="TimeReported" type="xs:dateTime" />
          <xs:element minOccurs="0" maxOccurs="1" name="MeterId" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="Event" nillable="true" type="eMeterEvent" />
          <xs:element minOccurs="1" maxOccurs="1" name="EventCode" type="xs:int" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

```

    </xs:sequence>
  </xs:complexType>
  <xs:simpleType name="eMeterEvent">
    <xs:restriction base="xs:string">
      <xs:enumeration value="PowerOut" />
      <xs:enumeration value="PowerIn" />
      <xs:enumeration value="Nominal" />
    </xs:restriction>
  </xs:simpleType>
</xs:schema>

```

9.1.4. Events

Characteristics

Message Type	HTTP POST
URI	api/adms/ami/v2/events
Source	External application
Body (Response)	JSON/XML
Type (Response)	MeterEventV2[]
Frequency	On demand

External applications use this message to send an array of meter events (power on/off) to the AMI server web service.

For the MeterEventV2 schema, see [Event](#).

9.1.5. VoltEvent

Characteristics

Message Type	HTTP POST
URI	api/adms/ami/v2/voltEvent
Source	External application
Body (Response)	JSON/XML
Type (Response)	MeterVoltEventV2
Frequency	On demand

External applications use this message to send meter voltage exception events (voltage sag/swell) to the AMI server web service.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
```

```

<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="MeterVoltEventV2" nillable="true" type="MeterVoltEventV2" />
  <xs:complexType name="MeterVoltEventV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="1" maxOccurs="1" name="QueryTimeStamp" type="xs:dateTime" />
          <xs:element minOccurs="0" maxOccurs="1" name="MeterId" type="xs:string" />
          <xs:element minOccurs="1" maxOccurs="1" name="EventType" type="xs:int" />
          <xs:element minOccurs="1" maxOccurs="1" name="VoltA" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="VoltB" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="VoltC" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="VarA" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="VarB" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="VarC" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="WattA" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="WattB" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="WattC" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="AmpA" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="AmpB" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="AmpC" nillable="true" type="xs:float" />
          <xs:element minOccurs="1" maxOccurs="1" name="TimeReported" type="xs:dateTime" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
</xs:schema>

```

9.1.6. VoltEvents

Characteristics

Message Type	HTTP POST
URI	api/adms/ami/v2/voltEvents
Source	External application
Body (Response)	JSON/XML
Type (Response)	MeterVoltEventV2[]
Frequency	On demand

External applications use this message to send an array of meter voltage exception events (voltage sag/swell) to the AMI server web service.

For the MeterVoltEventV2 schema, see [VoltEvent](#).

9.1.7. Ping

Characteristics

Message Type	HTTP POST
URI	api/adms/ami/v2/ping
Source	External application
Body	JSON/XML
Type	MeterPingResponseV2
Frequency	On demand

External applications use this message to send meter pings to the AMI server web service.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="MeterPingResponseV2" nillable="true" type="MeterPingResponseV2" />
  <xs:complexType name="MeterPingResponseV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="PingRequestId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="MeterStatuses" type="ArrayOfMeterStatusV2" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
```

```

    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="MeterStatusV2">
  <xs:complexContent mixed="false">
    <xs:extension base="MessageBaseV2">
      <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="RequestId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="MeterId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="Status" nillable="true" type="eAmiQueryResult"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="StatusCode" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="LastCommunicationTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="VoltA" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="VoltB" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="VoltC" nillable="true" type="xs:float" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:simpleType name="eAmiQueryResult">
  <xs:restriction base="xs:string">
    <xs:enumeration value="On" />
    <xs:enumeration value="Undetermined" />
    <xs:enumeration value="NotTried" />
    <xs:enumeration value="CommunicationError" />
    <xs:enumeration value="InvalidMeter" />
    <xs:enumeration value="Out" />
    <xs:enumeration value="Expired" />
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="ArrayOfMeterStatusV2">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="MeterStatusV2" nillable="true"
type="MeterStatusV2" />
  </xs:sequence>
</xs:complexType>

```

```
</xs:schema>
```

9.1.8. Status

Characteristics

Message Type	HTTP GET
URI	api/adms/ami/v2/Status
Source	External application
Body (Response)	JSON/XML
Type (Response)	Status
Frequency	On demand

External applications use this message to get the status of the AMI server web service.

For the Status schema, see [Status](#).

9.1.9. AmiMessages

Characteristics

Message Type	HTTP GET
URI	api/ext/ami/v2/AmiMessages?from={from}
Source	AMI server
Body (Response)	JSON/XML
Type (Response)	AmiMessageResponseV2
Frequency	Configurable time period
From	Date and time of the latest message from the external application

The AMI server uses Message Query Request and Message Query Response messages to get queued messages from the external application.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="AmiMessageResponseV2" nillable="true" type="AmiMessageResponseV2" />
  <xs:complexType name="AmiMessageResponseV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="PingResponses" type="ArrayOfMeterStatusV2" />
          <xs:element minOccurs="0" maxOccurs="1" name="VoltPingResponses"
type="ArrayOfMeterVoltStatusV2" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
</xs:schema>
```

```

        <xs:element minOccurs="0" maxOccurs="1" name="MeterEvents" type="ArrayOfMeterEventV2" />
        <xs:element minOccurs="0" maxOccurs="1" name="MeterVoltEvents"
type="ArrayOfMeterVoltEventV2" />
        <xs:element minOccurs="1" maxOccurs="1" name="UpTo" nillable="true" type="xs:dateTime" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:complexType name="MessageBaseV2">
    <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
        <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
        <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="MeterVoltEventV2">
    <xs:complexContent mixed="false">
        <xs:extension base="MessageBaseV2">
            <xs:sequence>
                <xs:element minOccurs="1" maxOccurs="1" name="QueryTimeStamp" type="xs:dateTime" />
                <xs:element minOccurs="0" maxOccurs="1" name="MeterId" type="xs:string" />
                <xs:element minOccurs="1" maxOccurs="1" name="EventType" type="xs:int" />
                <xs:element minOccurs="1" maxOccurs="1" name="VoltA" nillable="true" type="xs:float" />
                <xs:element minOccurs="1" maxOccurs="1" name="VoltB" nillable="true" type="xs:float" />
                <xs:element minOccurs="1" maxOccurs="1" name="VoltC" nillable="true" type="xs:float" />
                <xs:element minOccurs="1" maxOccurs="1" name="VarA" nillable="true" type="xs:float" />
                <xs:element minOccurs="1" maxOccurs="1" name="VarB" nillable="true" type="xs:float" />
                <xs:element minOccurs="1" maxOccurs="1" name="VarC" nillable="true" type="xs:float" />
                <xs:element minOccurs="1" maxOccurs="1" name="WattA" nillable="true" type="xs:float" />
                <xs:element minOccurs="1" maxOccurs="1" name="WattB" nillable="true" type="xs:float" />
                <xs:element minOccurs="1" maxOccurs="1" name="WattC" nillable="true" type="xs:float" />
                <xs:element minOccurs="1" maxOccurs="1" name="AmpA" nillable="true" type="xs:float" />
                <xs:element minOccurs="1" maxOccurs="1" name="AmpB" nillable="true" type="xs:float" />
                <xs:element minOccurs="1" maxOccurs="1" name="AmpC" nillable="true" type="xs:float" />
                <xs:element minOccurs="1" maxOccurs="1" name="TimeReported" type="xs:dateTime" />
            </xs:sequence>
        </xs:extension>
    </xs:complexContent>
</xs:complexType>

```

```

<xs:complexType name="MeterEventV2">
  <xs:complexContent mixed="false">
    <xs:extension base="MessageBaseV2">
      <xs:sequence>
        <xs:element minOccurs="1" maxOccurs="1" name="TimeReported" type="xs:dateTime" />
        <xs:element minOccurs="0" maxOccurs="1" name="MeterId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="Event" nillable="true" type="eMeterEvent" />
        <xs:element minOccurs="1" maxOccurs="1" name="EventCode" type="xs:int" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:simpleType name="eMeterEvent">
  <xs:restriction base="xs:string">
    <xs:enumeration value="PowerOut" />
    <xs:enumeration value="PowerIn" />
    <xs:enumeration value="Nominal" />
  </xs:restriction>
</xs:simpleType>
<xs:complexType name="MeterVoltStatusV2">
  <xs:complexContent mixed="false">
    <xs:extension base="MessageBaseV2">
      <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="RequestId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="MeterId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="VoltA" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="VoltB" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="VoltC" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="VarA" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="VarB" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="VarC" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="WattA" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="WattB" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="WattC" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="AmpA" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="AmpB" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="AmpC" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="LastCommunicationTime" nillable="true"
type="xs:dateTime" />
      </xs:sequence>
    </xs:extension>
  </xs:complexContent>
</xs:complexType>
<xs:complexType name="MeterStatusV2">
  <xs:complexContent mixed="false">
    <xs:extension base="MessageBaseV2">
      <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="1" name="RequestId" type="xs:string" />
        <xs:element minOccurs="0" maxOccurs="1" name="MeterId" type="xs:string" />
        <xs:element minOccurs="1" maxOccurs="1" name="Status" nillable="true" type="eAmiQueryResult"
/>
        <xs:element minOccurs="1" maxOccurs="1" name="StatusCode" type="xs:int" />
        <xs:element minOccurs="1" maxOccurs="1" name="LastCommunicationTime" type="xs:dateTime" />

```

```

        <xs:element minOccurs="1" maxOccurs="1" name="VoltA" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="VoltB" nillable="true" type="xs:float" />
        <xs:element minOccurs="1" maxOccurs="1" name="VoltC" nillable="true" type="xs:float" />
    </xs:sequence>
</xs:extension>
</xs:complexContent>
</xs:complexType>
<xs:simpleType name="eAmiQueryResult">
    <xs:restriction base="xs:string">
        <xs:enumeration value="On" />
        <xs:enumeration value="Undetermined" />
        <xs:enumeration value="NotTried" />
        <xs:enumeration value="CommunicationError" />
        <xs:enumeration value="InvalidMeter" />
        <xs:enumeration value="Out" />
        <xs:enumeration value="Expired" />
    </xs:restriction>
</xs:simpleType>
<xs:complexType name="ArrayOfMeterStatusV2">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="MeterStatusV2" nillable="true"
type="MeterStatusV2" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfMeterVoltStatusV2">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="MeterVoltStatusV2" nillable="true"
type="MeterVoltStatusV2" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfMeterEventV2">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="MeterEventV2" nillable="true"
type="MeterEventV2" />
    </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfMeterVoltEventV2">
    <xs:sequence>
        <xs:element minOccurs="0" maxOccurs="unbounded" name="MeterVoltEventV2" nillable="true"
type="MeterVoltEventV2" />
    </xs:sequence>
</xs:complexType>
</xs:schema>

```

9.1.10. Ping

Characteristics

Message Type	HTTP POST
URI	api/ext/ami/v2/Ping

Source	AMI server
Body	JSON/XML
Type	MeterPingRequestV2
Frequency	On demand

The AMI server uses this URL to send ping requests to the external application.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="MeterPingRequestV2" nillable="true" type="MeterPingRequestV2" />
  <xs:complexType name="MeterPingRequestV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="MeterIds" type="ArrayOfString" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
  <xs:complexType name="MessageBaseV2">
    <xs:sequence>
      <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
      <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
      <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
      <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
      <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfPropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="PropertyValue">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
      <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
    </xs:sequence>
  </xs:complexType>
  <xs:complexType name="ArrayOfString">
    <xs:sequence>
      <xs:element minOccurs="0" maxOccurs="unbounded" name="string" nillable="true" type="xs:string"
/>
    </xs:sequence>
  </xs:complexType>
</xs:schema>
```

9.1.11. VoltPing

Characteristics

Message Type	HTTP POST
URI	api/ext/ami/v2/VoltPing
Source	AMI server
Body	JSON/XML
Type	MeterPingRequestV2
Frequency	On demand

The AMI server uses this URL to send voltage ping requests to the external application.

For the MeterPingRequestV2 schema, see [Ping](#).

9.1.12. Control

Characteristics

Message Type	HTTP POST
URI	api/ext/ami/v2/Control/Disconnect api/ext/ami/v2/Control/Reconnect
Source	AMI server
Body	JSON/XML
Type	MeterControlRequestV2
Frequency	On demand

The AMI server uses this URL to request disconnecting or reconnecting the specified meters.

Message Schema

```
<?xml version="1.0" encoding="utf-8"?>
<xs:schema elementFormDefault="qualified" xmlns:xs="http://www.w3.org/2001/XMLSchema">
  <xs:element name="MeterControlRequestV2" nillable="true" type="MeterControlRequestV2" />
  <xs:complexType name="MeterControlRequestV2">
    <xs:complexContent mixed="false">
      <xs:extension base="MessageBaseV2">
        <xs:sequence>
          <xs:element minOccurs="0" maxOccurs="1" name="Action" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="MeterId" type="xs:string" />
          <xs:element minOccurs="0" maxOccurs="1" name="Reason" type="xs:string" />
        </xs:sequence>
      </xs:extension>
    </xs:complexContent>
  </xs:complexType>
```

```

<xs:complexType name="MessageBaseV2">
  <xs:sequence>
    <xs:element minOccurs="1" maxOccurs="1" name="Sequence" type="xs:unsignedInt" />
    <xs:element minOccurs="0" maxOccurs="1" name="MessageId" type="xs:string" />
    <xs:element minOccurs="1" maxOccurs="1" name="EventTime" type="xs:dateTime" />
    <xs:element minOccurs="1" maxOccurs="1" name="MessageTime" type="xs:dateTime" />
    <xs:element minOccurs="0" maxOccurs="1" name="AdditionalData" type="ArrayOfPropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="ArrayOfPropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="unbounded" name="PropertyValue" nillable="true"
type="PropertyValue" />
  </xs:sequence>
</xs:complexType>
<xs:complexType name="PropertyValue">
  <xs:sequence>
    <xs:element minOccurs="0" maxOccurs="1" name="PropertyName" type="xs:string" />
    <xs:element minOccurs="0" maxOccurs="1" name="Value" type="xs:string" />
  </xs:sequence>
</xs:complexType>
</xs:schema>

```

9.1.13. Status

Characteristics

Message Type	HTTP POST
URI	api/ext/ami/v2/Status
Source	AMI server
Body	JSON/XML
Type	MessageProcessingUpdate
Frequency	Triggered by AmiMessageResponseV2

The AMI server uses this URL to provide a list of errors to the external application after receiving a batch of messages. If no errors are encountered, the Status message contains only the number of messages processed.

For the MessageProcessingUpdate schema, see [Status](#)